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Gamma-Type Functions

Arithmetic and Differential Independence, Probability Laws, and Interpolation

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Abstract

We develop a common arithmetic framework for gamma-type functions and apply it to differential independence, probability laws, and interpolation. Normalized fractional iterated integrals form an arithmetic shuffle algebra: at nonzero algebraic arguments, their relations over the field of algebraic exponential values are exactly the shuffle relations. The resulting independent coordinates yield relative transcendence beyond the rational-shift polyexponential tower, reflection-normalized incomplete-gamma values, and Gaussian normalization and zero results. A coefficient-growth argument proves joint differential algebraic independence of incomplete-gamma order functions over a specified differential field of complete gamma functions. For truncated Gaussian samples, we classify algebraic rank-event probabilities and obtain an infinite algebraically independent comparison family whose probability transforms are differentially transcendental. For Bessel bridges, we classify algebraic optimal boundaries, stopping-time moment fields, exceptional moments, and joint reward–law arithmetic. A complementary sampling analysis gives exact prime residues, denominator growth, and stride-dependent summability for reciprocal-gamma approximants. Finally, we prove the Newton expansion proposed for the principal inverse gamma function and give a counterexample to the published rational-root criterion for factorial reduction. All essential arguments are included; classical inputs and unresolved arithmetic targets are identified. No irrationality or transcendence conclusion for the Euler–Mascheroni constant is obtained.

Consolidated research manuscript

Proofs, shared notation, and references in one self-contained source

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1 Introduction and logical organization

Arithmetic independence, differential independence, and numerical specialization are distinct questions. The purpose of this article is to organize a collection of gamma-related results around explicit transfers between them. One transfer proceeds from primitive obstructions to iterated integrals, from iterated integrals to E-functions, and from E-functions to algebraic independence of values. A second transfer compares the transcendence-degree growth of Taylor coefficients with differential dimension. Probability and stopping-time models supply natural objects on which these mechanisms can be tested. Interpolation and recurrence arithmetic provide a complementary route, including precise limitations of rational sampling.

The article consolidates ten predecessor research drafts. Repeated arithmetic lemmas have been replaced by a common proof, and the two stopping-time treatments have been integrated. Every theorem used from an earlier draft is either proved here or replaced by a precise reference to a classical published result. No companion manuscript, external calculation file, or external bibliography is needed to follow the arguments.

1.1 Main results and dependencies

1. **A shared arithmetic algebra.** Theorem 5.1 identifies all relations among the normalized iterated-integral values. Corollary 6.1 extracts the joint independence of the rational-shape logarithmic moment towers and records their shift reductions. The same algebra produces new values transcendental over the full depth-one field.

2. **Normalization and differential dimension.** Reflection removes complete-gamma normalizations from certain probability products. Conjugation controls a common real normalization for complex Gaussian values. Coefficient recovery at a possibly singular center then turns independent arithmetic arrays into relative functional differential independence.

3. **Rank probabilities.** For Gaussian densities truncated to $[0, 1]$ with positive algebraic precisions, rank-event algebraicity has an exact combinatorial classification. Distinct precisions also yield an infinite algebraically independent comparison sequence and differentially transcendental probability transforms.

4. **Stopping boundaries and laws.** For positive rational Bessel dimension d and reward exponent q, the radial optimal boundary is algebraic exactly when $q - d \in 2\mathbb{Z}_{\geq 0}$. At an algebraic barrier, the whole integer residual-time moment field has transcendence degree one, with algebraic moments characterized by Laguerre zeros. Joint reward–law arithmetic and even-power rewards are classified on the stated parameter families.

5. **Sampling and two explicit conjectures.** Reciprocal-gamma sampling has exact prime-residue and reduced-denominator bounds. Separately, Newton–Stieltjes interpolation proves the inverse gamma expansion proposed by Ulgenes. A polynomial continued fraction with irrational polynomial roots disproves the rational-root criterion for factorial reduction in the published 2024 formulation of Ben David et al. The distinction from its earlier formulation is retained.

Parts II and III use the shared arithmetic results of Part I. Part IV uses classical E-function specialization and the shared exponential-integral input. The sampling and interpolation arguments in Parts V and VI are largely independent of the shuffle-algebra construction.

1.2 Notation and the status of the claims

Throughout, $\overline{\mathbb{Q}}$ is the field of algebraic numbers, $\mathcal{E} = \overline{\mathbb{Q}}(e^{\beta} : \beta \in \overline{\mathbb{Q}})$, and $\mathcal{T}$ is the depth-one field defined in (2.1). In particular, $\mathcal{E}$ and $\mathcal{T}$ do not by definition contain π or arbitrary complete

gamma values. Probability and expectation are denoted by $\mathbb{P}$ and $\mathbb{E}$, so neither is confused with $\mathcal{E}$. The symbols $F_{a,q}$ and Ein_q retain their common meaning across the article. Other letters, including H, P, Q, and C, are defined locally in each mathematical setting. The order-variable differential field $\mathscr{B}$ is defined separately in Part III.

For an infinite family, algebraic or differential algebraic independence means independence of every finite subfamily. Differential transcendence is a property of a function; it does not imply transcendence of an arbitrarily chosen value. A theorem at nonzero algebraic arguments is not extended to transcendental arguments without a separate argument.

The manuscript supplies proofs from the stated classical inputs and distinguishes them from conditional criteria. Internal verification does not establish historical priority or replace independent peer review. The two explicitly identified conjectures are addressed in their stated formulations; no general resolution of a major transcendence problem is claimed. In particular, the arguments leave the arithmetic nature of the Euler–Mascheroni constant unresolved.

Part I
A common arithmetic shuffle algebra

2 The concrete relative-transcendence conclusion

Put

$$\mathcal{E} = \overline{\mathbb{Q}}(e^{\beta} : \beta \in \overline{\mathbb{Q}}), \qquad F_{a,q}(\lambda) = \sum_{n=0}^{\infty} \frac{(-\lambda)^n}{n!(n+a)^q}, \qquad \mathrm{Ein}_q(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} z^n}{n! n^q}.$$

Unless specified otherwise, $0 < a < 1$ is rational, $q \geq 1$ is an integer, and $\lambda \in \overline{\mathbb{Q}}^{\times}$. Define the entire depth-one field

$$\mathcal{T} = \mathcal{E}\big(F_{a,q}(\lambda), \mathrm{Ein}_q(\lambda) : a \in \mathbb{Q} \cap (0,1),\ q \geq 1,\ \lambda \in \overline{\mathbb{Q}}^{\times}\big). \tag{2.1}$$

For positive a, these functions encode the normalized incomplete-gamma integral and its shape derivatives:

$$F_{a,1}(\lambda) = \int_0^1 t^{a-1} e^{-\lambda t}\, dt, \qquad \partial_a^{q-1} F_{a,1}(\lambda) = (-1)^{q-1}(q-1)! F_{a,q}(\lambda).$$

Thus the base field includes the full rational-shape tower, not only a finite collection of selected constants. All statements about infinite families mean that every finite subfamily has the indicated independence property.

Theorem 2.1. *For $H_n = \sum_{k=1}^{n} 1/k$, the pre-existing dressed polyexponential value*

$$C = \mathrm{EL}_{1,1}(1) = \sum_{n=1}^{\infty} \frac{H_n}{n\, n!} = \int_0^1 \frac{e^t \,\mathrm{Ein}_1(t)}{t}\, dt \tag{2.2}$$

is transcendental over $\mathcal{T}$. Numerically, $C = 1.5030087998883739079\ldots$. More generally, for every fixed nonzero algebraic α, the families

$$\{\mathrm{EL}_{m,n}(\alpha) : m \geq n \geq 1\}, \qquad \{\mathrm{EL}_{m,n}(-\alpha) : m > n \geq 1\} \tag{2.3}$$

are jointly algebraically independent over $\mathcal{T}$. This also holds jointly for all positive real algebraic α.

The field in (2.1) can equivalently include every rational shape $a \notin \{0, -1, -2, \ldots\}$: the additional values already belong to it. Corollary 6.1 proves this reduction and supplies the common vertical-value theorem used in the later parts of this article.

3 The normalized fractional alphabet

Use the following countable alphabet, always restricted to a finite subalphabet when applying a finite-dimensional theorem:

letter	differential form	shape
0	dz/z	0
x_λ	$(e^{\lambda z} - 1)\, dz/z$	0
$y_{a,\lambda}$	$z^{a-1} e^{\lambda z}\, dz$	a

$(0 < a < 1,\ a \in \mathbb{Q},\ \lambda \in \overline{\mathbb{Q}}^{\times})$.

A word is called admissible when it is empty or its last letter is not 0. Use the outer-first convention for iterated integrals:

$$I_{\varnothing} = 1, \qquad I_{\ell v}(z) = \int_0^z \omega_\ell(t) I_v(t).$$

Choose the positive real branch initially; analytic continuation or ramification supplies the equivalent germ formulation. Every admissible integral converges at its lower endpoint. Let A_w be the sum of the shapes of its letters, and set

$$J_w(z) = z^{-A_w} I_w(z), \qquad J_\varnothing = 1. \tag{3.1}$$

Despite the fractional form of the individual integrals, J_w is an entire function of z with Taylor coefficients in $\overline{\mathbb{Q}}$.

Lemma 3.1 (Recursions and E-functions)**.** *Every admissible J_w is an E-function. For $D = d/dz$, its recursion is*

$$(zD + A_w)J_{0v} = J_v, \tag{3.2}$$

$$(zD + A_w)J_{x_\lambda v} = (e^{\lambda z} - 1)J_v, \tag{3.3}$$

$$(zD + A_w)J_{y_{a,\lambda} v} = e^{\lambda z} J_v. \tag{3.4}$$

Moreover, normalization preserves the ordinary shuffle product:

$$J_u J_v = \sum_{w \in u \sqcup\!\sqcup v} J_w, \tag{3.5}$$

where multiplicities of identical shuffled words are included.

Proof. Differentiating $I_w = z^{A_w} J_w$ gives (3.4); for a y letter, the identity $A_w = a + A_v$ cancels the fractional power exactly. If a right-hand side is $h(z) = \sum h_n z^n/n!$, the corresponding coefficients are $h_n/(n + A_w)$ whenever $A_w > 0$. For $A_w = 0$, the nonempty admissible solution has constant coefficient zero, and the formula is h_n/n for $n \geq 1$; in this case the right-hand side has constant coefficient zero.

These operations preserve the arithmetic estimates defining E-functions. For completeness, multiplication by $e^{\lambda z}$ replaces the exponential-series coefficients by $\sum_{k=0}^{n} \binom{n}{k} \lambda^k h_{n-k}$, so the bounds for every algebraic conjugate remain exponential. Division by $n + A$, with a fixed nonnegative rational A, preserves these bounds. If $A = r/M > 0$, the new common denominators through index n divide the old ones times

$$\operatorname{lcm}\{Mk + r : 0 \leq k \leq n\},$$

up to a fixed integer factor; this least common multiple divides $\operatorname{lcm}(1, \ldots, Mn + r)$ and is exponentially bounded in n. For $A = 0$, use $\operatorname{lcm}(1, \ldots, n)$ instead. Products of finitely many such bounds are still exponential. The functions are entire by the coefficient estimates, and are holonomic by the first-order equations with holonomic right-hand sides; alternatively, the explicit finite system in Section 5 gives this directly.

The ordinary iterated-integral shuffle identity is valid for the convergent integrals. Every shuffle w of u, v satisfies $A_w = A_u + A_v$, so dividing the identity by $z^{A_u + A_v}$ proves (3.5) without modifying any structure constant. □

4 Functional freeness after ramification

The required input from iterated-integral theory is the following precise criterion: if the coefficient forms $f_\ell(t)\,dt$ over a differential field K of characteristic zero have no nonzero constant linear combination in dK, then all word coefficients of a group-like solution of $S' = (\sum_\ell f_\ell \ell)S$ are linearly independent over K. This is the applicable implication of Theorem 1 in Deneufchâtel–Duchamp–Hoang Ngoc Minh–Solomon [1]. The actual theorem is not restricted to Fuchsian coefficient forms.

Lemma 4.1 (Primitive obstruction)**.** *Fix finitely many letters and any finitely many additional algebraic exponents. Let $M \geq 2$ clear the denominators of the shapes, put $z = t^M$, and let $K = \mathbb{C}(t, e^{\nu_1 t^M}, \dots, e^{\nu_d t^M})$, where $\nu_1, \dots, \nu_d$ form a $\mathbb{Z}$-basis for the additive group generated by the exponents in question. No nonzero complex linear combination of the pulled-back alphabet forms has a primitive in K.*

Proof. The pulled-back forms are

$$M\frac{dt}{t}, \qquad M\frac{e^{\lambda t^M}-1}{t}\,dt, \qquad Mt^{r_a-1}e^{\lambda t^M}\,dt, \quad r_a = Ma \in \{1, \dots, M-1\}.$$

The exponential characters $X_j = e^{\nu_j t^M}$ are algebraically independent over $\mathbb{C}(t)$. Indeed, an algebraic relation would give a finite rational linear combination of exponentials with distinct exponents; along a generic ray at infinity, their distinct exponential growth rates rule out such a relation. All differential fields here and their iterated-integral extensions can be realized as fields of analytic germs away from zero. Their constant field is therefore exactly $\mathbb{C}$. Embed $K = \mathbb{C}(t)(X_1, \dots, X_d)$ in an iterated Laurent-series field. Differentiation acts coefficientwise, with $D(X^m) = M\mu t^{M-1}X^m$ for $\mu = \sum_j m_j\nu_j$.

Suppose $g \in K$ is a primitive of a constant combination, with coefficients c_0 for 0, d_λ for x_λ, and $c_{a,\lambda}$ for $y_{a,\lambda}$. At every nonzero character weight μ outside the finite forcing support, its rational Laurent coefficient R would satisfy

$$R' + M\mu t^{M-1}R = 0.$$

There is no nonzero rational solution: a finite pole is impossible by its highest-order derivative term, and a nonzero polynomial is impossible by comparison of leading degrees. The zero-weight homogeneous coefficient is constant. Thus the expansion has finite support, apart from that constant.

At a forcing weight $\lambda \neq 0$, a rational coefficient must satisfy

$$R' + M\lambda t^{M-1}R = Md_\lambda/t + M\sum_a c_{a,\lambda}t^{r_a-1}. \tag{4.1}$$

No finite pole of R is possible: the right-hand side is regular away from zero and has at most a simple pole at zero, whereas differentiating a pole of R produces a pole of strictly higher order that the other left-hand term cannot cancel. Thus R is polynomial and $d_\lambda = 0$. The polynomial on the right has degree at most $M-2$. If $R \neq 0$, the left has degree $\deg R + M - 1 \geq M - 1$, a contradiction. Hence $R = 0$, and distinct a give distinct powers, forcing every $c_{a,\lambda} = 0$. Finally, the zero character requires $R_0' = M(c_0 - \sum d_\lambda)/t$; the residue of a rational derivative is zero, so $c_0 = 0$ as well. □

Proposition 4.2 (Faithful normalized shuffle algebra)**.** *For every finite alphabet and the field K in Lemma 4.1, the admissible functions $J_w(t^M)$ are linearly independent over K. Consequently the map from the abstract admissible shuffle algebra to the functions J_w is injective. With 0 the least letter in any total order, the J_l indexed by Lyndon words $l \neq 0$ are algebraically independent over K after ramification.*

Proof. Apply the stated iterated-integral independence criterion to the pulled-back alphabet, using Lemma 4.1. For precision, the solution at zero is interpreted tangentially. At every finite word truncation, the only logarithmic residue comes from $M\,dt/t$ on the letter 0; the other forms are holomorphic at zero. The normalized tangential solution has admissible word coefficients equal to the convergent integrals $I_w(t^M)$. Thus these coefficients are linearly independent over K. Multiplying the coefficient indexed by w by the nonzero element $t^{-MA_w} \in \mathbb{C}(t)$ preserves that linear independence and gives $J_w(t^M)$.

A nonempty word is Lyndon if it is strictly smaller, lexicographically, than each of its proper nonempty suffixes. The shuffle algebra in characteristic zero is a polynomial algebra on Lyndon words [2]. When 0 is least, all Lyndon words other than the one-letter word 0 end in a nonzero letter. The admissible shuffle subalgebra is the polynomial algebra on precisely these generators: shuffle regularization splits the full algebra as the polynomial extension of the admissible algebra by the letter 0. Injectivity and (3.5) now prove the last assertion. □

5 Arithmetic specialization without exceptional nonzero points

Theorem 5.1 (Arithmetic freeness)**.** *At every $\alpha \in \overline{\mathbb{Q}}^{\times}$, the numbers $J_w(\alpha)$ indexed by all admissible words are linearly independent over $\mathcal{E}$. Equivalently, every polynomial relation over $\mathcal{E}$ between their values is exactly a shuffle relation. In particular, all admissible Lyndon values are jointly algebraically independent over $\mathcal{E}$.*

Proof. Fix a finite alphabet, a word bound N, and a finite set B of algebraic starting exponents. Let Λ be the set of exponents attached to its x and y letters. For each admissible or empty word w of length at most N, include the E-functions

$$U_{\mu,w}(z) = e^{\mu z} J_w(z), \quad \mu = b + \lambda_1 + \cdots + \lambda_k, \quad b \in B, \quad 0 \le k \le N - |w|, \quad \lambda_j \in \Lambda.$$

Repeated indices (μ, w) are retained only once. The recursions in Lemma 3.1 give the closed system

$$\begin{aligned} U'_{\mu,\varnothing} &= \mu U_{\mu,\varnothing}, \\ U'_{\mu,0v} &= (\mu - A_{0v}/z) U_{\mu,0v} + U_{\mu,v}/z, \\ U'_{\mu,x_\lambda v} &= (\mu - A_{x_\lambda v}/z) U_{\mu,x_\lambda v} + (U_{\mu+\lambda,v} - U_{\mu,v})/z, \\ U'_{\mu,y_{a,\lambda} v} &= (\mu - A_{y_{a,\lambda} v}/z) U_{\mu,y_{a,\lambda} v} + U_{\mu+\lambda,v}/z. \end{aligned}$$

The coefficient matrix lies in $\overline{\mathbb{Q}}[z, 1/z]$; its only possible finite singularity is zero. The exponent budget closes because passing to a suffix decreases word length by one.

These coordinates are linearly independent over $\mathbb{C}(z)$. After ramification, a putative relation can first be grouped by w; Proposition 4.2 makes each exponential coefficient zero. Distinct exponential characters are linearly independent over $\mathbb{C}(t)$, so each original rational coefficient is zero. The linear E-function specialization theorem (equivalently, the linear case of Beukers's refinement of Siegel–Shidlovskii [3, Corollary 1.4]) therefore makes the values $U_{\mu,w}(\alpha)$ linearly independent over $\overline{\mathbb{Q}}$ at every nonzero algebraic α. Its nonsingularity hypothesis holds because the displayed system has no other finite pole.

Finally suppose a finite relation $\sum_w c_w J_w(\alpha) = 0$ has $c_w \in \mathcal{E}$. Clear the common denominator in this finitely generated subfield of $\mathcal{E}$ and expand each resulting numerator as a finite algebraic linear combination of exponential values. Combine equal exponents. Choose B to contain β/α for all exponents e^β that occur. The resulting relation is a $\overline{\mathbb{Q}}$-linear relation between the independent $U_{\mu,w}(\alpha)$, hence all its coefficients vanish. This proves independence over $\mathcal{E}$. Shuffle multiplication and the polynomial Lyndon presentation then give exactly the remaining assertions. □

6 Canonical vertical values and rational shifts

Corollary 6.1 (The common vertical-value theorem)**.** *Let*

$$\mathcal{E} = \overline{\mathbb{Q}}(e^\beta : \beta \in \overline{\mathbb{Q}}), \qquad F_{a,q}(\lambda) = \sum_{n \ge 0} \frac{(-\lambda)^n}{n!(n+a)^q}, \qquad \mathrm{Ein}_q(\lambda) = \sum_{n \ge 1} \frac{(-1)^{n-1}\lambda^n}{n! n^q}.$$

The canonical family

$$\{F_{a,q}(\lambda) : a \in \mathbb{Q} \cap (0,1),\ q \geq 1, \lambda \in \overline{\mathbb{Q}}^{\times}\} \ \cup\ \{\mathrm{Ein}_q(\lambda) : q \geq 1,\ \lambda \in \overline{\mathbb{Q}}^{\times}\}$$

is jointly algebraically independent over $\mathcal{E}$. Here and throughout, q is an integer, and independence of an infinite family means independence of each finite subfamily.

For every $a \in \mathbb{Q} \setminus \{0, -1, -2, \ldots\}$ and $q \geq 1$, the exact shift identity is

$$\lambda F_{a+1,q}(\lambda) = aF_{a,q}(\lambda) - F_{a,q-1}(\lambda), \qquad F_{a,0}(\lambda) = e^{-\lambda}. \tag{6.1}$$

The positive-integer starting values are

$$F_{1,1}(\lambda) = \frac{1 - e^{-\lambda}}{\lambda}, \qquad F_{1,q}(\lambda) = \frac{\mathrm{Ein}_{q-1}(\lambda)}{\lambda} \quad (q \geq 2). \tag{6.2}$$

Consequently adjoining all such rational shapes does not enlarge the field $\mathcal{T}$ in (2.1). Values at $\lambda = 0$ are a^{-q} and add no transcendental generators.

Proof. At $z = 1$ the exact vertical-word dictionary is

$$J_{0^{q-1}y_{a,-\lambda}}(1) = F_{a,q}(\lambda), \qquad J_{0^{q-1}x_{-\lambda}}(1) = -\,\mathrm{Ein}_q(\lambda). \tag{6.3}$$

Each additional zero integration divides the exponential-series coefficient by $n+a$, or by n in the x case. With 0 least in the alphabet, every displayed word is Lyndon, and distinct canonical indices give distinct words. Theorem 5.1 at 1 therefore proves the asserted joint algebraic independence. It also identifies $\mathcal{T}$ precisely as the field generated over $\mathcal{E}$ by these vertical Lyndon coordinates.

To prove (6.1), compare coefficients of $(-\lambda)^n/n!$. For $n \geq 1$ the coefficient on either side is $-n/(n+a)^q$; for $n = 0$ both sides vanish. This includes $q = 1$ under the convention $F_{a,0} = e^{-\lambda}$. Reindexing the defining series proves (6.2).

For a noninteger rational shape, the shift recurrence reduces finite positive integer shifts to its representative in $(0, 1)$. Negative shifts are reduced by solving for $F_{a,q}$ and inducting on q; division by a is harmless because these shapes are noninteger. Positive integer shapes start from (6.2) and then use the same recurrence. This proves the field assertion. Finally, setting $\lambda = 0$ in the defining series gives a^{-q}. □

7 The depth-two arithmetic layer

For the dressed functions of Aminov–Arnaudo [4], the defining recursion is

$$\begin{aligned} z\mathrm{EL}'_{1,s_2,\ldots,s_r}(z) &= -e^z \mathrm{EL}_{s_2,\ldots,s_r}(-z) && (r > 1), \\ z\mathrm{EL}'_{s_1,\ldots,s_r}(z) &= \mathrm{EL}_{s_1-1,s_2,\ldots,s_r}(z) && (s_1 > 1), \end{aligned}$$

with $\mathrm{EL}_n(z) = \sum_{k\geq 1} z^k/(k!k^n)$ and zero initial values. These definitions give

$$\mathrm{EL}_{m,n}(\alpha) = -J_{0^{m-1}x_\alpha 0^{n-1}x_{-\alpha}}(1) - J_{0^{m+n-1}x_{-\alpha}}(1). \tag{7.1}$$

Only shape-zero letters occur here, so $I_w = J_w$ for these words.

Proof of Theorem 2.1. Choose 0 least and $x_\alpha < x_{-\alpha}$. For $m > n$, both words $0^{m-1}x_{\pm\alpha}0^{n-1}x_{\mp\alpha}$ are Lyndon: their initial run of zeros is longer than the initial zero run of any suffix starting in the second block, while a suffix starting inside the first block has fewer initial zeros. Suffixes starting in a nonzero letter are larger. For $m = n$, the positive orientation is Lyndon by the chosen ordering of $x_\alpha, x_{-\alpha}$. These mixed Lyndon words are distinct and are not vertical words. Theorem 5.1 therefore makes their values jointly algebraically independent over the field generated

by all vertical ones, namely $\mathcal{T}$. The final term in (7.1) belongs to $\mathcal{T}$; translating the independent coordinates by these base-field elements proves (2.3). For positive real algebraic α, the pairs $\{\alpha, -\alpha\}$ are disjoint, so all the indicated orderings can be chosen simultaneously.

For $\alpha = m = n = 1$, the dictionary is $C = -J_{x_1x_{-1}}(1) - J_{0x_{-1}}(1)$, proving the first assertion. To verify the familiar-looking form (2.2), the coefficient identity

$$e^t \operatorname{Ein}_1(t) = \sum_{n\geq 1} H_n \frac{t^n}{n!}$$

follows from $\sum_{k=1}^{n}(-1)^{k-1}\binom{n}{k}/k = H_n$. Its termwise integral after division by t gives the stated sum. □

For each fixed pair $\pm\alpha$, total weight $m + n = N$ supplies $N - 1$ independent new depth-two coordinates; weights at most W supply $W(W-1)/2$. The complementary orientations reduce by Lyndon shuffle reduction to the selected coordinates and vertical products. Thus the displayed family is also a transcendence basis over $\mathcal{T}$ for the field generated by all depth-two dressed values at that pair. Indeed, Lyndon shuffle reduction preserves the number of occurrences of every letter. A word containing one letter of each sign reduces only to a mixed Lyndon coordinate or to products of two vertical coordinates. The admissible mixed Lyndon words are exactly the two orientations with the first zero run longer, together with the preferred orientation when the zero runs have equal length.

Exact relation algorithm. For any finite collection of the dressed functions, replace each by its finite regularized word expansion, expand products by shuffle, and collect word coefficients. A polynomial relation over $\mathcal{E}$ at any fixed nonzero algebraic point holds if and only if this finite word polynomial is zero. The same algorithm works when the normalized fractional iterated integrals are included. This is a finite identity test and, through polynomial elimination in finitely many Lyndon coordinates, a description of the full relation ideal of a fixed finite tuple. It does not assert that the original alternating dressed family alone is linearly closed under products.

8 Scope of the common arithmetic framework

The functional independence criterion and the Lyndon presentation used above are classical. The construction here combines the rational-shape and exponential-integral alphabets, normalizes their fractional powers, and transfers the resulting full shuffle algebra to algebraic-point values. Ramification is used solely to establish functional independence. The specialization step takes place in the original variable z; no assertion that e^{t^M} is an E-function is used.

The conclusion is relative transcendence over the explicitly defined field $\mathcal{T}$. In particular, it says more than ordinary transcendence of the harmonic-factorial constant C: it cannot be algebraic over any finite collection of the canonical depth-one values defined here. The statement does not adjoin π or complete gamma values to $\mathcal{T}$, and proves no separate claim about Euler's constant.

Aminov–Arnaudo [4] define the dressed functions used here. Related arithmetic results for integrals involving Kummer functions appear in Gorelov–Voronov [5]. The proof in this article is independent of that work. Their Theorem 2 concerns fixed-parameter integrated Kummer functions and derivatives in the argument. The precise adjacent family at exponential parameter zero has the expansion

$$V_{\lambda,\nu,0}(z) = 1 + \sum_{n\geq 1} \frac{\nu z^n}{(\nu+n)(\lambda+1)\cdots(\lambda+n)}.$$

Termwise differentiation near $(\lambda, \nu) = (0, 0)$ gives

$$\mathrm{EL}_{1,1}(z) = -\,\partial_\lambda \partial_\nu V_{\lambda,\nu,0}(z)|_{\lambda=\nu=0}\,.$$

Those parameter derivatives at a degenerate parameter are not among the derivatives in the argument appearing in their theorem. Their statement therefore does not directly supply the relative result over the all-orders field $\mathcal{T}$. A complete comparison of overlapping subfamilies remains part of the priority assessment. No claim of being the first theorem on nested polyexponential values, or of resolving a named prior transcendence conjecture, is made. The preceding sections supply the common arithmetic framework for the applications developed in the remaining parts of this article.

Part II
Incomplete gamma normalization and shape derivatives

9 Reflection-normalized incomplete-Gamma values

Let $P(a,x)=\Gamma(a)^{-1}\int_0^x t^{a-1}e^{-t}\,dt$ for $a,x>0$. This part determines the arithmetic nature of concrete combinations of these probabilities by eliminating their complete-Gamma normalization before specialization. For example, the three exact real numbers

$$\frac{\operatorname{erf}(2)}{2\operatorname{erf}(1)},\qquad \frac{2P(1/3,1)P(2/3,1)}{\sqrt{3}\operatorname{erf}(1)^2},\qquad \frac{P(1/4,2)P(3/4,2)}{\sqrt{2}\operatorname{erf}(1)^2} \tag{9.1}$$

are jointly algebraically independent, even over

$$\mathcal{E}=\overline{\mathbb{Q}}(e^{\beta}:\beta\in\overline{\mathbb{Q}}). \tag{9.2}$$

In particular, each is transcendental. The conclusions below apply simultaneously to infinitely many shape parameters, algebraic cuts, and shape derivatives. For an infinite family, algebraic independence always means independence of every finite subfamily with distinct indices.

Reflection cancels the complete-Gamma factors exactly. The surviving coordinates are logarithmic moments of e^{-xt} controlled by the common arithmetic theorem. The consequences below follow by explicit rational changes of these independent coordinates; the Gaussian normalization results then add a conjugation argument.

The starting point is classical: nonintegral shifted polyexponentials already occur in Shidlovskii's algebraic-independence theory [6, 7], and incomplete-Gamma transcendence is studied in [8]. Related multiparameter integrals are treated in [5]. The common arithmetic theorem supplies the joint moment input; we now derive its exact relation calculus and explicit reflection-normalized consequences. The proof uses no unproved assertion about complete Gamma values or Euler's constant.

10 Reflection-normalized probabilities and shape jets

Write

$$\mathcal{E}=\overline{\mathbb{Q}}(e^{\beta}:\beta\in\overline{\mathbb{Q}}),\qquad F_{a,q}(z)=\sum_{n\ge0}\frac{(-z)^n}{n!(n+a)^q}\quad(a\in\mathbb{Q}\cap(0,1),\ q\ge1).$$

The arithmetic input for this section is the following nonresonant multipoint theorem, recorded in Corollary 11.1:

> The family $\{F_{a,q}(\alpha):a\in\mathbb{Q}\cap(0,1),\ q\ge1,\ \alpha\in\overline{\mathbb{Q}}^{\times}\}$ is algebraically independent over $\mathcal{E}$, in the finite-subset sense.

Its one-shift specialization belongs to the classical theory of Shidlovskii. The consequences below are proved directly from the stated joint version; no assumption about complete Gamma values, Euler's constant, or an intersection of period rings is used.

For $u,x>0$, put

$$\gamma_<(u,x)=\int_0^x t^{u-1}e^{-t}\,dt,\qquad P(u,x)=\frac{\gamma_<(u,x)}{\Gamma(u)}.$$

Thus P is the regularized lower incomplete-Gamma function and $P(1/2,1)=\operatorname{erf}(1)$. Set

$$B=F_{1/2,1}(1)^2>0.$$

Theorem 10.1 (Reflection-normalized probability-jet independence)**.** *For $a \in \mathbb{Q} \cap (0, 1/2)$, $\alpha \in \overline{\mathbb{Q}}_{>0}$, and $|s| < a$, define*

$$\mathcal{R}_{a,\alpha}(s) = \frac{P(a+s,\alpha)P(1-a-s,\alpha)}{\alpha \sin(\pi(a+s))\,\mathrm{erf}(1)^2}, \qquad C_{a,\alpha,k} = [s^k]\mathcal{R}_{a,\alpha}(s). \tag{10.1}$$

For complex u with $\Re u > 0$, $F_{u,q}$ is defined by the same convergent series. The functions in the displayed formula are continued analytically from positive real parameters; the displayed disk keeps both shape parameters in the right half-plane and contains no zero of the sine denominator. Then

$$\mathcal{R}_{a,\alpha}(s) = \frac{F_{a+s,1}(\alpha)F_{1-a-s,1}(\alpha)}{B}, \tag{10.2}$$

and

$$C_{a,\alpha,k} = \frac{1}{B}\sum_{j=0}^{k}(-1)^j F_{a,j+1}(\alpha)F_{1-a,k-j+1}(\alpha). \tag{10.3}$$

The complete family

$$\{C_{a,\alpha,k} : a \in \mathbb{Q} \cap (0,1/2),\ \alpha \in \overline{\mathbb{Q}}_{>0},\ k \geq 0\}$$

is algebraically independent over $\mathcal{E}$. Moreover, every coefficient is real and has the strict sign

$$(-1)^k C_{a,\alpha,k} > 0. \tag{10.4}$$

Proof. Termwise integration gives $\gamma_<(u,\alpha) = \alpha^u F_{u,1}(\alpha)$. Euler's reflection formula [9, §5.5] and $\Gamma(1/2)^2 = \pi$ give

$$P(u,\alpha)P(1-u,\alpha) = \frac{\alpha\sin(\pi u)}{\pi}F_{u,1}(\alpha)F_{1-u,1}(\alpha), \qquad \mathrm{erf}(1)^2 = \frac{B}{\pi}.$$

This proves (10.2). In particular the complete Gamma factors, the power of the cut, and the common factor π cancel exactly; no numerical algebraic-independence assertion about any of these constants is required.

Uniformly on compact subsets of $|s| < a$,

$$F_{a+s,1}(\alpha) = \sum_{j\geq 0}(-1)^j F_{a,j+1}(\alpha)s^j, \qquad F_{1-a-s,1}(\alpha) = \sum_{j\geq 0} F_{1-a,j+1}(\alpha)s^j.$$

Multiplication proves (10.3); these are Taylor coefficients, so the k-th derivatives are $k!C_{a,\alpha,k}$.

We prove independence in an arbitrary finite collection of blocks (a,α) and orders. Enlarge the collection, if necessary, to include all orders from zero to its maximum in every block. Write

$$X_{a,\alpha,j} = F_{a,j+1}(\alpha), \qquad Y_{a,\alpha,j} = F_{1-a,j+1}(\alpha).$$

Adjoin to $\mathcal{E}$ the anchor $F_{1/2,1}(1)$ and all the finitely many Y's involved, obtaining a field L. The arithmetic input says that all the X's in the finite collection are algebraically independent over L. There is no overlap between the three sets of coordinates: their rational shapes lie respectively below $1/2$, above $1/2$, and at $1/2$. For each block,

$$C_{a,\alpha,k} = \frac{(-1)^k Y_{a,\alpha,0}}{B}X_{a,\alpha,k} + \frac{1}{B}\sum_{j<k}(-1)^j X_{a,\alpha,j}Y_{a,\alpha,k-j}.$$

The coefficient of the new coordinate $X_{a,\alpha,k}$ is nonzero. These equations form an invertible triangular affine change of coordinates over L. Thus all the selected C's are algebraically independent over L, hence over $\mathcal{E}$. Removing any extra orders that were adjoined proves the stated finite-subset result.

For the signs, let $t = 1/2 - a > 0$ and set $u = \log(x/y)$. Differentiating the convergent double integral gives

$$k!BC_{a,\alpha,k} = \int_0^1 \int_0^1 (xy)^{-1/2} e^{-\alpha(x+y)} u^k e^{-tu}\, dx\, dy.$$

The weight preceding $u^k e^{-tu}$ is positive and symmetric in x, y. Pair the two triangles $x > y$ and $x < y$. On $x > y$, one has $u > 0$, and the paired factor is

$$u^k(e^{-tu} + (-1)^k e^{tu}).$$

It is positive for even k and negative for odd k. Each triangle has positive measure, so the signs are strict. This proves (10.4). □

Corollary 10.2 (Normalized moments and logarithmic jets)**.** *Define*

$$M_{a,\alpha,k} = \frac{C_{a,\alpha,k}}{C_{a,\alpha,0}} \quad (k \geq 1), \qquad \kappa_{a,\alpha,k} = [s^k] \log \frac{\mathcal{R}_{a,\alpha}(s)}{\mathcal{R}_{a,\alpha}(0)} \quad (k \geq 1).$$

In each of the following two families all entries are jointly algebraically independent over $\mathcal{E}$:

$$\{C_{a,\alpha,0}\}_{a,\alpha} \cup \{M_{a,\alpha,k}\}_{a,\alpha,k\geq 1}, \qquad \{C_{a,\alpha,0}\}_{a,\alpha} \cup \{\kappa_{a,\alpha,k}\}_{a,\alpha,k\geq 1}.$$

These are two alternative coordinate systems; taking their union would create relations and is not asserted to preserve independence. In particular every normalized logarithmic derivative of positive order is transcendental over $\mathcal{E}$.

Proof. In every finite jet range, the changes

$$(C_0, C_1, \ldots, C_m) \longleftrightarrow (C_0, M_1, \ldots, M_m) \longleftrightarrow (C_0, \kappa_1, \ldots, \kappa_m)$$

are birational over $\mathbb{Q}$, since $C_0 \neq 0$ and

$$\kappa_k = M_k + Q_k(M_1, \ldots, M_{k-1}), \qquad Q_k \in \mathbb{Q}[T_1, \ldots, T_{k-1}].$$

The inverse is obtained by exponentiating the formal logarithmic series. Apply the theorem block by block, then pass to finite subsets. □

For example, suppressing the cut α in the right-hand sides,

$$\kappa_{a,\alpha,1} = \frac{F_{1-a,2}}{F_{1-a,1}} - \frac{F_{a,2}}{F_{a,1}},$$
$$\kappa_{a,\alpha,2} = \frac{F_{a,3}}{F_{a,1}} - \frac{1}{2}\left(\frac{F_{a,2}}{F_{a,1}}\right)^2 + \frac{F_{1-a,3}}{F_{1-a,1}} - \frac{1}{2}\left(\frac{F_{1-a,2}}{F_{1-a,1}}\right)^2.$$

Equivalently,

$$\kappa_{a,\alpha,1} = \partial_u \log P(u,\alpha)|_{u=a} - \partial_u \log P(u,\alpha)|_{u=1-a} - \pi\cot(\pi a).$$

The first two logarithmic coefficients satisfy $\kappa_{a,\alpha,1} < 0$ and $\kappa_{a,\alpha,2} > 0$. For the second assertion, the double-integral representation identifies $2\kappa_2$ with the variance of the nonconstant random variable $\log(x/y)$ under the corresponding positive probability density. No sign claim is made for higher logarithmic coefficients.

Corollary 10.3 (Compatible error-function ratios)**.** *For positive algebraic $r \neq 1$, set*

$$E_r = \frac{\operatorname{erf}(r)}{\operatorname{erf}(1)} = \frac{rF_{1/2,1}(r^2)}{F_{1/2,1}(1)}.$$

The entire family $\{E_r : r \in \overline{\mathbb{Q}}_{>0}, r \neq 1\}$ is algebraically independent over $\mathcal{E}$. It remains jointly algebraically independent with the complete family of reflection-normalized coefficients $C_{a,\alpha,k}$, or with either one of the two alternative coordinate systems in the preceding corollary.

Proof. The extra input coordinates $F_{1/2,1}(r^2)$, with $r > 0$ and $r \neq 1$, are distinct from the anchor and from every coordinate used in the reflection-normalized blocks. After adjoining the anchor and the complementary-shape coordinates as above, the change from these extra input coordinates to E_r is diagonal and invertible. The same triangular-coordinate proof therefore applies jointly. □

11 The shared arithmetic input

Recall $F_{a,0}(z) = e^{-z}$ and the regularized functions

$$\mathrm{Ein}_q(z) = \sum_{n\geq 1} \frac{(-1)^{n-1} z^n}{n^q n!} \quad (q \geq 1), \qquad \mathrm{Ein}_0(z) = 1 - e^{-z}.$$

The index q is a positive integer recording logarithmic moment order.

Corollary 11.1 (Joint fractional and regularized moment independence)**.** *The family*

$$\{F_{a,q}(\lambda) : a \in \mathbb{Q} \cap (0,1),\ q \geq 1, \lambda \in \overline{\mathbb{Q}}^{\times}\} \ \cup\ \{\mathrm{Ein}_q(\lambda) : q \geq 1,\ \lambda \in \overline{\mathbb{Q}}^{\times}\}$$

is jointly algebraically independent over $\mathcal{E}$. Repeated indices are included once, and the assertion means independence of every finite subfamily.

Proof. This is Corollary 6.1, obtained from the common arithmetic theorem, Theorem 5.1. Its proof provides the functional independence and ordinary-point specialization once for all applications in this part. □

12 Positive rational shapes: exact relations

For arbitrary $a \in \mathbb{Q}_{>0}$ the same series defines $F_{a,q}$. Direct coefficient comparison gives

$$zF_{a+1,q}(z) = aF_{a,q}(z) - F_{a,q-1}(z), \qquad q \geq 1,$$

where $F_{a,0} = e^{-z}$. At the integral representative,

$$F_{1,1}(z) = \frac{1 - e^{-z}}{z}, \qquad F_{1,q}(z) = \frac{\mathrm{Ein}_{q-1}(z)}{z} \quad (q \geq 2),$$

with removable values at zero. Iterating the shift formula at a nonzero algebraic λ therefore reduces every positive rational shape to fractional representatives in $(0,1)$ or to the regularized coordinates, together with an element of $\mathcal{E}$.

Corollary 12.1 (Exact finite relation ideal after shift reduction)**.** *Let $U_1, \ldots, U_n$ be any finite collection of positive rational-shape moments at nonzero algebraic parameters. Reduce them by the preceding identities to*

$$U = b + AV, \qquad b \in \mathcal{E}^n, \quad A \in \mathrm{Mat}_{n\times h}(\overline{\mathbb{Q}}),$$

where the coordinates of V are distinct elements of the independent basis in Corollary 11.1. Then the kernel of evaluation $\mathcal{E}[T_1, \ldots, T_n] \longrightarrow \mathbb{C}$, $T_i \mapsto U_i$, is generated by

$$\ell^{\mathsf{T}}(T - b), \qquad \ell^{\mathsf{T}} A = 0.$$

One may take a basis of this left nullspace over $\overline{\mathbb{Q}}$. In particular,

$$\mathrm{trdeg}_{\mathcal{E}}\, \mathcal{E}(U_1, \ldots, U_n) = \mathrm{rank}(A).$$

Proof. The basis coordinates V are algebraically independent over $\mathcal{E}$. An invertible linear change of variables reduces AV to a coordinate projection of rank $\operatorname{rank}(A)$. Its polynomial ideal is precisely the linear ideal of its image. Translation by b produces the displayed affine equations and no additional polynomial relations. □

Example 12.2 (A complete four-coordinate relation ideal)**.** Fix $\lambda \in \overline{\mathbb{Q}}^{\times}$ and put

$$U = (F_{1/2,1}(\lambda), F_{1/2,2}(\lambda), F_{3/2,1}(\lambda), F_{3/2,2}(\lambda)).$$

Writing $V = (F_{1/2,1}(\lambda), F_{1/2,2}(\lambda))$, the reduction is

$$U = \begin{pmatrix} 0 \\ 0 \\ -e^{-\lambda}/\lambda \\ 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1/(2\lambda) & 0 \\ -1/\lambda & 1/(2\lambda) \end{pmatrix} V.$$

Its rank is two. Consequently the entire ideal in $\mathcal{E}[T_1, T_2, T_3, T_4]$ is generated by the two explicit equations

$$\lambda T_3 - \tfrac{1}{2}T_1 + e^{-\lambda} = 0, \qquad \lambda T_4 - \tfrac{1}{2}T_2 + T_1 = 0.$$

Every polynomial relation follows from these equations; the first two coordinates are algebraically independent.

13 Explicit constants and the Euler-constant connection

13.1 Nine independent elementary logarithmic integrals

For $a > 0$, $q \geq 1$, and $x > 0$, termwise integration yields

$$F_{a,q}(x) = \frac{1}{(q-1)!} \int_0^1 t^{a-1} (-\log t)^{q-1} e^{-xt}\, dt. \tag{13.1}$$

The substitution $t = u^m$ gives, for $a = r/m$,

$$F_{r/m,q}(x) = \frac{m^q}{(q-1)!} \int_0^1 u^{r-1} (-\log u)^{q-1} e^{-xu^m}\, du. \tag{13.2}$$

Thus the following table describes nine exact integrals with elementary integrands. By Corollary 11.1, all nine are jointly algebraically independent over $\mathcal{E}$.

Shape a	$F_{a,1}(1)$	$F_{a,2}(1)$	$F_{a,3}(1)$
1/3	2.422533546419	8.516293965308	26.613445734703
1/2	1.493648265625	3.623761905289	7.732228218711
2/3	1.049688491642	1.949597732837	3.182354741914

The decimals identify the values; the independence assertion concerns the exact integrals, not the rounded numbers. Since $e \in \mathcal{E}$, these nine numbers together with e have transcendence degree ten over $\overline{\mathbb{Q}}$.

13.2 Reflected probability coefficients

For the notation of Theorem 10.1, the next table supplies three blocks of shape coefficients. All nine entries in the three C-columns are jointly algebraically independent over $\mathcal{E}$.

a	x	$C_{a,x,0}$	$C_{a,x,1}$	$C_{a,x,2}$
1/3	1	1.139812873291	−1.889967700256	8.535171780942
1/3	2	0.736365068175	−1.283786942203	5.809549164116
1/4	1	1.373538554303	−3.994719960590	18.726952443553

For example, the first coefficient in the table equals

$$C_{1/3,1,0} = \frac{9\left(\int_0^1 e^{-u^3}\,du\right)\left(\int_0^1 ue^{-u^3}\,du\right)}{4\left(\int_0^1 e^{-u^2}\,du\right)^2}. \tag{13.3}$$

This gives a concrete transcendental quotient of elementary definite integrals without any remaining complete-Gamma factor.

The three introductory constants in (9.1) are approximately

$$0.59055495933201584204, \qquad 1.13981287329096915814, \qquad 0.89528246545027779218.$$

Their joint independence follows from the compatible error-function corollary and distinct reflected blocks. Independence also holds if any finite number of positive-order shape coefficients of the two reflected blocks is included.

For the first block, the logarithmic Taylor coefficients are

$$\kappa_{1/3,1,1} = -1.658138580940584963 62\ldots, \qquad \kappa_{1/3,1,2} = 6.113510176946401213 59\ldots.$$

The second derivative of the logarithm is $2\kappa_{1/3,1,2}$; the factor of two distinguishes derivatives from Taylor coefficients.

13.3 Three independent combinations containing Euler's constant

Let $\gamma = \lim_{n\to\infty}(\sum_{j=1}^{n} j^{-1} - \log n)$ and define the convergent Gaussian tail moments

$$T_k = \int_1^\infty e^{-u^2}(\log u)^k\,du \qquad (k \ge 0). \tag{13.4}$$

Splitting the Gamma integral at one and substituting $t = u^2$ in its upper part gives the analytic identity, for $|s| < 1/2$,

$$F_{1/2+s,1}(1) = \Gamma(1/2+s) - 2\int_1^\infty e^{-u^2}u^{2s}\,du. \tag{13.5}$$

Differentiation under the integral is justified uniformly on compact subdisks by Gaussian decay. Therefore

$$F_{1/2,k+1}(1) = \frac{(-1)^k}{k!}\left(\Gamma^{(k)}(1/2) - 2^{k+1}T_k\right). \tag{13.6}$$

All expressions on the right, for $k \ge 0$, are jointly algebraically independent over $\mathcal{E}$ by the proved moment theorem.

Corollary 13.1 (Explicit Euler-constant combinations)**.** *The following three real numbers are jointly algebraically independent over $\mathcal{E}$:*

$$D_0 = \sqrt{\pi} - 2T_0, \tag{13.7}$$

$$D_1 = \sqrt{\pi}(\gamma + 2\log 2) + 4T_1, \tag{13.8}$$

$$D_2 = \sqrt{\pi}\left((\gamma + 2\log 2)^2 + \frac{\pi^2}{2}\right) - 8T_2. \tag{13.9}$$

In particular, each is transcendental, and

$$\operatorname{trdeg}_{\mathcal{E}}\,\mathcal{E}(\gamma, \pi, \log 2, T_0, T_1, T_2) \ge 3. \tag{13.10}$$

Proof. The classical special values

$$\Gamma(1/2) = \sqrt{\pi}, \quad \psi(1/2) = -\gamma - 2\log 2, \quad \psi'(1/2) = \pi^2/2$$

give $\Gamma' = \Gamma\psi$ and $\Gamma'' = \Gamma(\psi^2 + \psi')$; see [9, §§5.4,5.15]. Equation (13.6) identifies (D_0, D_1, D_2) with $(F_{1/2,1}(1), F_{1/2,2}(1), 2F_{1/2,3}(1))$. Nonzero rational rescaling preserves algebraic independence. Adjoining $\sqrt{\pi}$ to the field in (13.10) is an algebraic extension and puts all three independent quantities inside it, proving the lower bound. □

This corollary is a statement about the displayed combinations. It does not isolate the arithmetic nature of γ from the tail moments. The independent combinations and reflection-normalized constants are unconditional consequences of the shared arithmetic theorem.

13.4 Certified evaluation of the displayed constants

For $x > 0$, $a > 0$, and fixed $q \geq 1$, the terms of the series for $F_{a,q}(x)$ are alternating, with absolute ratio

$$\frac{x}{n+1}\left(\frac{n+a}{n+1+a}\right)^q \leq \frac{x}{n+1}.$$

If $N + 1 \geq x$, the truncation through $n = N$ satisfies the rigorous bound

$$\left| F_{a,q}(x) - \sum_{n=0}^{N} \frac{(-x)^n}{n!(n+a)^q} \right| \leq \frac{x^{N+1}}{(N+1)!(N+1+a)^q}. \tag{13.11}$$

For rational inputs the two consecutive truncations give rational enclosures, so quotient values with positive denominators can also be enclosed by rational interval arithmetic. This provides a direct way to certify as many displayed digits as desired.

Numerical evaluation serves only to identify the displayed constants; the independence conclusions follow from the exact proofs.

14 Gaussian values under a real normalization

The real normalization in

$$c_0 = \operatorname{erf}(1) = \frac{2}{\sqrt{\pi}} \int_0^1 e^{-t^2}\, dt$$

cannot simply be cancelled in a transcendence argument. Pairing complex values with their conjugates instead gives unconditional independence and zero classifications, while retaining the precise distinction between normalized and unnormalized values.

The arithmetic input is classical E-function theory, as organized in Theorem 5.1 and Corollary 6.1. The zero conclusions also follow from Delaygue's theorem on entire rational-parameter hypergeometric values [10, Theorem 1.4], applied to $F_{a,1}(w) = a^{-1}{}_1F_1(a; a+1; -w)$ together with nonalgebraicity. The results are therefore presented as explicit normalization consequences, without a claim to resolve a recognized open problem.

14.1 The first logarithmic layer of the shared theorem

The notation here is the same as in the reflection calculation:

$$F_{a,1}(z) = \sum_{n\geq 0} \frac{(-z)^n}{n!(n+a)}, \qquad a \in \mathbb{Q} \cap (0,1).$$

Direct differentiation gives

$$zF'_{a,1}(z) + aF_{a,1}(z) = e^{-z}. \tag{14.1}$$

Corollary 14.1. *The family* $\{F_{a,1}(\lambda) : a \in \mathbb{Q} \cap (0,1),\ \lambda \in \overline{\mathbb{Q}}^{\times}\}$ *is algebraically independent over* $\mathcal{E}$*.*

Proof. Take $q = 1$ in Corollary 6.1. No assumption about π, complete Gamma values, or Euler's constant relative to $\mathcal{E}$ is involved. □

15 Conjugation and a real normalization

Lemma 15.1 (Descent through one real scalar)**.** *Let* $B \subseteq \mathbb{C}$ *be a field stable under complex conjugation. Suppose*

$$u_1, \dots, u_n, \overline{u_1}, \dots, \overline{u_n}$$

are algebraically independent over B*. For every nonzero real number* c*, the numbers* $u_1/c, \dots, u_n/c$ *are algebraically independent over* $B(c)$*.*

Proof. It is equivalent to prove independence of the u_j over $B(c)$. Suppose this fails. Then

$$\operatorname{trdeg}_{B(c)} B(c, u_1, \dots, u_n) \le n - 1.$$

Conjugation preserves $B(c)$, so the same bound holds for the conjugate tuple. Subadditivity consequently gives

$$\operatorname{trdeg}_{B(c)} B(c, u_1, \dots, u_n, \overline{u_1}, \dots, \overline{u_n}) \le 2n - 2.$$

Since adjoining c has transcendence degree at most one, the transcendence degree of this field over B is at most $2n-1$. This contradicts the assumed independence of the $2n$ displayed numbers over B. Finally, multiplication by the nonzero element $c \in B(c)$ preserves algebraic independence. □

Define the conjugation-stable field

$$B_{\mathrm{real}} = \mathcal{E}\big(F_{a,1}(x) : a \in \mathbb{Q} \cap (0,1), x \in \overline{\mathbb{Q}} \cap \mathbb{R}^{\times}\big).$$

Every adjoined generator is real; $\mathcal{E}$ is itself stable under conjugation.

Theorem 15.2 (Joint independence in an open quadrant)**.** *The complete family*

$$\Big\{\operatorname{erf}(z) : z \in \overline{\mathbb{Q}},\ \Re z > 0,\ \Im z > 0\Big\}$$

is algebraically independent over $B_{\mathrm{real}}(\sqrt{\pi})$*. In particular it is algebraically independent over*

$$\mathcal{E}\big(\pi, \operatorname{erf}(r), \operatorname{erfi}(r) : r \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}\big).$$

Proof. For any complex z, power-series integration yields

$$U(z) := \sqrt{\pi}\operatorname{erf}(z) = 2\int_0^z e^{-t^2}\,dt = zF_{1/2,1}(z^2).$$

Take distinct algebraic points $z_1, \dots, z_n$ in the open first quadrant. Their squares are distinct points of the open upper half-plane. Their conjugate squares are distinct points of the lower half-plane. These $2n$ arguments are therefore pairwise distinct and disjoint from all real arguments used to generate B_{real}. Corollary 14.1 proves joint algebraic independence of $U(z_j)$ and $U(\overline{z_j}) = \overline{U(z_j)}$ over B_{real}. (Every prospective relation uses only finitely many generators of the base field.) Apply the preceding lemma with $c = \sqrt{\pi}$.

For the stated smaller base field, use

$$\operatorname{erf}(r) = \frac{rF_{1/2,1}(r^2)}{\sqrt{\pi}}, \qquad \operatorname{erfi}(r) = \frac{rF_{1/2,1}(-r^2)}{\sqrt{\pi}}.$$

Both belong to $B_{\mathrm{real}}(\sqrt{\pi})$, as does π. □

Corollary 15.3. *For any distinct algebraic $z_1, \dots, z_n$ in the open first quadrant,*

$$\pi, \operatorname{erf}(z_1), \dots, \operatorname{erf}(z_n)$$

are algebraically independent over $\overline{\mathbb{Q}}$. In particular $\pi, \operatorname{erf}(1+i), \operatorname{erf}(2+i)$ are algebraically independent.

Proof. The theorem gives independence of the error-function values over $\overline{\mathbb{Q}}(\pi)$; combine this with the transcendence of π over $\overline{\mathbb{Q}}$. □

The same argument works for any choice of algebraic points whose squares and conjugate squares are pairwise distinct. The quadrant is only a convenient canonical choice. It does not include the real axis, where $U(z) = \overline{U(z)}$ and the hypothesis of the descent lemma fails.

16 Incomplete Gamma functions and zero classification

For this section, extend the series definition of $F_{a,1}$ to every $a \in \mathbb{Q} \setminus \mathbb{Z}$; none of its denominators then vanishes.

Lemma 16.1 (Extension of the shape parameter)**.** *For every fixed $a \in \mathbb{Q} \setminus \mathbb{Z}$, the values $F_{a,1}(\lambda)$, at distinct nonzero algebraic arguments, are algebraically independent over $\mathcal{E}$.*

Proof. Coefficient comparison gives

$$zF_{a+1,1}(z) = aF_{a,1}(z) - e^{-z}.$$

Choose $b \in \mathbb{Q} \cap (0,1)$ such that $a - b \in \mathbb{Z}$. Repeatedly applying this identity upwards or downwards yields

$$F_{a,1}(z) = A_a(z)F_{b,1}(z) + B_a(z), \qquad A_a(z) \in \mathbb{Q}(z), \quad B_a(z) \in \mathbb{Q}(z, e^{-z}).$$

Every intermediate shape is nonintegral. Each multiplicative factor in A_a is of the form c/z or z/c, with nonzero rational c. Consequently $A_a(\lambda) \neq 0$ at every nonzero algebraic λ, and $B_a(\lambda) \in \mathcal{E}$. These invertible affine changes of individual coordinates preserve algebraic independence. □

Let $D = \mathbb{C} \setminus (-\infty, 0]$, and use the principal branch of z^a on D. For $a \in \mathbb{Q} \setminus \mathbb{Z}$, define

$$G_a(z) = z^a F_{a,1}(z), \quad P(a,z) = \frac{G_a(z)}{\Gamma(a)}, \quad Q(a,z) = 1 - P(a,z), \quad \Gamma(a,z) = \Gamma(a) - G_a(z).$$

These are the usual meromorphic continuation in the shape parameter of the lower and upper incomplete gamma identities. In particular $\Gamma(a)$ is a nonzero real number for the real nonintegral shapes under consideration.

Theorem 16.2 (Nonreal values at a fixed rational shape)**.** *Fix $a \in \mathbb{Q} \setminus \mathbb{Z}$. For every finite set of distinct algebraic numbers $z_1, \dots, z_m$ in the open upper half-plane, each of the following three families, considered separately, is algebraically independent over $\mathcal{E}(\Gamma(a))$:*

$$\{P(a,z_j)\}_{j=1}^m, \qquad \{Q(a,z_j)\}_{j=1}^m, \qquad \{\Gamma(a,z_j)\}_{j=1}^m.$$

The union of these three families is not asserted independent, since $Q = 1 - P$ and $\Gamma(a,z) = \Gamma(a)Q(a,z)$. The same statement holds with the lower half-plane in place of the upper half-plane. In particular, $\Gamma(a,z) \neq 0$ whenever $z \in D$ is nonreal and algebraic.

Proof. The $2m$ arguments $z_j, \overline{z_j}$ are distinct and nonzero. The shape lemma gives algebraic independence over $\mathcal{E}$ of the corresponding $F_{a,1}$-values. The factors z_j^a and $\overline{z_j}^{\,a}$ are nonzero algebraic numbers, so the corresponding G_a-values remain independent. The real coefficients of $F_{a,1}$, together with the principal-branch convention, give

$$G_a(\overline{z}) = \overline{G_a(z)}.$$

Apply conjugation descent with $c = \Gamma(a)$ to obtain independence of the P-values. The other two assertions follow by invertible affine changes over $\mathcal{E}(\Gamma(a))$. □

Corollary 16.3 (Zeros at nonintegral rational shapes)**.** *For every $a \in \mathbb{Q} \setminus \mathbb{Z}$, all zeros in D of the principal upper incomplete gamma function $z \mapsto \Gamma(a, z)$ are transcendental numbers.*

Proof. A nonreal algebraic zero is excluded by the theorem. A positive real zero is excluded by the convergent positive integral

$$\Gamma(a, x) = \int_x^\infty t^{a-1} e^{-t}\, dt > 0 \qquad (x > 0),$$

which is valid for every real a because its lower endpoint is strictly positive. These are all possible locations in D. □

The two boundary values on the negative real cut are nonzero as well. Indeed, for $r > 0$,

$$\Gamma(a, -r \pm i0) = \Gamma(a) - r^a e^{\pm i\pi a} F_{a,1}(-r),$$

where $F_{a,1}(-r) \in \mathbb{R}$. If this expression vanished, its imaginary part, together with $\sin(\pi a) \neq 0$, would imply $F_{a,1}(-r) = 0$. Its real part would then equal the nonzero number $\Gamma(a)$, a contradiction. No claim about other sheets for negative shapes is needed.

For comparison, if $n \geq 1$ is an integer, then

$$\Gamma(n, z) = (n-1)! e^{-z} \sum_{k=0}^{n-1} \frac{z^k}{k!}.$$

Thus all its zeros are algebraic; for $n = 1$ there are no zeros. This gives the following exact dichotomy for *positive rational* shapes: integer shapes have only algebraic zeros, whereas noninteger shapes have only transcendental zeros. Both conclusions concern locations of zeros; they do not assert existence of zeros for every shape.

16.1 Complementary error-function zeros

The direct Gaussian version avoids any branch ambiguity. If z is algebraic with both real and imaginary parts nonzero, then the two numbers

$$U = zF_{1/2,1}(z^2), \qquad \overline{U} = \overline{z} F_{1/2,1}(\overline{z}^{\,2})$$

are independent over $\mathcal{E}$, because $z^2 \neq \overline{z}^{\,2}$. Conjugation descent with $c = \sqrt{\pi}$ shows that $\operatorname{erf}(z) = U/\sqrt{\pi}$ is transcendental over $\mathcal{E}(\sqrt{\pi})$.

Corollary 16.4 (All complementary Gaussian zeros)**.** *Every zero of* $\operatorname{erfc}(z)$ *is transcendental. Every zero of the Faddeeva function $w(z) = e^{-z^2} \operatorname{erfc}(-iz)$ is transcendental as well.*

Proof. An algebraic zero off the two coordinate axes would give the forbidden algebraic value $\operatorname{erf}(z) = 1$. On the real axis, $-1 < \operatorname{erf}(x) < 1$ at every finite x. On the imaginary axis, $\operatorname{erf}(iy)$ is purely imaginary. Thus no algebraic zero exists anywhere. The exponential factor in w never vanishes, and multiplication of a nonzero transcendental number by i preserves transcendence. □

NIST DLMF, Section 7.13(ii) [9], records an infinite sequence of erfc-zeros in the sector $\pi/2 < \arg z < 3\pi/4$, with their conjugates providing the remaining zeros. Its Table 7.13.2 gives the first member, ordered by increasing absolute value in that sector, as

$$z_1 \simeq -1.3548101281 + 1.9914668428\,i.$$

The zero theorem proves this complex number is transcendental. It does not prove that each of its real and imaginary parts is transcendental, nor does it prove algebraic independence among different zeros.

Remark 16.5 (Branches and existence). The upper-Gamma theorem concerns the principal branch and its two boundary values, and does not assert the existence of zeros for every shape. It must not be used to identify every erfc zero with a principal-branch zero of $\Gamma(1/2, z^2)$: the square map can cross the branch cut. The entire-function proof for erfc given above has no such branch substitution.

Corollary 16.6 (Algebraic target values off the axes). *If $b \in \overline{\mathbb{Q}}$ has both nonzero real and imaginary parts, every solution of* $\operatorname{erf}(z) = b$ *is transcendental. The same conclusion holds for real algebraic b with $|b| \geq 1$.*

Proof. An algebraic solution off the axes contradicts the Gaussian theorem. On the real axis the value lies in $(-1, 1)$; on the imaginary axis it is purely imaginary. These observations exclude the remaining algebraic possibilities for the specified targets. □

17 The global relation ideal after common normalization

Proposition 17.1 (At most one global relation). *Let k be a field, let $(X_j)_{j\in J}$ be algebraically independent over k, and let $c \neq 0$ belong to a common field extension. Put $Y_j = X_j/c$, and let*

$$I = \{P \in k[T_j : j \in J] : P((Y_j)) = 0\}.$$

Then either $I = 0$, or $I = (P)$ for a single irreducible polynomial P in finitely many variables. Moreover, I contains no nonzero homogeneous polynomial. In the nonzero case, every normalized variable outside the finite support S of P is jointly algebraically independent over $k(c, (Y_j)_{j\in S})$.

Proof. For every finite $A \subseteq J$, adjoining c and rescaling gives

$$k(c, (Y_j)_{j\in A}) = k(c, (X_j)_{j\in A}).$$

Thus $\operatorname{trdeg}_k k((Y_j)_{j\in A}) \geq |A| - 1$. The kernel I_A of evaluation on the finite polynomial ring $k[T_j : j \in A]$ is prime and has height at most one.

If $I \neq 0$, choose a nonzero member and an irreducible factor P that also vanishes. Write S for its finite support. For every finite $A \supseteq S$, P remains irreducible in $k[T_j : j \in A]$. The nonzero height-one prime (P) lies in I_A, whose height is at most one; hence $I_A = (P)$. Applying this to the union of S with the support of any member of I proves $I = (P)$.

If a homogeneous polynomial Q of degree d vanished on the Y_j, then $Q((X_j)) = c^d Q((Y_j)) = 0$, contrary to independence.

For the last claim, homogenize the equality $P((X_j/c)_{j\in S}) = 0$ in c by multiplying by a sufficiently large power of c. The resulting polynomial in c over $k((X_j)_{j\in S})$ is nonzero: each of its coefficients is a homogeneous component of P evaluated on algebraically independent X_j. Therefore c is algebraic over $k((X_j)_{j\in S})$. Independence of all remaining X_j persists under this algebraic extension; rescaling by c gives their asserted normalized independence. □

Corollary 17.2 (A single exceptional value)**.** *In the setting of the proposition, at most one Y_j can be algebraic over k. If one is algebraic over k, all other Y_j are jointly algebraically independent over $k(c)$.*

Proof. Two algebraic normalized values would make their ratio algebraic, whereas $Y_j/Y_l = X_j/X_l$ is transcendental over k. If Y_j is algebraic, use its minimal polynomial for P in the proposition; all other Y_l are independent over $k(c, Y_j)$ and hence over $k(c)$. □

For the Gaussian application take one representative of each class $z \sim -z$ in $\overline{\mathbb{Q}}^\times$, $X_z = \sqrt{\pi}\,\mathrm{erf}(z)$, $c = \sqrt{\pi}$, and $k = \mathcal{E}$. Corollary 14.1 establishes independence of these X_z. These are structural consequences of that input; they do not establish the transcendence of the particular real value $\mathrm{erf}(1)$, and no historical priority claim is made.

18 The real anchor and its conditional consequences

Write $A = F_{1/2,1}(1) = \sqrt{\pi}\,c_0$. The arithmetic theorem proves that A is transcendental over $\mathcal{E}$, but it does not prove the same statement for c_0.

Proposition 18.1 (An exact consequence of the opposite hypothesis)**.** *If $c_0 = \mathrm{erf}(1)$ is algebraic, then π is transcendental over $\mathcal{E}$. In particular e and π are algebraically independent. Moreover, π together with all the other $\mathrm{erf}(z)$ values, choosing one representative of each class $z \sim -z$ in $\overline{\mathbb{Q}}^\times$ and excluding the class of 1, is algebraically independent over $\mathcal{E}$.*

Proof. The positive number c_0 is nonzero. If it is algebraic, then $\pi = A^2/c_0^2$, which is transcendental over $\mathcal{E}$ because A is. For the last assertion, the unnormalized coordinates $U(z) = zF_{1/2,1}(z^2)$ are jointly independent over $\mathcal{E}$, and $\sqrt{\pi} = U(1)/c_0$. Dividing all other independent coordinates by $U(1)/c_0$ is an invertible rational change after retaining $U(1)$. Replacing $U(1)$ by $\pi = U(1)^2/c_0^2$ changes only an algebraic extension, so preserves the asserted transcendence degree. □

This is a conditional consequence of the algebraicity hypothesis, not a contradiction. It therefore does not settle c_0. Conjugation descent cannot be applied to the pair $A, \overline{A}$, since they are equal.

There is also a precise connection to the arithmetic of period rings. Let $\mathbf{E}$ denote the ring of all E-function values at algebraic points; this ring is different from the exponential field $\mathcal{E}$ used above. If c_0 were algebraic, $\pi = A^2/c_0^2$ would belong to $\mathbf{E}$. The number π is also a G-function value, for example $4\arctan(1)$. The conjectural equality $\mathbf{E} \cap \mathbf{G} = \overline{\mathbb{Q}}$ would therefore exclude this possibility. Fischler–Rivoal explicitly treat that equality as a conjecture [11]; it is not an input to any theorem here.

Part III
Coefficient growth and differential independence

19 Incomplete gamma functions over a gamma differential field

Throughout, $D = d/ds$ and $K\langle f_i\rangle_D$ denotes the smallest differential field containing K and the indicated functions. Differential algebraic independence means algebraic independence of all derivatives of all the indicated functions. An infinite family is independent when every finite subfamily is independent.

For $\lambda \in \mathbb{C}^\times$, define

$$H_\lambda(s) = \sum_{n\geq 0} \frac{(-\lambda)^n}{n!(n+s)}. \tag{19.1}$$

The series converges normally on compact subsets avoiding the nonpositive integers. It defines a meromorphic function of s, with residue $(-\lambda)^n/n!$ at $s = -n$. For positive real λ,

$$H_\lambda(s) = \int_0^1 t^{s-1}e^{-\lambda t}\,dt = \lambda^{-s}\gamma(s,\lambda), \qquad \Re s > 0. \tag{19.2}$$

Meromorphic continuation gives the corresponding identity elsewhere. For complex λ, a fixed logarithm specifies the factor λ^s and the matching branch of the incomplete gamma function.

The coefficient field in our main statement is

$$\mathscr{B} = \mathbb{C}(s)\left\langle \Gamma(cs+b) : c \in \mathbb{C}^\times,\ b \in \tfrac{1}{2}\mathbb{Z}\right\rangle_D. \tag{19.3}$$

We may work with meromorphic germs at zero. Each element of $\mathscr{B}$ uses only finitely many of the displayed generators, derivatives, and complex constants. No independence of these gamma generators is asserted.

Theorem 19.1 (Independence relative to the gamma field)**.** *The family*

$$\left\{H_\lambda(s+r) : \lambda \in \overline{\mathbb{Q}}^\times, \quad r \in \mathbb{Q}\cap[0,1)\right\} \tag{19.4}$$

is jointly differentially algebraically independent over $\mathscr{B}$.

Thus a polynomial in finitely many derivatives of distinct members of (19.4), with coefficients that are rational differential expressions in finitely many gamma functions from (19.3), cannot vanish unless it is the zero polynomial over that actual coefficient field. This includes nonlinear relations of arbitrary finite order.

All other rational order shifts are controlled by

$$\lambda H_\lambda(s+r+1) - (s+r)H_\lambda(s+r) + e^{-\lambda} = 0. \tag{19.5}$$

The following statement says precisely that there are no further relations.

Theorem 19.2 (Complete differential relation ideal)**.** *Let $\mathcal{R} = \mathscr{B}\{Y_{\lambda,r} : \lambda \in \overline{\mathbb{Q}}^\times, r \in \mathbb{Q}\}$ be the differential polynomial ring. For the evaluation map*

$$\mathrm{ev} : \mathcal{R} \longrightarrow \mathbb{C}((s)), \qquad Y_{\lambda,r} \longmapsto H_\lambda(s+r),$$

the kernel is the differential ideal generated by

$$\lambda Y_{\lambda,r+1} - (s+r)Y_{\lambda,r} + e^{-\lambda}, \qquad \lambda \in \overline{\mathbb{Q}}^\times,\ r \in \mathbb{Q}. \tag{19.6}$$

In particular, the quotient is a free differential polynomial ring over $\mathscr{B}$, with one generator for each pair $(\lambda, r \bmod \mathbb{Z})$.

These are functional statements. One concrete consequence is joint differential algebraic independence of $\Gamma(s)$ and $\Gamma(s,1)$ over $\mathbb{C}(s)$. More generally, the normalized gamma distribution functions at distinct nonzero algebraic cuts are jointly differentially algebraically independent; see Section 24. The results do not assert transcendence of Euler's constant or of an unresolved numerical value at a fixed order.

Relation with earlier work. Hölder's theorem establishes differential transcendence of the complete gamma function. Johnson–Reinhart–Rubel and Arreche studied the two-variable incomplete gamma function; Arreche's theorem concerns ∂_s-transcendence over $\mathbb{C}(s,x)$ [12, 13]. It does not by its statement alone imply joint independence at a specified collection of fixed endpoints, relative to (19.3). The broad difference-equation results of Adamczewski–Dreyfus–Hardouin [14] are also relevant to individual hypertranscendence. Our focus is the joint relative statement and the complete relation ideal. We make no claim here that these statements have no antecedent in the wider differential-algebraic literature.

Coefficient recursions for differentially algebraic series are classical. For their arithmetic history and recent developments, see Krattenthaler–Rivoal [15]. We give the needed formal argument explicitly and use it to compare coefficient transcendence growth with differential dimension. The arithmetic input is the vertical specialization in 6.1 of the common arithmetic shuffle theorem, Theorem 5.1. The standard E-function specialization theorems used there are not claimed as new.

20 Coefficient growth and differential dimension

Fix a subfield $k \subseteq \mathbb{C}$. The expansion center in this section is zero.

Definition 20.1 (Coefficient transcendence growth)**.** For a finite tuple $f = (f_1, \dots, f_m)$ of formal power series, write $f_i(s) = \sum_{n\geq 0} f_{i,n} s^n$ and set

$$a_k(f;N) = \operatorname{trdeg}_k k(f_{i,n} : 1 \leq i \leq m,\ 0 \leq n \leq N), \tag{20.1}$$

$$\rho_k(f) = \limsup_{N\to\infty} \frac{a_k(f;N)}{N+1}. \tag{20.2}$$

For Laurent series include all their finitely many negative-degree coefficients as well. This changes the rank by at most a constant relative to discarding the principal parts.

The rate lies between 0 and m. It depends on k and on the expansion center; coordinate invariance is not claimed. Extending k by finitely many arbitrary complex constants changes $a_k(f;N)$ by at most a constant and hence leaves the rate unchanged. For a finite tuple of rational differential expressions in f, coefficients through degree N depend on coefficients of f through degree $N+C$ for a fixed C, together with finitely many constants. This follows from formal division and differentiation; a nonzero denominator has a finite valuation and nonzero leading coefficient.

Lemma 20.2 (Coefficient recovery at a singular center)**.** *Let $f, u_1, \dots, u_\ell \in \mathbb{C}[[s]]$, and let P be a polynomial in s and finitely many derivatives of these series, with coefficients in a field $k_0 \subseteq \mathbb{C}$. Suppose $P(f,u) = 0$ and its evaluated linearization in f is nonzero:*

$$L = \sum_{j=0}^{r} A_j(s) D^j \neq 0, \qquad A_j(s) = \left(\frac{\partial P}{\partial Y^{(j)}}\right)(f,u).$$

There exist integers $n_0, C \geq 0$ and a field $k_1 \supseteq k_0$ of finite transcendence degree over k_0 such that every coefficient of f through degree N belongs to

$$k_1([s^n]u_i : 1 \leq i \leq \ell,\ 0 \leq n \leq N + C).$$

The extension k_1/k_0 is generated by finitely many initial coefficients and finitely many coefficients of the linearization.

Proof. Increase r to bound all derivative orders appearing in P. Set

$$d = \min_{A_j \neq 0} (\operatorname{ord}_0 A_j - j), \qquad I(T) = \sum_{\operatorname{ord}_0 A_j - j = d} [s^{j+d}] A_j(s)\, (T)_j, \tag{20.3}$$

where $(T)_j = T(T-1)\cdots(T-j+1)$ and $(T)_0 = 1$. The polynomial I is nonzero: its summand with largest j has strictly largest degree and nonzero leading coefficient.

Write $f = \sum_{n \geq 0} f_n s^n$ and $f_{<n} = \sum_{v<n} f_v s^v$. Taylor expansion of the polynomial P in its f-jet variables, about the actual solution, gives for $n > r$

$$P(f_{<n}, u) = -L(f - f_{<n}) + O(s^{2(n-r)}). \tag{20.4}$$

Every omitted term contains at least two derivatives of $f - f_{<n}$; the remaining factors are power series. If $n > d + 2r$, comparison of the coefficient at degree $n + d$ yields

$$[s^{n+d}] P(f_{<n}, u) = -I(n) f_n. \tag{20.5}$$

Indeed, $L(s^v)$ has valuation at least $v + d$, so terms with $v > n$ do not contribute, and terms with $v = n$ give exactly $I(n) f_n$.

Only finitely many nonnegative integers are zeros of I. Choose n_0 beyond them and beyond all preceding bounds. The left side of (20.5) uses only f_v with $v < n$ and coefficients of the u_i through degree $n + d + r$. Adjoin the coefficients of I and $f_0, \ldots, f_{n_0 - 1}$ to k_0. Taking $C \geq \max(0, d + r)$, induction proves the assertion. □

Theorem 20.3 (Differential dimension bounds coefficient growth)**.** *Let $f = (f_1, \ldots, f_m)$ and $g = (g_1, \ldots, g_\ell)$ be finite tuples in $\mathbb{C}((s))$. Put*

$$F = \mathbb{C}(s)\langle g \rangle_D, \qquad q = \operatorname{dtrdeg}_F F\langle f \rangle_D.$$

For some fixed integer $C \geq 0$,

$$a_k(f; N) \leq q(N + C + 1) + a_k(g; N + C) + O(1). \tag{20.6}$$

Consequently

$$\rho_k(f) \leq q + \rho_k(g), \qquad q \geq \lceil \rho_k(f) - \rho_k(g) \rceil. \tag{20.7}$$

More precisely,

$$q \geq \left\lceil \limsup_{N \to \infty} \frac{a_k(f; N) - a_k(g; N)}{N + 1} \right\rceil. \tag{20.8}$$

Proof. Subtract the finite principal parts, whose coefficients are absorbed into a finitely generated constant field. We can therefore assume all germs are power series. A maximal differentially independent subset of the f_i is a differential transcendence basis for $F\langle f \rangle_D / F$. Reorder it as $f_1, \ldots, f_q$. Every remaining f_i is differentially algebraic over $F\langle f_1, \ldots, f_q \rangle_D$.

For each $i > q$, choose a nonzero differential relation of minimal total degree in the differential indeterminate representing f_i, over that actual coefficient field. In characteristic zero, one of its nonzero partial derivatives has lower total degree. Its evaluation cannot vanish by minimality. Express the finitely many coefficient field elements as rational functions of derivatives of g and of $f_1, \ldots, f_q$, and clear their nonzero denominators. The resulting polynomial has a nonzero evaluated linearization in f_i.

Lemma 20.2 applies using only g and the chosen basis as auxiliary series. It does not use any of the other dependent f_j. Taking the maximum of the finitely many cutoffs, every coefficient

of every f_i through degree N is generated by a fixed finite set of complex numbers, the q basis streams through degree $N+C$, and the coefficients of g through that degree. The former streams contribute at most $q(N+C+1)$ generators. This proves (20.6). Division by $N+1$ and the limsup give (20.7). Since g is a finite tuple, $a_k(g;N+C)-a_k(g;N)\le \ell C$; this also proves (20.8). □

Corollary 20.4 (Independence over a base of rate less than one)**.** *Suppose all coefficients of $f_1,\dots,f_m\in\mathbb{C}[[s]]$ are jointly algebraically independent over k. If a finite tuple g of Laurent germs satisfies $\rho_k(g)<1$, then the f_i are jointly differentially algebraically independent over $\mathbb{C}(s)\langle g\rangle_D$. The conclusion also holds over the field generated by an infinite background family if every finite subfamily has joint rate less than one.*

Proof. The hypothesis gives $a_k(f;N)=m(N+1)$, hence $\rho_k(f)=m$. Theorem 20.3 gives $q>m-1$. Since q is an integer and $q\le m$, one has $q=m$. A putative relation over an infinite background uses only finitely many of its generators. □

Remark 20.5 (Sharp threshold and joint growth). The universal threshold 1 cannot be included: the background $g=f_1$ has rate 1 and gives the relation $Y_1-g=0$. It is essential to bound joint growth, rather than the rate of each background function separately. For a series f with independent coefficients, its even and odd parts each have rate $1/2$, but together recover f and have joint rate 1.

21 Arithmetic coefficient arrays for the incomplete gamma family

Recall the exponential field $\mathcal{E}$ and the canonical functions $F_{a,q}$ and Ein_q from 6.1.

Proposition 21.1 (The coefficient independence input)**.** *All numbers $F_{a,q}(\lambda)$ and $\mathrm{Ein}_q(\lambda)$, with distinct index tuples, $a\in\mathbb{Q}\cap(0,1)$, $q\ge 1$, and $\lambda\in\overline{\mathbb{Q}}^{\times}$, are jointly algebraically independent over $\mathcal{E}$.*

Proof. The vertical-word identities in 6.1 express these numbers as distinct canonical Lyndon coordinates, up to nonzero algebraic factors. Theorem 5.1 makes every finite collection of those coordinates algebraically independent over $\mathcal{E}$. An algebraic relation in the displayed infinite family would use only finitely many coordinates, so the finite assertion proves the claim. □

For the canonical shifts, introduce holomorphic germs

$$h_{\lambda,r}(s)=\begin{cases} H_\lambda(s)-s^{-1}, & r=0,\\ H_\lambda(s+r), & 0<r<1.\end{cases} \tag{21.1}$$

Expanding the denominators in (19.1), with absolute convergence near zero, gives

$$[s^n]h_{\lambda,0}(s)=(-1)^{n+1}\,\mathrm{Ein}_{n+1}(\lambda), \tag{21.2}$$

$$[s^n]h_{\lambda,r}(s)=(-1)^n F_{r,n+1}(\lambda)\quad(0<r<1). \tag{21.3}$$

Thus for any m distinct canonical labels, the tuple h satisfies

$$a_{\mathcal{E}}(h;N)=m(N+1). \tag{21.4}$$

This is arithmetic information about every finite coefficient array, not a numerical approximation or an assertion based on a finite search for polynomial relations.

22 The gamma background has coefficient rate at most one half

The classical expansion at zero is

$$\Gamma(s) = \frac{1}{s}\exp\left(-\gamma s + \sum_{n\geq 2}\frac{(-1)^n\zeta(n)}{n}s^n\right), \qquad |s|<1. \tag{22.1}$$

Here γ is Euler's constant. Since $\zeta(2n)\in\mathbb{Q}\pi^{2n}$, the coefficients of $\Gamma(s)-s^{-1}$ through degree N belong to

$$\mathcal{E}(\gamma,\pi,\zeta(3),\zeta(5),\dots,\zeta(2\lfloor N/2\rfloor+1)). \tag{22.2}$$

In particular their transcendence degree over $\mathcal{E}$ is at most $\lfloor N/2\rfloor+2$. The formulas are classical; see [9, §§5.7, 25.6]. Only an upper bound is used. No independence hypothesis on odd zeta values or on γ is needed.

Proposition 22.1 (A common coefficient pool)**.** *Every finite tuple g of elements of $\mathscr{B}$ has*

$$a_{\mathcal{E}}(g;N)\leq \tfrac{1}{2}N+O(1). \tag{22.3}$$

The implied constant can depend on the tuple.

Proof. For finitely many scales $c_1,\dots,c_v$, substituting c_is into (22.1) introduces only those finitely many complex constants. The coefficients of all $\Gamma(c_is)$ use the same pool of odd zeta values. The duplication formula gives

$$\Gamma\left(\tfrac{1}{2}+cs\right)=\sqrt{\pi}\,e^{-2cs\log 2}\frac{\Gamma(1+2cs)}{\Gamma(1+cs)}. \tag{22.4}$$

Its coefficients introduce only $\log 2$, $\sqrt{\pi}$, and the same zeta pool. Recurrence $\Gamma(z+1)=z\Gamma(z)$ reduces every half-integer translate to an integer or a half-integer center; negative shifts merely introduce rational factors in s.

Each entry of g is a rational differential expression in a finite selection of such generators. Formal differentiation and division require coefficients through degree at most $N+C$ for a fixed C. Therefore all coefficients counted on the left of (22.3) lie in a field generated by a fixed finite set of complex numbers and the odd zeta values of index at most $N+C+1$. There are at most $N/2+O(1)$ of the latter. Counting generators proves the bound even if they satisfy unknown relations. □

It matters that these are *common* generators for the whole finite tuple. Counting separately for each gamma function would give an unnecessarily large bound and would not establish the main theorem. Arbitrary rational shifts of the complete gamma function, such as $\Gamma(s+1/3)$, are not included in (19.3); the present argument does not assign their joint coefficient pool the same bound.

23 Proof of the relative independence and the relation ideal

Proof of Theorem 19.1. Take any m distinct canonical labels and their regularized germs h in (21.1). Their coefficients are jointly algebraically independent over $\mathcal{E}$ by Proposition 21.1 and (21.2)–(21.3). A differential relation over $\mathscr{B}$ would involve a finite tuple of background generators. Proposition 22.1 gives joint rate at most $1/2$ for this tuple. Applying Corollary 20.4 with $k=\mathcal{E}$ rules out the relation. Adding the rational principal part $1/s$ back to the zero-shift germs preserves differential independence. □

For clarity, the contradiction can be written without rates. A nontrivial relation would reduce the differential dimension of the m germs to at most $m-1$, so Theorem 20.3 gives

$$m(N+1) \le (m-1)(N+C+1) + \tfrac{1}{2}(N+C) + O(1).$$

The two sides have incompatible linear growth. The proof remains valid although the coefficient field contains every complex constant: a particular polynomial relation uses only finitely many of them.

Proof of Theorem 19.2. Termwise manipulation of the normally convergent series yields

$$\lambda H_\lambda(s+1) = sH_\lambda(s) - e^{-\lambda}.$$

Translation gives (19.5), so the differential ideal J generated by (19.6) lies in $\ker(\mathrm{ev})$.

Write each $r \in \mathbb{Q}$ uniquely as $r = r_0 + n$ with $0 \le r_0 < 1$ and $n \in \mathbb{Z}$. The recurrences express $Y_{\lambda,r}$ modulo J as

$$A_{\lambda,r}(s)Y_{\lambda,r_0} + B_{\lambda,r}(s), \qquad A_{\lambda,r} \in \mathbb{C}(s)^\times, \quad B_{\lambda,r} \in \mathbb{C}(s).$$

This works in both directions: λ and each $s+r$ are invertible in the coefficient field. Differentiating these expressions also eliminates all derivatives of noncanonical symbols. There is no cyclic consistency condition, since each shift orbit is a copy of the integer line. Substitution of the displayed expressions defines an inverse to the natural map from the free differential polynomial ring on the canonical symbols to $\mathcal{R}/J$.

By Theorem 19.1, evaluation of that free ring is injective. Hence $\ker(\mathrm{ev}) = J$. The use of a coefficient field is essential here; an analogous statement over a polynomial coefficient ring would require localization or saturation. □

24 Consequences and the precise scope of the result

Lemma 24.1 (Differentially algebraic changes of base)**.** *If L/K is a differentially algebraic extension inside a common differential overfield, a finite differentially independent family over K stays differentially independent over L.*

Proof. For a family f of size m, the compositum $L\langle f\rangle$ is differentially algebraic over $K\langle f\rangle$. Additivity of differential transcendence degree in towers gives $\mathrm{dtrdeg}_L\, L\langle f\rangle = m$. A relation uses only finitely many elements of L, so no finiteness assumption on L/K is needed. □

Corollary 24.2 (Lower, upper, and normalized incomplete gamma)**.** *Choose one logarithm for each nonzero algebraic λ and set*

$$L_\lambda(s) = e^{s\log\lambda}H_\lambda(s), \qquad U_\lambda(s) = \Gamma(s) - L_\lambda(s), \qquad P_\lambda(s) = L_\lambda(s)/\Gamma(s).$$

Each of the three families $(L_\lambda)_\lambda$, $(U_\lambda)_\lambda$, and $(P_\lambda)_\lambda$ is separately jointly differentially algebraically independent over $\mathscr{B}$. The same holds over any differentially algebraic extension of $\mathscr{B}$ in a common differential overfield.

Proof. Adjoin the required factors $e^{s\log\lambda}$. They satisfy $Dy = (\log\lambda)y$ and hence form a differentially algebraic extension. Apply Lemma 24.1 and then the invertible affine changes of coordinates displayed in the statement. □

For positive real cuts, these are the usual lower incomplete gamma, upper incomplete gamma, and regularized lower gamma functions. Their *union* is not claimed to be independent: for example, $L_\lambda + U_\lambda = \Gamma$.

Corollary 24.3 (Complete and fixed-cut upper gamma)**.** *The two functions* $\Gamma(s)$ *and* $\Gamma(s,1)$ *are jointly differentially algebraically independent over* $\mathbb{C}(s)$*. More generally,* $\Gamma(s)$ *together with the upper incomplete gamma functions at any finite set of distinct nonzero algebraic cuts is jointly differentially algebraically independent over* $\mathbb{C}(s)$*.*

Proof. Hölder's theorem gives differential transcendence of Γ; see also the historical discussion in [14]. Theorem 19.1 and Corollary 24.2 give independence of the remaining functions over a field containing $\mathbb{C}(s)\langle\Gamma(s)\rangle_D$. The resulting differential transcendence degree is the total number of functions. At cut 1, the identity is simply $\Gamma(s,1) = \Gamma(s) - H_1(s)$. □

The general dimension estimate in Theorem 20.3 is the transferable part of the proof: it applies to any finite coefficient arrays and any background admitting a coefficient-growth bound. For a different special-function family, one must establish those two inputs; no general arithmetic freeness assumption is automatic.

What has been settled within this framework. Theorems 19.1 and 19.2 determine the complete set of differential polynomial relations for this entire rational-shift family over the specified gamma field. They hold for arbitrary finite derivative orders and without conjectures on zeta values. Their proof does not depend on numerical fitting or a finite-order computation.

What is not implied. Specializing the order s to one number need not preserve functional independence. Nothing here proves that γ, $\operatorname{erf}(1)$, or the Gompertz constant is transcendental. In the growth comparison, every fixed finite collection of coefficients contributes only $O(1)$; thus the method deliberately loses information about any prescribed single coefficient. This part establishes the displayed functional statements, not a resolution of those numerical problems. Historical priority and suitability for a particular journal require further external assessment.

25 Gaussian rank probabilities and the probability model

For $\lambda > 0$, let X_λ have density

$$f_\lambda(t) = \frac{e^{-\lambda t^2}}{Z_\lambda}\mathbf{1}_{[0,1]}(t), \qquad Z_\lambda = \int_0^1 e^{-\lambda t^2}\,dt.$$

This is a centered Gaussian conditioned to lie in $[0,1]$. Rank words retain the population labels after sorting observations; they forget the individual labels of observations from the same population. This distinction is essential: an event concerning only the order of two identically distributed individuals can have probability $1/2$.

The first result classifies all algebraic rank-event probabilities in this model. The second concerns the entire sequence of comparisons between one observation and larger independent samples. For example, when X has precision one and Y has precision two,

n	$\mathbb{P}(\max(X_1,\dots,X_n) < Y)$
1	0.43630479028317338234…
2	0.26488636152732716938…
3	0.18546407010629428595…

Here the subscript on each sample X_i indexes an independent copy of the precision-one population; Y has precision two. The full comparison sequence is jointly algebraically independent. The decimal values merely illustrate the probabilities; none is used as evidence for the proof.

The final conclusion is a statement about functions: the natural probability transforms satisfy no nonzero polynomial equation involving the variable, the function, and finitely many

derivatives, even with arbitrary complex coefficients. It makes no assertion about the arithmetic of a particular transform value such as $H(1)$.

26 The arithmetic input in Gaussian coordinates

Retain the fields $\mathcal{E}$ and $\mathcal{T}$ and the canonical functions $F_{a,q}$ and Ein_q from 6.1. For a positive algebraic precision λ, put

$$\omega_\lambda(t) = e^{-\lambda t^2}\,dt, \qquad Z_\lambda = \int_0^1 e^{-\lambda t^2}\,dt = \frac{1}{2}F_{1/2,1}(\lambda) \in \mathcal{T}.$$

Use outer-first iterated integrals: $I_{\ell v}(t) = \int_0^t \omega_\ell(u) I_v(u)$ and $I_\varnothing = 1$. When $z = t^2$, each Gaussian letter becomes $\frac{1}{2}z^{-1/2}e^{-\lambda z}\,dz$. Thus, for every word w of length r, $I_w(1)$ equals 2^{-r} times the corresponding normalized fractional-shuffle value in Theorem 5.1.

Lemma 26.1 (Gaussian shuffle specialization)**.** *Fix finitely many distinct positive algebraic precisions. For any total order on their letters, the values $I_l(1)$ indexed by Lyndon words l of length at least two are jointly algebraically independent over $\mathcal{T}$. Every polynomial identity over $\mathcal{E}$ between Gaussian word values is a shuffle identity. The only Gaussian-letter Lyndon coordinates that belong to the vertical tower are the one-letter values Z_λ.*

Proof. The substitution above embeds the Gaussian alphabet into the canonical fractional alphabet with letters $y_{1/2,-\lambda}$. Its Lyndon words of length at least two are distinct from every vertical word $0^{q-1}y_{a,\mu}$ and $0^{q-1}x_\mu$. Theorem 5.1, together with the field identification in 6.1, makes these mixed Lyndon coordinates independent over the field generated by all vertical coordinates, which is $\mathcal{T}$. Multiplication by the nonzero rational factors 2^{-r} changes no independence statement. The shuffle identities are preserved by this scaling. □

27 All finite rank events: the distinct-precision case

Take $d \geq 2$ populations with pairwise distinct positive algebraic precisions $\lambda_1, \ldots, \lambda_d$. Population j has density

$$f_j(t) = Z_{\lambda_j}^{-1} e^{-\lambda_j t^2} \mathbf{1}_{[0,1]}(t).$$

Take $n_j \geq 1$ independent observations from population j, all observations mutually independent. Let $\mathcal{W}_{\boldsymbol{n}}$ be the finite set of words having n_j occurrences of letter j. The observed rank word records population labels in *decreasing* order of the observations; ties have probability zero. This direction matches the outer-first integral convention. For $w \in \mathcal{W}_{\boldsymbol{n}}$,

$$\mathbb{P}(W = w) = \frac{\prod_j n_j!}{\prod_j Z_{\lambda_j}^{n_j}} I_w(1). \tag{27.1}$$

Indeed, one ordered simplex contributes the normalized word integral; there are $\prod_j n_j!$ interchangeable labelings within the populations.

Theorem 27.1 (Arithmetic classification of rank payoffs)**.** *Let $c_w \in \overline{\mathbb{Q}}$ be fixed payoffs on $\mathcal{W}_{\boldsymbol{n}}$. Then*

$$\mathbb{E}[c_W] \text{ is algebraic over } \mathcal{T} \quad \Longleftrightarrow \quad c_w \text{ is constant on } \mathcal{W}_{\boldsymbol{n}}.$$

In the constant case the expectation is that common payoff. In every other case it is transcendental over $\mathcal{T}$. Consequently every nonempty proper rank event $S \subsetneq \mathcal{W}_{\boldsymbol{n}}$ has probability transcendental over $\mathcal{T}$.

Proof. The nonzero prefactor in (27.1) belongs to $\mathcal{T}$, so it suffices to consider $P = \sum_w c_w I_w(1)$. Present the abstract shuffle algebra as the polynomial algebra on Lyndon words. The polynomial P has multidegree $\boldsymbol{n}$, meaning exactly n_j occurrences of each letter j. By Lemma 26.1, every mixed Lyndon coordinate is algebraically independent over $\mathcal{T}$. Therefore P can be algebraic over $\mathcal{T}$ only if its polynomial expansion contains no mixed Lyndon coordinate. It must then be a polynomial in the one-letter coordinates alone. Here the one-letter coordinates are themselves algebraically independent over $\mathcal{E}$, by 6.1, so a nonzero coefficient polynomial in them cannot vanish upon evaluation. Multihomogeneity forces

$$P = c \prod_j Z_{\lambda_j}^{n_j} \qquad (c \in \overline{\mathbb{Q}}).$$

The word expansion of that product is

$$\prod_j Z_{\lambda_j}^{n_j} = \Big(\prod_j n_j!\Big) \sum_{w \in \mathcal{W}_{\boldsymbol{n}}} I_w(1).$$

Faithfulness of the word basis makes all c_w equal. Conversely a constant payoff plainly has the stated expectation. Taking $c_w = \mathbf{1}_S(w)$ proves the event assertion. □

For two populations this applies, for example, to any nontrivial exact rejection region determined solely by the two-sample rank word. There is a sharp arithmetic contrast with equal precisions: under identical distributions each rank word has rational probability $\binom{n_1+n_2}{n_1}^{-1}$. The theorem concerns exact probabilities, not a limitation on numerical approximation. For example, the Mann–Whitney rank statistic [16]

$$U_{m,n} = \sum_{i=1}^{m} \sum_{j=1}^{n} \mathbf{1}_{\{X_i < Y_j\}}$$

has transcendental expectation, and every tail probability $\mathbb{P}(U_{m,n} \geq k)$, $1 \leq k \leq mn$, is transcendental over $\mathcal{T}$ when the two positive algebraic precisions differ. Each tail is a nonempty proper subset of the population-label rank words. This applies to exact rejection probabilities under the stated alternative model; it does not change the usual rational rank law under identical distributions.

28 Repeated precisions: a complete combinatorial criterion

Now allow precisions to repeat among d labeled populations. Partition their labels into classes $C_1, \ldots, C_r$ of equal precision, and put $N_j = \sum_{i \in C_j} n_i$. For an original rank word w, let $\bar{w}$ be the merged word that replaces each population label by its precision class. For every merged word v, the number of refinements is the same integer

$$D = \prod_{j=1}^{r} \frac{N_j!}{\prod_{i \in C_j} n_i!}.$$

For an event S of original rank words, write

$$b_v = \#\{w \in S : \bar{w} = v\}.$$

Theorem 28.1 (Balanced-fiber criterion)**.** *The probability of S is algebraic over $\mathcal{T}$ if and only if the integer b_v is independent of the merged rank word v. In that case $\mathbb{P}(S) = b_v/D$ is rational. Otherwise $\mathbb{P}(S)$ is transcendental over $\mathcal{T}$.*

Proof. Conditional on a merged rank word v, its D refinements have equal probability: observations within each equal-precision class have identical densities, and permuting their population labels preserves their joint law. Therefore

$$\mathbb{P}(S) = \sum_v \frac{b_v}{D} \mathbb{P}(\bar{W} = v).$$

If $r \geq 2$, the merged populations have distinct precisions, so Theorem 27.1 applies to the rational payoffs b_v/D. If $r = 1$, there is a single merged word; the criterion is automatic, and every rank event has probability $|S|/D$. □

Thus there are no algebraic irrational rank-event probabilities in this model: all algebraic cases are exactly the balanced-fiber cases.

29 An infinite independent family of comparison probabilities

Let X have precision λ and Y precision μ, with $\lambda \neq \mu$ positive algebraic. Let $X_1, X_2, \ldots$ be independent copies of X, independent of Y, and define

$$p_0 = 1, \qquad p_n = \mathbb{P}\{\max(X_1, \ldots, X_n) < Y\} \quad (n \geq 1).$$

If $F_X(t) = Z_\lambda^{-1} \int_0^t e^{-\lambda u^2}\, du$, then

$$p_n = \mathbb{E}[F_X(Y)^n] = \frac{n!}{Z_\lambda^n Z_\mu} I_{BA^n}(1), \tag{29.1}$$

where A is the X letter and B the Y letter.

Theorem 29.1. *The numbers $p_1, p_2, \ldots$ are jointly algebraically independent over $\mathcal{T}$. In particular the comparison-probability sequence satisfies no nonzero algebraic recurrence over $\mathcal{T}(n)$ involving finitely many consecutive terms and valid for all sufficiently large integers n.*

Proof. Choose the letter order $B < A$. Every word BA^n, $n \geq 1$, is Lyndon: all its proper nonempty suffixes start in A. These words are distinct mixed words. Apply Lemma 26.1 and the nonzero base-field scalings in (29.1). For the recurrence statement, clear denominators and fix an integer n at which at least one nonzero polynomial coefficient remains nonzero. Such an integer exists arbitrarily far out. The recurrence would give a nonzero polynomial relation between a finite subfamily of the independent numbers, a contradiction. □

30 Why independent coefficients force differential transcendence

The coefficient-recovery argument of Lemma 20.2 also handles arbitrary complex coefficients in a proposed differential equation for a single probability transform.

Lemma 30.1 (Finite coefficient transcendence degree)**.** *If a formal power series $f(s) = \sum_{n\geq 0} a_n s^n \in \mathbb{C}[[s]]$ is differentially algebraic over $\mathbb{C}(s)$, then the field $\overline{\mathbb{Q}}(a_0, a_1, \ldots)$ has finite transcendence degree over $\overline{\mathbb{Q}}$. In fact all its coefficients belong to a field generated by finitely many complex constants over $\overline{\mathbb{Q}}$.*

Proof. Choose a nonzero differential polynomial relation of minimal total degree over $\mathbb{C}(s)$ and clear its denominators. In characteristic zero, a nonzero partial derivative has lower total degree, so minimality ensures that its evaluation does not vanish. The evaluated linearization is therefore nonzero. Let k_0 be the field generated over $\overline{\mathbb{Q}}$ by the finitely many complex coefficients of this relation. Apply Lemma 20.2 with no auxiliary series: it puts all coefficients of f in a field generated over k_0 by finitely many initial coefficients and linearization coefficients. This is a field generated by finitely many complex constants over $\overline{\mathbb{Q}}$. □

The coefficient principle is classical: Ostrowski's 1920 Theorem 10 gives the relevant obstruction for convergent power series [17, 18]. The shared coefficient-recovery lemma supplies the direct finite-generation argument, including potentially singular initial points.

Theorem 30.2 (Differentially transcendental probability transforms)**.** *For the rank variable $U = F_X(Y) \in [0,1]$, both*

$$M(s) = \mathbb{E}[e^{sU}] = \sum_{n\geq 0} \frac{p_n}{n!} s^n, \qquad R(s) = \mathbb{E}\left[\frac{1}{1-sU}\right] = \sum_{n\geq 0} p_n s^n \quad (|s|<1)$$

are differentially transcendental over $\mathbb{C}(s)$. The moment generating function M is entire of exponential type at most one.

Proof. The positive-index coefficients of either series are algebraically independent over $\mathcal{T}$, hence over $\overline{\mathbb{Q}}$, by Theorem 29.1. Their coefficient fields have infinite transcendence degree. Lemma 30.1 rules out any algebraic differential equation with complex coefficients. Since $0 \leq U \leq 1$, the entire extension of M follows from dominated termwise integration, with $|M(s)| \leq e^{|s|}$. □

For equal precisions, the probability integral transform instead gives $F_X(Y)$ uniform on $[0,1]$. In that case

$$p_n = \frac{1}{n+1}, \qquad M(s) = \frac{e^s - 1}{s}, \qquad R(s) = -\frac{\log(1-s)}{s},$$

with removable values at zero. Both transforms are differentially algebraic. Thus equality versus inequality of the positive algebraic precisions exactly separates these two differential-algebraic behaviors.

31 A simple nonlinear kernel whose finite Laplace transform is not DA

The choice $\lambda = 1$, $\mu = 2$ yields a particularly explicit analytic example. Put

$$A(t) = \int_0^t e^{-u^2}\, du, \quad a = A(1), \qquad H(s) = \int_0^1 \exp(sA(t) - 2t^2)\, dt.$$

Corollary 31.1 (An entire Gaussian transform)**.** *The function H is entire and differentially transcendental over $\mathbb{C}(s)$. Every positive-index Taylor coefficient of H is algebraically independent of all the others over $\mathcal{T}$. Furthermore*

$$H(s) = \int_0^a e^{sx} h(x)\, dx,$$

where h is positive and analytic on $[0,a]$ and satisfies

$$h(x)h''(x) = -2, \qquad h(0) = 1, \qquad h'(0) = 0. \tag{31.1}$$

Thus a finite-interval Laplace transform of this explicitly specified differentially algebraic kernel is differentially transcendental.

Proof. With A and B now denoting the Gaussian letters of precisions one and two, termwise expansion gives exactly

$$[s^n]H(s) = \frac{1}{n!}\int_0^1 A(t)^n e^{-2t^2}\, dt = I_{BA^n}(1).$$

The mixed Lyndon independence and Lemma 30.1 prove the arithmetic and differential claims. The bound $|H(s)| \leq Z_2 e^{a|s|}$ proves entireness.

The function A has positive derivative on the interval, so its inverse $t = t(x)$ is analytic there. Substituting $x = A(t)$ yields $h(x) = e^{-t(x)^2}$, since $dx = e^{-t^2}dt$. Now $t'(x) = e^{t(x)^2} = 1/h(x)$, whence $h'(x) = -2t(x)$ and $h''(x) = -2/h(x)$. The stated initial conditions follow from $t(0) = 0$. The kernel extends analytically to a neighborhood of the interval because A' has no zero at its real points. □

32 Scope and prior arithmetic results

The arguments classify exact probabilities and transforms of familiar rank constructions. They do not assert joint algebraic independence of all probabilities within one fixed sample size: shuffle products create additional relations. The infinite family singled out in Theorem 29.1 avoids those dependencies.

The dependence on finite truncation and algebraic precisions is explicit. The theorem does not classify arbitrary untruncated Gaussian comparison probabilities or extend the coefficient field by π and complete gamma values. Functional differential transcendence of H does not imply transcendence of $H(1)$.

The proof uses the classical iterated-integral criterion of Deneufchâtel et al. [1], the Lyndon description of the shuffle algebra [2], and Beukers's arithmetic specialization theorem [3]. Ostrowski's coefficient criterion supplies the final differential-transcendence transfer. The required specializations and coefficient-recovery argument are proved in the common sections; the cited classical input theorems are not claimed as new. Existing work on integrated Kummer functions includes related individual Gaussian integrals [5]. The contribution developed here is the rank-event classification and simultaneous all-orders arithmetic, together with the resulting probability transforms. Historical priority has not been exhaustively established. No previously named open conjecture is claimed to be resolved by these Gaussian rank results.

Part IV
Optimal stopping: boundaries, moments, and rewards

33 Bessel stopping laws and their moment fields

Let $d = 2b > 0$, and let Q be the squared Bessel bridge from zero to zero on the time interval $[0, 1]$:

$$dQ_t = \left(2b - \frac{2Q_t}{1-t}\right) dt + 2\sqrt{Q_t}\, dW_t, \qquad Q_0 = 0. \tag{33.1}$$

Using the squared process covers all positive dimensions, including those for which its square root is not an ordinary semimartingale. For a positive scaled barrier x, define

$$\tau_x = \inf\{0 \le t < 1 : Q_t \ge 2x(1-t)\}, \qquad \inf \varnothing = 1, \qquad \mu_m(x) = \mathbb{E}(1-\tau_x)^m. \tag{33.2}$$

Thus the corresponding radial boundary is $\sqrt{2x}\sqrt{1-t}$. Throughout, $\overline{\mathbb{Q}}$ denotes the field of all algebraic numbers.

We use Kummer's function and the generalized Laguerre polynomials in the normalizations

$$\mathrm{M}(a, b, z) = {}_1F_1(a; b; z) = \sum_{n\ge 0} \frac{(a)_n}{(b)_n} \frac{z^n}{n!}, \tag{33.3}$$

$$L_n^{(\alpha)}(z) = \sum_{j=0}^{n} \binom{n+\alpha}{n-j} \frac{(-z)^j}{j!}. \tag{33.4}$$

The binomial coefficients in (33.4) are defined by finite products, so this formula also applies at negative integer parameters.

Theorem 33.1 (The entire moment field)**.** *Let $b \in \mathbb{Q}_{>0}$ and $x \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}$. For every integer $m \ge 1$,*

$$\mu_m(x) \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad L_{m-1}^{(1-b)}(-x) = 0. \tag{33.5}$$

The complete moment field satisfies

$$\overline{\mathbb{Q}}(\mu_1(x), \mu_2(x), \ldots) = \overline{\mathbb{Q}}(\mathrm{M}(1, b, x)), \qquad \mathrm{trdeg}_{\overline{\mathbb{Q}}} = 1. \tag{33.6}$$

More explicitly, there are polynomials $P_m, Q_m \in \mathbb{Q}[z]$, with $Q_m(z) = L_{m-1}^{(1-b)}(-z)$, such that

$$\mu_m(x) = \frac{\mu_1(x)}{P_m(x)\mu_1(x) + Q_m(x)}. \tag{33.7}$$

If $Q_m(x) = 0$, then $P_m(x) \ne 0$ and $\mu_m(x) = 1/P_m(x)$. Otherwise $\overline{\mathbb{Q}}(\mu_m(x)) = \overline{\mathbb{Q}}(\mu_1(x))$.

Theorem 33.2 (Every positive exceptional barrier)**.** *Fix $b \in \mathbb{Q}_{>0}$ and an integer $m \ge 2$. The polynomial $L_{m-1}^{(1-b)}(-x)$ has a positive zero if and only if $(2-b)_{m-1} < 0$. If such a zero exists, it is unique and algebraic. There are no positive exceptional barriers when $b \le 2$. When $b > 2$, every positive exceptional barrier lies in $(0, b-2]$; equality occurs only for $m = 2$.*

For example, at dimension $d = 6$ and scaled barrier $x = 1$,

$$\mu_1(1) = \frac{1}{2(e-2)} \quad \text{is transcendental,} \qquad \mu_2(1) = \frac{1}{2} \quad \text{is rational.} \tag{33.8}$$

The theorem therefore distinguishes actual algebraic exceptions; it does not merely provide a sufficient condition for transcendence.

We also study the optimal reward

$$U_{d,q} = \sup_{0\le\tau\le 1} \mathbb{E}Q_\tau^{q/2}, \qquad q > 0. \tag{33.9}$$

Hobson and Liu [19] identify its optimal square-root boundary. For positive rational d, q, this boundary is algebraic exactly when $q-d \in 2\mathbb{Z}_{\ge 0}$; a complete proof of that arithmetic classification is supplied in Section 37.

Theorem 33.3 (The optimal reward and its stopping law)**.** *Let $d = 2b \in \mathbb{Q}_{>0}$ and $q = d + 2k$, where $k \in \mathbb{Z}_{\ge 0}$. Write τ^* for the optimal stopping time in (33.9), and put $\mu_m^* = \mathbb{E}(1-\tau^*)^m$. Every μ_m^*, $m \ge 1$, is transcendental, and*

$$\operatorname{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(U_{d,q}, \mu_1^*, \mu_2^*, \ldots) = \begin{cases} 1, & d \in 2\mathbb{Z}_{>0}, \\ 2, & d \notin 2\mathbb{Z}_{>0}. \end{cases} \tag{33.10}$$

In the second case, $U_{d,q}$ and each individual μ_m^* are algebraically independent over $\overline{\mathbb{Q}}$.

The moment field itself has transcendence degree one in both cases. It is the addition of the optimal reward that distinguishes the two dimension classes. This statement concerns the joint arithmetic of the reward and the entire infinite moment sequence.

Theorem 33.4 (All even powers in rational dimension)**.** *For every $d \in \mathbb{Q}_{>0}$ and every integer $m \ge 1$, the optimal reward $U_{d,2m}$ is transcendental.*

34 The stopping-time transform

The time change

$$Y_s = \frac{e^s}{2} Q_{1-e^{-s}} \tag{34.1}$$

turns (33.1) into the normalized square-root diffusion

$$dY_s = (b - Y_s)\, ds + \sqrt{2Y_s}\, dB_s, \qquad Y_0 = 0. \tag{34.2}$$

Its generator is $\mathcal{L}f(y) = yf''(y) + (b-y)f'(y)$. If $T_x = \inf\{s \ge 0 : Y_s = x\}$, then $T_x = -\log(1-\tau_x)$, with the natural convention at infinity.

Lemma 34.1. *For every $b, x, s > 0$,*

$$\mathbb{E}e^{-sT_x} = \frac{1}{\mathrm{M}(s, b, x)}. \tag{34.3}$$

Consequently $\mu_m(x) = 1/\mathrm{M}(m, b, x)$ for every integer $m \ge 1$.

Proof. Kummer's equation is

$$zf'' + (b - z)f' - af = 0, \qquad f(z) = \mathrm{M}(a, b, z). \tag{34.4}$$

For $a = s$, the function $h(y) = \mathrm{M}(s, b, y)/\mathrm{M}(s, b, x)$ is analytic on a neighborhood of $[0, x]$, satisfies $\mathcal{L}h = sh$, and has $h(x) = 1$. Its denominator is positive by (33.3). Itô's formula applied to $e^{-s(t\wedge T_x)}h(Y_{t\wedge T_x})$ gives a martingale: the stochastic integrand is bounded on each bounded time interval because Y is stopped in $[0, x]$. Therefore

$$h(0) = \mathbb{E}[e^{-sT_x}; T_x \le t] + \mathbb{E}[e^{-st}h(Y_t); T_x > t].$$

The second term tends to zero since h is bounded on $[0, x]$. Letting $t \to \infty$ proves (34.3). This argument uses the analytic solution directly in Itô's formula and remains valid when $0 < b < 1$; no selection of a solution by boundedness at zero is required. □

35 Kummer specialization and a two-value dichotomy

We use two classical forms of E-function specialization. A vector of E-functions satisfying a homogeneous differential system over $\overline{\mathbb{Q}}(z)$ has the same transcendence degree after evaluation at a nonzero algebraic ordinary point as it has functionally over $\overline{\mathbb{Q}}(z)$. We also use Beukers's refinement [3, Corollary 1.4]: functional linear independence of the components implies linear independence of their values at every such ordinary point. The latter statement gives the precise exclusion of exceptional algebraic evaluation points needed below.

For rational $a, b > 0$, $\mathrm{M}(a, b, z)$ is an E-function. Here is the arithmetic verification. In its exponential-series normalization the coefficients are $c_n = (a)_n/(b)_n \in \mathbb{Q}$ and grow at most polynomially. Outside the finitely many primes dividing the fixed denominators of a, b, divisibility counts for each prime power in the two arithmetic progressions differ by at most one. A common denominator for $c_0, \dots, c_N$ consequently divides a fixed integer to the power $O(N)$ times $\mathrm{lcm}(1, \dots, \lfloor CN \rfloor)$, for a fixed C. Its logarithm is $O(N)$. Equation (34.4) gives holonomicity. These are the required E-function conditions; derivatives remain E-functions.

Set $H_b(z) = \mathrm{M}(1, b, z)$. Directly from its power series,

$$zH_b' = (z + 1 - b)H_b + b - 1. \tag{35.1}$$

Lemma 35.1. *For rational $b > 0$ and nonzero algebraic x, $H_b(x)$ is transcendental.*

Proof. The functions $1, H_b$ are linearly independent over $\overline{\mathbb{Q}}(z)$: H_b is an entire nonpolynomial, so it cannot be rational. They satisfy the closed homogeneous system

$$\frac{d}{dz}\begin{pmatrix} 1 \\ H_b \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ (b-1)/z & 1 + (1-b)/z \end{pmatrix}\begin{pmatrix} 1 \\ H_b \end{pmatrix}.$$

Its only possible finite pole is zero. Beukers's theorem makes $1, H_b(x)$ linearly independent over $\overline{\mathbb{Q}}$, which is exactly the asserted transcendence. □

Lemma 35.2 (A two-value dichotomy)**.** *Let $b \in \mathbb{Q}_{>0}$ and $x \in \overline{\mathbb{Q}} \setminus \{0\}$. If $b \notin \mathbb{Z}$, then $e^x, H_b(x)$ are algebraically independent over $\overline{\mathbb{Q}}$. If $b \in \mathbb{Z}_{>0}$, then*

$$\overline{\mathbb{Q}}(e^x, H_b(x)) = \overline{\mathbb{Q}}(e^x)$$

has transcendence degree one.

Proof. Suppose first that b is nonintegral. Choose $N \geq 1$ with $Nb \in \mathbb{Z}$, and use the differential field $K = \mathbb{C}(z^{1/N}, u)$, where $u = e^z$ and $u' = u$. This is a rational function field in u over $\mathbb{C}(z^{1/N})$. On a fixed local branch away from zero, put

$$V = z^{b-1}u^{-1}H_b, \qquad V' = (b-1)z^{b-2}u^{-1}.$$

We claim that V' has no primitive in K. If it had one, Laurent expansion in u at $u = 0$ would give a coefficient $r \in \mathbb{C}(z^{1/N})$ satisfying

$$r' - r = (b-1)z^{b-2}. \tag{35.2}$$

A finite pole away from zero is impossible: differentiation would produce a pole of strictly greater order with no matching term on the right. Thus r is a finite Laurent polynomial in $z^{1/N}$. Comparison of highest exponents makes its largest exponent $b - 2$. The smallest exponent must be zero, since otherwise its derivative would produce an unmatched smaller exponent. The coefficient recursion in (35.2) connects exponents differing by one; the chain from the highest exponent must terminate at zero. This forces $b - 2 \in \mathbb{Z}_{\geq 0}$, contrary to the hypothesis.

If V were algebraic over K, the normalized trace from $K(V)$ to K would give a primitive of V' in K; differentiation commutes with trace in characteristic zero. This contradiction proves that V is transcendental over K. Hence $e^z, H_b(z)$ are algebraically independent over $\mathbb{C}(z)$. The E-function vector $1, e^z, H_b(z)$ satisfies a homogeneous system whose only possible finite pole is zero. The Siegel–Shidlovskii theorem transfers transcendence degree two to every nonzero algebraic x.

For integer $b = 1$, $H_b = e^z$. For integer $b \geq 2$, its series gives

$$H_b(z) = (b-1)!z^{1-b}\left(e^z - \sum_{j=0}^{b-2}\frac{z^j}{j!}\right). \tag{35.3}$$

At a nonzero algebraic x the coefficient of e^x is nonzero algebraic. The field equality follows, and Lindemann's theorem gives transcendence degree one. □

36 Contiguous relations and the whole moment field

Fix $b \in \mathbb{Q}_{>0}$. The contiguous relation

$$\mathrm{M}(m+1, b, z) = \mathrm{M}(m, b, z) + \frac{z}{m}\mathrm{M}_z(m, b, z) \tag{36.1}$$

and (35.1) give

$$\mathrm{M}(m, b, z) = P_m(z) + Q_m(z)H_b(z), \tag{36.2}$$

where the rational polynomials are defined by

$$P_1 = 0, \qquad Q_1 = 1, \tag{36.3}$$

$$P_{m+1} = P_m + \frac{z}{m}P'_m + \frac{b-1}{m}Q_m, \tag{36.4}$$

$$Q_{m+1} = \left(1 + \frac{1-b+z}{m}\right)Q_m + \frac{z}{m}Q'_m. \tag{36.5}$$

Comparison with the finite formula (33.4) yields

$$Q_m(z) = L_{m-1}^{(1-b)}(-z) = \sum_{j=0}^{m-1}\binom{m-b}{m-1-j}\frac{z^j}{j!}. \tag{36.6}$$

In particular,

$$\deg Q_m = m-1, \qquad [z^{m-1}]Q_m = \frac{1}{(m-1)!}, \qquad Q_m(0) = \frac{(2-b)_{m-1}}{(m-1)!}. \tag{36.7}$$

Proof of Theorem 33.1. Lemma 34.1 and (36.2) show

$$\mu_m(x)^{-1} = P_m(x) + Q_m(x)H_b(x).$$

Here $P_m(x), Q_m(x)$ are algebraic and $H_b(x)$ is transcendental by Lemma 35.1. If $Q_m(x) \neq 0$, the right side, and hence its reciprocal, is transcendental; the same identity recovers $H_b(x)$ from $\mu_m(x)$. If $Q_m(x) = 0$, positivity of $\mathrm{M}(m, b, x)$ gives $P_m(x) > 0$, and $\mu_m(x) = 1/P_m(x)$ is algebraic. Since $\mu_1(x) = 1/H_b(x)$, these facts prove (33.5), (33.6), and (33.7). □

For reference, the first nontrivial identity is

$$\mathrm{M}(2, b, z) = (b-1) + (z+2-b)H_b(z), \qquad \mu_2(x) = \frac{\mu_1(x)}{(b-1)\mu_1(x) + x + 2 - b}.$$

At $b = 3, x = 1$, formula (35.3) gives $H_3(1) = 2(e-2)$, proving (33.8).

Proof of Theorem 33.2. Write $Q = Q_m$, with $m \geq 2$. Its differential equation is

$$zQ'' + (z + 2 - b)Q' - (m - 1)Q = 0. \tag{36.8}$$

A positive zero is simple, since a simultaneous zero of Q, Q' would force the solution of this ordinary differential equation to vanish identically. There cannot be two positive zeros. Indeed, between two consecutive zeros the polynomial has constant sign, and hence either a positive local maximum or a negative local minimum. At such a critical point (36.8) gives $Q'' = (m - 1)Q/z$, contradicting the required sign of the second derivative.

The leading coefficient is positive by (36.7). If $Q(0) \neq 0$, there is therefore one positive zero exactly when $Q(0) < 0$. If $Q(0) = 0$, then $b = k \in \{2, \dots, m\}$, and the finite coefficient formula gives

$$L_{m-1}^{(1-k)}(-z) = z^{k-1}\frac{(m-k)!}{(m-1)!}L_{m-k}^{(k-1)}(-z) > 0 \qquad (z > 0). \tag{36.9}$$

The last inequality follows because every coefficient of the final polynomial is positive. This proves the exact existence criterion. When $b \leq 2$, (36.6) has nonnegative coefficients and a positive leading coefficient, so there is no positive zero.

For the upper bound when $b > 2$, put $z = w + b - 2$. The recurrence becomes

$$Q_{m+1}(w + b - 2) = \frac{w + m - 1}{m}Q_m(w + b - 2) + \frac{w + b - 2}{m}\frac{d}{dw}Q_m(w + b - 2).$$

Starting with $Q_1 = 1$ and $Q_2(w + b - 2) = w$, it follows inductively that these polynomials have nonnegative coefficients in w. For $m \geq 3$ their constant terms are strictly positive: this first holds for $m = 3$, whose constant term is $(b - 2)/2$, and is preserved by the recurrence. Thus no zero exceeds $b - 2$, and equality is possible only for $m = 2$. □

37 Optimal Bessel boundaries and their arithmetic classification

For $d, q > 0$, Hobson and Liu [19, Corollary 3.1 and Theorem 3.1] solve (33.9). In the present notation, let $a = q/2$, $b = d/2$. Their power series (3.5) is $\psi(Y) = \mathrm{M}(a, b, Y/2)$, and their boundary equation (3.7) becomes

$$x\mathrm{M}_z(a, b, x) = a\mathrm{M}(a, b, x). \tag{37.1}$$

Its positive solution $x = x_{a,b}$ gives the optimal rule $\tau^* = \tau_{x_{a,b}}$ and the initial reward

$$U_{d,q} = \frac{(2x_{a,b})^a}{\mathrm{M}(a, b, x_{a,b})}. \tag{37.2}$$

For $d = 1$ this is the reflected Brownian-bridge problem with payoff $|X_\tau|^q$ studied by Ekström and Wanntorp [20]. We use the stochastic optimality theorem from these sources. The arithmetic classification needed here follows independently below.

Lemma 37.1. *For positive real a, b, the functions $f = \mathrm{M}(a, b, z)$ and f' are linearly dependent over $\mathbb{C}(z)$ if and only if $a - b \in \mathbb{Z}_{\geq 0}$.*

Proof. Dependence is equivalent to rationality of f'/f. Every zero of f is a pole of f'/f, with its positive multiplicity as residue, so f then has only finitely many zeros. The polynomial-growth bound for $(a)_n/(b)_n$ makes f entire of order at most one. Hadamard factorization gives $f = e^{cz+\delta}P(z)$ for constants c, δ and a nonzero polynomial P. Substituting into (34.4) yields

$$zP'' + ((2c - 1)z + b)P' + ((c^2 - c)z + bc - a)P = 0.$$

If $r = \deg P$, the coefficient of degree $r+1$ forces $c = 0$ or 1. The coefficient of degree r then forces $a = -r$ when $c = 0$, which is impossible, or $a = b + r$ when $c = 1$. Conversely, Kummer's transformation gives $\mathrm{M}(b+r, b, z) = e^z \mathrm{M}(-r, b, -z)$, an exponential times a polynomial. Its logarithmic derivative is rational. □

Theorem 37.2 (All zeros of the Kummer boundary equation)**.** *For $a, b \in \mathbb{Q}_{>0}$, put*

$$G_{a,b}(z) = z\mathrm{M}_z(a, b, z) - a\mathrm{M}(a, b, z).$$

This function has a unique positive zero $x_{a,b}$. If $a - b \notin \mathbb{Z}_{\geq 0}$, every complex zero of $G_{a,b}$ is transcendental. If $a - b \in \mathbb{Z}_{\geq 0}$, every complex zero is algebraic. In particular,

$$x_{a,b} \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad a - b \in \mathbb{Z}_{\geq 0}. \tag{37.3}$$

Proof. Write $f(z) = \sum c_n z^n$ with $c_n > 0$. For $z > 0$, the probabilities $p_n(z) = c_n z^n / f(z)$ define a nonconstant random variable N on the nonnegative integers. Its mean $R(z) = zf'(z)/f(z)$ satisfies $zR'(z) = \mathrm{Var}_z(N) > 0$. Also $R(z) \to 0$ as $z \downarrow 0$. For each fixed K, the probability of $N \leq K$ tends to zero as $z \to \infty$, by comparison of its degree-K numerator with $c_{K+1} z^{K+1}$. Thus $R(z) \to \infty$, and it crosses a once.

If $a - b \notin \mathbb{Z}_{\geq 0}$, Lemma 37.1 gives functional linear independence of f, f'. Their system is

$$\frac{d}{dz}\begin{pmatrix} f \\ f' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ a/z & 1 - b/z \end{pmatrix}\begin{pmatrix} f \\ f' \end{pmatrix}.$$

They are E-functions, and every nonzero algebraic point is ordinary. Since $G_{a,b}(0) = -a \neq 0$, any algebraic zero would be such an ordinary point. Beukers's theorem precludes the nontrivial linear relation $zf'(z) - af(z) = 0$ there. Thus every complex zero is transcendental in this parameter range. If $a = b + k$, write

$$f(z) = e^z S_k(z), \qquad S_k(z) = \mathrm{M}(-k, b, -z) \in \mathbb{Q}[z]. \tag{37.4}$$

Here $G_{a,b}$ is e^z times the nonzero polynomial $z(S_k + S_k') - (b+k)S_k$, of degree $k+1$. Hence all its complex zeros, including its positive zero, are algebraic. □

For later use, every coefficient of S_k is positive. If $k = 0$, $x_{b,b} = b$. If $k > 0$, the logarithmic derivative of a positive degree-k polynomial gives $0 < zS_k'/S_k < k$ for $z > 0$. At its boundary root,

$$x_{b+k,b} + \frac{x_{b+k,b} S_k'(x_{b+k,b})}{S_k(x_{b+k,b})} = b + k, \qquad x_{b+k,b} > b. \tag{37.5}$$

In particular every algebraic optimal boundary in this family has $x \geq b$.

37.1 Reflected Brownian-bridge normalization

Let X be a Brownian bridge pinned at zero at time 1; starting from (t, x), it satisfies

$$dX_s = -\frac{X_s}{1-s}\, ds + dW_s, \qquad X_t = x.$$

For $q > 0$, define

$$V_q(t, x) = \sup_{t \leq \tau \leq 1} \mathbb{E}_{t,x} |X_\tau|^q.$$

Ekström and Wanntorp [20, Theorem 3.2 and equation (3.8)] give the optimal stopping rule

$$\tau_q^* = \inf\{s \geq t : |X_s| \geq B_q \sqrt{1-s}\},$$

where B_q is the unique positive solution of

$$B_q\mathcal{H}_q'(B_q) = q\mathcal{H}_q(B_q), \qquad \mathcal{H}_q(y) = \int_0^\infty u^{q-1}e^{-u^2/2}(e^{yu}+e^{-yu})\,du. \tag{37.6}$$

Their functions G_q+F_q equal $\mathcal{H}_q$ with these conventions. The integral converges for all complex y, and differentiation and termwise integration are justified uniformly on compact sets by Gaussian domination. Its exact normalization is

$$\mathcal{H}_q(0) = 2^{q/2}\Gamma(q/2), \qquad \frac{\mathcal{H}_q(y)}{\mathcal{H}_q(0)} = \mathrm{M}\left(\frac{q}{2},\frac{1}{2},\frac{y^2}{2}\right), \tag{37.7}$$

with M normalized as in (33.3). Indeed, the integral of $u^{q+2k-1}e^{-u^2/2}$ is $2^{q/2+k-1}\Gamma(q/2+k)$, while $(2k)! = 4^k k!(1/2)_k$. Thus, writing $z_q = B_q^2/2$, the boundary equation is

$$2z_q\mathrm{M}_z(q/2,1/2,z_q) - q\mathrm{M}(q/2,1/2,z_q) = 0. \tag{37.8}$$

The gamma normalization in (37.7) cancels entirely from this equation. No arithmetic assertion concerning π or gamma values is needed.

The notation is consistent with the squared-bridge formulation: $V_q(0,0) = U_{1,q}$ and $z_q = x_{q/2,1/2}$.

Corollary 37.3 (Reflected Brownian bridge)**.** *For every positive rational q, the optimal stopping constant B_q in (37.6) is algebraic if and only if q is a positive odd integer. In every other positive rational case it is transcendental.*

Proof. Apply Theorem 37.2 with $a = q/2, b = 1/2$ and use $B_q = \sqrt{2x_{q/2,1/2}}$. □

For $q = 2m+1$ put $S_m(z) = \mathrm{M}(-m,1/2,-z)$. An exact polynomial for z_q is

$$2z(S_m + S_m') - (2m+1)S_m = 0.$$

In particular $B_1 = 1$, $B_3 = 3^{1/4}$, and B_5 is the positive root of $B^6+5B^4-15B^2-15=0$. The following decimal values are illustrative numerical computations, not ingredients in the proof.

q	B_q	Arithmetic classification
1/2	0.934809739781851	transcendental
1	1	algebraic
3/2	1.072192974430715	transcendental
2	1.150257957204158	transcendental
3	1.316074012952492	algebraic
4	1.481459062236451	transcendental
5	1.637429086210873	algebraic
6	1.782187725478474	transcendental

37.2 Dimension, excursion, and algebraic-boundary rewards

With $B_{d,q} = \sqrt{2x_{q/2,d/2}}$ and $Z_{d,q} = B_{d,q}^2$, the complete rational-parameter classification is

$$Z_{d,q} \in \overline{\mathbb{Q}} \iff B_{d,q} \in \overline{\mathbb{Q}} \iff q-d \in 2\mathbb{Z}_{\geq 0}. \tag{37.9}$$

For example, $B_{d,d} = \sqrt{d}$ and $B_{d,d+2} = [d(d+2)]^{1/4}$. For a Brownian excursion, $(d,q) = (3,1)$, Hobson and Liu's constant $C = B_{3,1} \simeq 1.50339538$ is transcendental. It satisfies their excursion boundary equation

$$C = 2e^{-C^2/2}\int_0^C e^{t^2/2}\,dt.$$

On the exceptional rays $q = d + 2k$, write $b = d/2$ and $x = x_{b+k,b}$. Equations (37.2) and (37.4) give

$$U_{d,q} = \frac{(2x)^{b+k}}{e^x S_k(x)}.$$

The numerator and $S_k(x)$ are nonzero algebraic numbers, and $x > 0$ is algebraic. Lindemann's theorem therefore makes this reward transcendental. The next result strengthens this conclusion to a joint statement about the reward and its stopping law.

38 The joint arithmetic of the reward and its stopping law

Proof of Theorem 33.3. Put $x = x_{b+k,b}$. By Theorem 37.2, x is positive algebraic, and (37.5) gives $x \geq b$. If $b \leq 2$, Theorem 33.2 gives $Q_m(x) \neq 0$ for every m. If $b > 2$, every positive zero of Q_m is at most $b - 2 < x$. Thus the same conclusion holds, and Theorem 33.1 gives

$$\overline{\mathbb{Q}}(\mu_m^*) = \overline{\mathbb{Q}}(H_b(x)) \qquad \text{for every } m \geq 1. \tag{38.1}$$

In particular all these moments are transcendental.

Equations (37.2) and (37.4) show

$$U_{d,q} = c\, e^{-x}, \qquad c = \frac{(2x)^{b+k}}{S_k(x)} \in \overline{\mathbb{Q}} \setminus \{0\}.$$

The positive rational power is algebraic because $2x > 0$ is algebraic; also $S_k(x) > 0$. Hence

$$\overline{\mathbb{Q}}(U_{d,q}, \mu_1^*, \mu_2^*, \ldots) = \overline{\mathbb{Q}}(e^x, H_b(x)).$$

Lemma 35.2 proves the exact transcendence degrees in (33.10). When b is nonintegral, the same lemma, together with (38.1), proves algebraic independence of $U_{d,q}$ and each μ_m^*. □

Two simple cases illustrate the distinction. For $(d, q) = (1, 1)$, the optimal scaled barrier is $x = 1/2$, and

$$U_{1,1} = e^{-1/2}, \qquad \mu_1^* = \frac{1}{\mathrm{M}(1, 1/2, 1/2)}.$$

These numbers are algebraically independent, and the same holds between $U_{1,1}$ and any other positive integer residual-time moment. For $(d, q) = (2, 2)$, one has $x = 1$ and

$$U_{2,2} = 2/e = 2\mu_1^*.$$

The field generated by this reward and all its integer moments has transcendence degree one. In both examples the stopping boundary itself is algebraic.

39 Even-power rewards in every rational dimension

The same contiguous polynomials prove Theorem 33.4, now including transcendental optimal boundaries. We require an exact elimination identity.

Lemma 39.1. *Fix $b \in \mathbb{Q}_{>0}$ and $m \geq 1$, and let P_m, Q_m be given by (36.3)–(36.5). Define*

$$A_m = zP_m' - mP_m + (b-1)Q_m, \tag{39.1}$$
$$D_m = zQ_m' + (z + 1 - b - m)Q_m. \tag{39.2}$$

Then

$$z\mathrm{M}_z(m,b,z) - m\mathrm{M}(m,b,z) = A_m + D_m H_b, \tag{39.3}$$

$$P_m D_m - Q_m A_m = C_m, \qquad C_m = \frac{(1-b)(2-b)_{m-1}}{(m-1)!}. \tag{39.4}$$

Moreover $\deg D_m = m$*, with leading coefficient* $1/(m-1)!$*.*

Proof. The first identity follows by differentiating (36.2) and using (35.1). For the second, work on a local branch and put $v_0 = e^z z^{1-b}$ and $v_m = Q_m v_0$. The function $v_1 = v_0$ solves Kummer's equation with first parameter 1. The operator $f \mapsto f + (z/m)f'$ takes any solution with first parameter m to a solution with first parameter $m+1$, as is checked by differentiating (34.4). Applied to $Q_m v_0$, it gives exactly the recurrence (36.5). Thus v_m is a solution with first parameter m.

Write $f = \mathrm{M}(m,b,z)$ and $W = f v_m' - f' v_m$. Direct substitution gives

$$P_m D_m - Q_m A_m = zW/v_0.$$

Abel's identity for (34.4) gives $W = c_0 e^z z^{-b}$, so this expression is constant. Its value is most simply obtained directly at zero: $P_m(0) + Q_m(0) = f(0) = 1$, and $D_m(0) = (1-b-m)Q_m(0)$, $A_m(0) = -mP_m(0) + (b-1)Q_m(0)$. Consequently

$$(P_m D_m - Q_m A_m)(0) = (1-b)Q_m(0) = C_m.$$

Finally the term zQ_m in (39.2) supplies the asserted leading coefficient and degree. □

Proof of Theorem 33.4. Put $b = d/2$ and $x = x_{m,b}$. If $b \in \{1,\dots,m\}$, then $m-b$ is a nonnegative integer. Theorem 37.2 and (37.4) give $U_{d,2m} = ce^{-x}$ with positive algebraic c, x. Lindemann's theorem proves transcendence.

Otherwise x is transcendental by Theorem 37.2, and $C_m \neq 0$. At this boundary (39.3) gives $A_m(x) + D_m(x)H_b(x) = 0$. The factor $D_m(x)$ cannot vanish: otherwise $A_m(x)$ would also vanish, contrary to (39.4). Eliminating $H_b(x)$ yields

$$\mathrm{M}(m,b,x) = \frac{C_m}{D_m(x)}, \qquad U_{d,2m} = \frac{2^m}{C_m} x^m D_m(x). \tag{39.5}$$

The latter expression is a rational polynomial in x of degree $2m$, with nonzero leading coefficient. If its value were algebraic, x would satisfy a nonzero polynomial over $\overline{\mathbb{Q}}$, a contradiction. This proves the theorem. □

For $m = 1$, $D_1(z) = z - b$ and $C_1 = 1 - b$. Thus

$$U_{d,2} = \frac{2x_{1,b}(x_{1,b} - b)}{1-b} \qquad (b \neq 1), \qquad U_{2,2} = 2/e.$$

At $d = 1$, writing $B = \sqrt{2x_{1,1/2}}$, this specializes to $U_{1,2} = B^2(B^2 - 1)$.

39.1 All positive integer Brownian-bridge rewards

Corollary 39.2. *For the reflected Brownian bridge,* $V_q(0,0)$ *is transcendental for every positive integer* q*. In particular,*

$$V_1(0,0) = e^{-1/2}, \qquad V_2(0,0) = B_2^2(B_2^2 - 1).$$

Proof. If q is odd, then $q - 1 \in 2\mathbb{Z}_{\geq 0}$, so the algebraic-boundary reward argument above applies with $d = 1$. For even q, use Theorem 33.4 with $d = 1$. The first displayed value uses $B_1 = 1$ and $\mathrm{M}(1/2, 1/2, z) = e^z$; the second is the $b = 1/2, m = 1$ case of (39.5). □

40 Exponential-integral specialization for selection thresholds

Corollary 40.1 (The three exponential-integral values)**.** *For every nonzero algebraic α, the numbers $e^{\alpha}, \mathrm{Ein}(\alpha), \mathrm{Ein}(-\alpha)$ are algebraically independent over $\overline{\mathbb{Q}}$, where* $\mathrm{Ein} = \mathrm{Ein}_1$ *in the tower normalization.*

Proof. Theorem 6.1 makes $\mathrm{Ein}_1(\alpha)$ and $\mathrm{Ein}_1(-\alpha)$ algebraically independent over the global exponential field $\mathcal{E} = \overline{\mathbb{Q}}(e^{\beta} : \beta \in \overline{\mathbb{Q}})$; the two arguments are distinct because $\alpha \neq 0$. In particular they are algebraically independent over $\overline{\mathbb{Q}}(e^{\alpha})$. Since e^{α} is transcendental by Lindemann's theorem, the displayed triple is algebraically independent over $\overline{\mathbb{Q}}$. □

41 Five classical thresholds in sequential selection

The next application concerns constants defined by established selection problems. It concerns the optimal thresholds themselves, as opposed to the optimal success probabilities or expected rewards. Put

$$\mathrm{Ein}(z) = \sum_{n\geq 1} \frac{(-1)^{n-1} z^n}{n\, n!}, \qquad J(z) = -\,\mathrm{Ein}(-z) = \sum_{n\geq 1} \frac{z^n}{n\, n!}.$$

For real $x > 0$,

$$\mathrm{Ein}(x) = \int_0^x \frac{1 - e^{-t}}{t}\, dt, \qquad J(x) = \int_0^x \frac{e^t - 1}{t}\, dt.$$

The apparent singularities of the integrands at zero are removable.

41.1 Probabilistic definitions and the threshold equations

All full-information problems below observe independent samples from a known continuous distribution. Applying its distribution function reduces the observations to independent uniform variables on $(0, 1)$.

1. The Gilbert–Mosteller constant c_B governs the full-information best-choice problem: choose one observation to maximize the probability that it is the largest of all observations. With m future observations, the optimal record threshold is $1 - c_B/m + o(m^{-1})$; its defining equation is $J(c_B) = 1$ [21, 22].

2. The recall-duration constant c_R governs maximizing the time for which a selected observation remains the current record when a previously seen observation may be recalled. Its threshold is $1 - c_R/m + o(m^{-1})$, where $\mathrm{Ein}(c_R) = 1$ [23, Theorem 5 and the following asymptotic calculation].

3. The duration constant c_D gives the corresponding threshold $1 - c_D/m + o(m^{-1})$ when recall is prohibited. Ferguson, Hardwick, and Tamaki's equation (3.8) is

$$\int_0^{c_D} e^t (1 - \mathrm{Ein}(t))\, dt = 0. \tag{41.1}$$

   See [23, Theorem 4 and equation (3.8)].

4. The uniform-horizon duration constant c_U belongs to the version in which the total number of observations is uniform on $\{1, \ldots, n\}$ and is initially unknown. If the last available record is selected, its holding time ends at the actual terminal observation. The limiting Poisson threshold is $c_U/(1 - t)$ [24, Theorem 2.1(c), equation (1)].

5. The fixed-threshold constant λ_F maximizes the asymptotic duration reward among rules that select the first observation exceeding one fixed level b. Under the scaling $n(1-b) \to \lambda$, the limiting reward is

$$f(\lambda) = \sum_{k\geq 1} \frac{H_k}{k+1} e^{-\lambda} \frac{\lambda^k}{k!} = \frac{\mathrm{Ein}(\lambda) - e^{-\lambda} J(\lambda)}{\lambda}, \qquad H_k = \sum_{j=1}^{k} \frac{1}{j}. \tag{41.2}$$

This restricted optimization problem is stated in [24, Section 4, Remark].

Here and below, shifting the finite-horizon index m by one has no effect on the displayed first-order asymptotic coefficients. The five constants are the unique positive solutions of

$$J(c_B) = 1, \tag{41.3}$$
$$\mathrm{Ein}(c_R) = 1, \tag{41.4}$$
$$e^{c_D} \mathrm{Ein}(c_D) - J(c_D) = e^{c_D} - 1, \tag{41.5}$$
$$e^{-c_U} J(c_U) + (c_U - 1)\,\mathrm{Ein}(c_U) = 2\big(e^{-c_U} + c_U - 1\big), \tag{41.6}$$
$$e^{\lambda_F} \mathrm{Ein}(\lambda_F) - (1+\lambda_F) J(\lambda_F) = 0. \tag{41.7}$$

Equation (41.5) follows from (41.1) by differentiation of $e^x(1-\mathrm{Ein}(x))-1+J(x)$, or by integration by parts. Equation (41.6) is Tamaki's equation after substituting $J(-x) = -\mathrm{Ein}(x)$, and (41.7) is the critical-point equation of (41.2).

Constant	Problem	Approximate value
c_B	Best choice	0.804352
c_R	Duration with recall	1.345017
c_D	Duration without recall	2.119824
c_U	Uniform random horizon	3.692536
λ_F	Best fixed-threshold duration policy	2.839699

41.2 Existence, uniqueness, and the restricted maximizer

We give the elementary real analysis so that no numerical root-finding is needed to identify which solution is meant. Both Ein and J are strictly increasing from zero to infinity on $(0,\infty)$; this proves existence and uniqueness in (41.3) and (41.4). For the other equations define

$$\begin{aligned} D(x) &= e^x(1-\mathrm{Ein}(x)) - 1 + J(x), \\ U(x) &= e^{-x} J(x) + (x-1)\,\mathrm{Ein}(x) - 2(e^{-x} + x - 1), \\ F(x) &= e^x\,\mathrm{Ein}(x) - (1+x) J(x). \end{aligned}$$

Direct differentiation gives

$$\begin{aligned} D'(x) &= e^x(1-\mathrm{Ein}(x)), \\ U''(x) &= e^{-x}(J(x)-1), \\ F''(x) &= e^x(\mathrm{Ein}(x)-1). \end{aligned}$$

Also $D(0) = 0$ and $U(0) = U'(0) = F(0) = F'(0) = 0$, interpreted by entire continuation. The function D increases until c_R and then decreases. Once $\mathrm{Ein}(x) \geq 2$, one has $D'(x) \leq -e^x$, so $D(x) \to -\infty$. Consequently D has exactly one positive zero.

The inequality

$$0 < J(x) \leq e^x - 1$$

follows by integrating $(e^t-1)/t \le e^t$. It implies

$$U'(x) = \mathrm{Ein}(x) - e^{-x}J(x) + e^{-x} - 1 \ge \mathrm{Ein}(x) - 2 + 2e^{-x} \longrightarrow \infty,$$
$$F'(x) = e^x\,\mathrm{Ein}(x) - J(x) - e^x + 1 \ge e^x(\mathrm{Ein}(x) - 2) + 2 \longrightarrow \infty.$$

Since U'' changes sign exactly once, from negative to positive, U' has exactly one positive zero. Thus U first decreases and then increases to infinity, giving exactly one positive zero of U. The same argument using F'' proves the corresponding assertion for F. In particular, the zero solutions at the origin in (41.5)–(41.7) are excluded by the word "positive."

Finally, if $N(x) = \mathrm{Ein}(x) - e^{-x}J(x)$, then $N'(x) = e^{-x}J(x)$, so

$$f'(x) = -\frac{e^{-x}}{x^2}F(x).$$

The preceding sign analysis shows that f increases up to λ_F and decreases after it. Hence λ_F is its unique global maximizer on $(0,\infty)$.

41.3 Transcendence transfer

By Corollary 40.1, for every nonzero algebraic α, the numbers

$$1,\ \mathrm{Ein}(\alpha),\ J(\alpha) \quad \text{are linearly independent over} \quad K_\alpha = \overline{\mathbb{Q}}(e^\alpha).$$

Here $J(\alpha) = -\,\mathrm{Ein}(-\alpha)$.

Theorem 41.1 (Transcendence of the five selection thresholds)**.** *Each of $c_B, c_R, c_D, c_U, \lambda_F$ is transcendental.*

Proof. Assume that one of these positive constants, denoted by x, is algebraic. Then $x \neq 0$ and $e^x, e^{-x} \in K_x$. Each of equations (41.3)–(41.7) is a nonzero K_x-linear relation among $1, \mathrm{Ein}(x), J(x)$. For (41.3) and (41.4), respectively, the coefficients of $J(x)$ and $\mathrm{Ein}(x)$ are 1. For (41.5) and (41.7), the coefficient of $\mathrm{Ein}(x)$ is $e^x \neq 0$; for (41.6), the coefficient of $J(x)$ is $e^{-x} \neq 0$. Every case contradicts Corollary 40.1. □

This is an application of algebraic-point independence to established optimal-stopping equations. It proves individual transcendence of all five thresholds. It does not assert their joint algebraic independence, or the transcendence of the corresponding optimal rewards. For example, $f(\lambda_F) \approx 0.426317$ is a value at a transcendental argument; the algebraic-point theorem does not classify it.

42 Scope of the stopping applications

The stochastic optimality formulas used in these applications are inputs from the cited probability literature. Their arithmetic consequences are proved here: the complete boundary classification, the transcendence of the stated optimal rewards, the exact moment fields and Laguerre exceptions, and the five selection thresholds. The direct transform proof and the common Kummer specialization argument serve all the Bessel and Brownian applications; the selection application uses the shared vertical independence theorem.

The moment-field theorem does not assert algebraic independence among all moments: their fractional-linear relations are explicit. The joint reward theorem concerns $q - d \in 2\mathbb{Z}_{\ge 0}$; the even-power theorem adds rewards at transcendental optimal boundaries. These results do not classify rewards for every rational pair (d, q). The Brownian family uses the symmetric reward $|X_\tau|^q$; its $q = 1$ boundary is 1, distinct from Shepp's one-sided boundary $0.839923675\ldots$. The latter is not classified by this argument. Priority for the arithmetic deductions has not been exhaustively verified; no resolution of a previously named transcendence conjecture is asserted by these stopping applications.

Part V
Reciprocal gamma sampling and arithmetic cost

43 Arithmetic sampling of the reciprocal gamma function

The preceding independence results concern finite values and functions. We now examine the arithmetic cost of reconstructing a connection constant from rational samples. Let

$$\gamma = \lim_{n\to\infty}(H_n - \log n), \qquad H_n = \sum_{k=1}^{n} \frac{1}{k}.$$

For the fixed normalization

$$g(z) = \frac{1}{\Gamma(1+z)}, \qquad g(n) = \frac{1}{n!} \quad (n \geq 0), \qquad g'(0) = \gamma, \tag{43.1}$$

the samples are rational, but controlling the derivative requires both analytic error estimates and reduced denominator bounds. The Newton–Laguerre identity is known [25, Theorem 2.1 and Corollary 2.6]; we justify the identification below and prove exact prime residues, growing-window denominator bounds, fixed-scale bounds, and an integer-stride summability transition. These results do not establish irrationality or transcendence of γ; see also the surveys and approximation work in [26, 27].

Notation and principal results

For $r \in \mathbb{Q}$, $\mathrm{den}(r)$ is its positive reduced denominator, with $\mathrm{den}(0) = 1$. Normalize $v_p(p) = 1$. Set $d_N = \mathrm{lcm}(1, \ldots, N)$ and

$$L_n(x) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{x^k}{k!}, \qquad R_N(x) = -\sum_{n=1}^{N} \frac{L_n(x)}{n}. \tag{43.2}$$

All implicit constants associated with a fixed rational scale or filter may depend on that scale or filter. They never depend on N.

The central conclusions are:

1. If $x = a/b > 0$ is reduced, $\sqrt{N} < p \leq N$, and $p \nmid ab$, then, with $m = \lfloor N/p \rfloor$,

$$p^{m+1} R_N(x) \equiv -\frac{x^m}{m\, m!} \pmod{p}.$$

   In particular $v_p(R_N(x)) = -(m+1)$.

2. For arbitrary integer weights with sum one and window $w_N = o(\sqrt{N})$, the reduced denominator of $\sum_{j=0}^{w_N} c_{N,j} R_{N+j}(1)$ remains at least $\exp((\frac{1}{2} - o(1))N \log N)$.

3. For fixed distinct positive rational scales and fixed nonzero integer weights, every nontrivial combination $\sum_j c_j R_N(x_j)$ has denominator at least $\exp(\frac{1}{3} N \log N - O(N))$.

4. For samples $x^j/(sj)!$, integer stride $s \geq 2$ destroys ordinary Newton convergence away from the terminating sample points. The derivative series is Abel summable only for $s = 2$ among these strides.

The constants $1/2$ and $1/3$ are proved lower bounds, not claimed optimal denominator exponents.

44 Rational sampling and an exact tail

We first establish the analytic identification carefully. Formal agreement at integers alone would not justify an interpolation formula or its differentiation.

Proposition 44.1 (Newton identification)**.** *For every fixed $x > 0$, the series*

$$\sum_{n=0}^{\infty}(-1)^n L_n(x)\binom{z}{n} = \frac{x^z}{\Gamma(1+z)} \tag{44.1}$$

converges absolutely and locally uniformly in $\Re z > -1/4$. Consequently,

$$-\sum_{n=1}^{\infty}\frac{L_n(x)}{n} = \gamma + \log x. \tag{44.2}$$

Proof. The classical fixed-x Laguerre estimate is $L_n(x) = O_x(n^{-1/4})$; a fuller expansion is used below. The bound for the generalized binomial coefficient on compact subsets of the indicated half-plane gives normal absolute convergence. One may use $(-1)^n\binom{z}{n} = \Gamma(n-z)/(\Gamma(-z)\Gamma(n+1))$; the apparent exception at nonnegative integers is removable.

For $-1/4 < \Re z < 0$, the beta integral gives

$$(-1)^n\binom{z}{n} = \frac{\int_0^1 t^{n-z-1}(1-t)^z\,dt}{\Gamma(-z)\Gamma(1+z)}.$$

Absolute convergence justifies interchanging sum and integral. The generating function, also obtained directly from the finite definition, is

$$\sum_{n\ge 0} L_n(x)t^n = \frac{1}{1-t}\exp\left(-\frac{xt}{1-t}\right), \quad |t| < 1. \tag{44.3}$$

After substituting $u = t/(1-t)$, the resulting integral is

$$\frac{1}{\Gamma(-z)\Gamma(1+z)}\int_0^{\infty} u^{-z-1}e^{-xu}\,du = \frac{x^z}{\Gamma(1+z)}.$$

Analytic continuation within the half-plane proves (44.1). Normal convergence permits differentiation at zero. Since $\frac{d}{dz}\binom{z}{n}\Big|_{z=0} = (-1)^{n-1}/n$ for $n \ge 1$, the derivative is (44.2). □

Lemma 44.2 (Finite rational form)**.** *For $N \ge 1$,*

$$R_N(x) = -H_N + \sum_{k=1}^{N}(-1)^{k-1}\binom{N}{k}\frac{x^k}{k\,k!}. \tag{44.4}$$

If $x = a/b$ is rational in lowest terms, then

$$\operatorname{den}(R_N(x)) \mid b^N N! d_N. \tag{44.5}$$

Proof. Interchange the two finite sums in (43.2) and use

$$\sum_{n=k}^{N}\frac{1}{n}\binom{n}{k} = \frac{1}{k}\sum_{n=k}^{N}\binom{n-1}{k-1} = \frac{1}{k}\binom{N}{k}.$$

Each denominator in (44.4) divides $b^N N! d_N$. □

Proposition 44.3 (Oscillatory tail)**.** *For fixed $x > 0$, put $A(x) = e^{x/2}/(\sqrt{\pi}\, x^{3/4})$. Then*

$$R_N(x) - \gamma - \log x = -A(x)N^{-3/4}\sin(2\sqrt{Nx} - \pi/4) + O_x(N^{-5/4}). \tag{44.6}$$

Proof. The classical Laguerre expansion [9, 18.15.14–18.15.16] specialized to parameter zero reads

$$L_n(x) = \frac{e^{x/2}}{\sqrt{\pi}x^{1/4}} n^{-1/4}\left(\cos\theta_n + \beta_x n^{-1/2}\sin\theta_n + O_x(n^{-1})\right),$$

where $\theta_n = 2\sqrt{nx} - \pi/4$ and $\beta_x = (4x^2 - 24x + 3)/(48\sqrt{x})$. By (44.2), the desired error equals $\sum_{n>N} L_n(x)/n$.

For $f(t) = t^{-5/4}\cos(2\sqrt{xt} - \pi/4)$, Euler–Maclaurin summation with its second-derivative remainder gives

$$\sum_{n>N} f(n) = \int_N^\infty f(t)\,dt + O_x(N^{-5/4}),$$

because $f(N) = O(N^{-5/4})$, $f'(N) = O_x(N^{-7/4})$, and $\int_N^\infty |f''(t)|dt = O_x(N^{-5/4})$. Substitution $u = \sqrt{t}$ followed by integration by parts gives

$$\int_N^\infty f(t)\,dt = -x^{-1/2}N^{-3/4}\sin\theta_N + O_x(N^{-5/4}).$$

The next, explicitly oscillatory term has tail $O_x(N^{-5/4})$ by the same argument; the remaining absolute tail is $\sum_{n>N} O_x(n^{-9/4}) = O_x(N^{-5/4})$. This proves the formula. Keeping the second oscillatory term matters: the first-order Laguerre error alone is insufficient for this conclusion. □

45 Exact prime residues after reduction

The raw denominator in (44.5) is an upper bound. The next theorem prevents unproved claims that cancellation makes it small.

Theorem 45.1 (Large-prime residue)**.** *Let $x \in \mathbb{Q}$ and $\sqrt{N} < p \le N$ be prime with $p \nmid \mathrm{den}(x)$. Set $m = \lfloor N/p \rfloor$. In the local ring $\mathbb{Z}_{(p)}$,*

$$p^{m+1}R_N(x) \equiv -\frac{x^m}{m\,m!} \pmod{p}. \tag{45.1}$$

If also p does not divide the numerator of x, then

$$v_p(R_N(x)) = -(m+1). \tag{45.2}$$

Proof. Write $N = mp+r$, $0 \le r < p$. Since $N < p^2$, we have $1 \le m < p$. In (44.4), $v_p(k!) = \lfloor k/p \rfloor$. The term $k = mp$ is the unique term whose denominator $k\,k!$ has valuation $m+1$; every other term has denominator valuation at most m. Also $v_p(H_N) \ge -1$. The binomial coefficient of the distinguished term is a unit, since

$$\binom{mp+r}{mp} \equiv 1 \pmod{p}.$$

This congruence follows immediately by comparing coefficients in $(1+T)^{mp+r} \equiv (1+T^p)^m(1+T)^r \pmod{p}$.

The product of the nonmultiples of p in each full block of p consecutive integers is -1 modulo p, so

$$\frac{(mp)!}{p^m} \equiv (-1)^m m! \pmod{p}.$$

Multiplying the distinguished summand by p^{m+1} therefore gives

$$\frac{(-1)^{mp-1}x^{mp}}{m(-1)^m m!} \equiv -\frac{x^m}{m\,m!} \pmod{p}.$$

The same expression is valid for $p = 2$, where signs coincide. All the other terms vanish after reduction. A nonzero residue gives (45.2). □

Lemma 45.2 (Elementary prime-weight estimate)**.** *Uniformly for $2 \le y \le N$,*

$$\sum_{y<p\le N} \left(\left\lfloor \frac{N}{p} \right\rfloor + 1 \right) \log p = N \log(N/y) + O(N). \tag{45.3}$$

Proof. The elementary Chebyshev bound $\vartheta(t) = \sum_{p\le t} \log p = O(t)$ and the factorial identity imply

$$\log(N!) = N \sum_{p\le N} \frac{\log p}{p} + O(N).$$

Indeed, replacing $\lfloor N/p \rfloor$ by N/p costs $O(N)$; the total contribution from prime powers p^r, $r \ge 2$, is $O(N)$ because $\sum_p \log p/[p(p-1)]$ converges. Stirling's elementary logarithmic estimate yields $\sum_{p\le t} \log p/p = \log t + O(1)$. Apply this at N and y. The floor and extra-one errors are $O(\vartheta(N))$. No prime number theorem is required. □

Corollary 45.3 (Factorial-scale reduced denominator)**.** *For each fixed positive rational x, with $q_N(x) = \mathrm{den}(R_N(x))$,*

$$\frac{1}{2} N \log N - O_x(N) \le \log q_N(x) \le N \log N + O_x(N). \tag{45.4}$$

Proof. All sufficiently large primes are coprime to the numerator and denominator of x. Sum (45.2) over $p > \sqrt{N}$ and apply Lemma 45.2. The upper bound follows from (44.5), $\log d_N = O(N)$, and Stirling's estimate. □

Example 45.4. For $x = 1$ and $N = 20$, primes above $\sqrt{20}$ occur in the reduced denominator with exponents

$$v_5(q_{20}) = 5, \quad v_7(q_{20}) = 3, \quad v_{11}(q_{20}) = v_{13}(q_{20}) = v_{17}(q_{20}) = v_{19}(q_{20}) = 2.$$

These are exact valuations, not numerical estimates or valuations of an unreduced common denominator.

46 The square-root scale for integer filters

Write $R_N = R_N(1)$. A filter with weights summing to one retains the target whenever its error tends to zero. The next assertion concerns its denominator even when its coefficients are too large to ensure convergence.

Theorem 46.1 (Growing-window denominator rigidity)**.** *Let w_N be a sequence of nonnegative integers with $w_N = o(\sqrt{N})$. For arbitrary integers $c_{N,0}, \dots, c_{N,w_N}$ satisfying $\sum_j c_{N,j} = 1$, put*

$$T_N = \sum_{j=0}^{w_N} c_{N,j} R_{N+j}, \qquad q_N = \mathrm{den}(T_N).$$

Then

$$\log q_N \ge \left(\frac{1}{2} - o(1) \right) N \log N. \tag{46.1}$$

There is no bound on the size or signs of the coefficients in this denominator assertion.

Proof. Take $\sqrt{N+w_N} < p \le N$ for which the interval $(N, N+w_N]$ contains no multiple of p. Then $\lfloor (N+j)/p \rfloor = m$ is independent of j, and Theorem 45.1 gives

$$p^{m+1} T_N \equiv -\frac{\sum_j c_{N,j}}{m\, m!} = -\frac{1}{m\, m!} \not\equiv 0 \pmod{p}.$$

Thus $v_p(T_N) = -(m+1)$ regardless of coefficient size.

A crossing prime must divide $\prod_{j=1}^{w_N}(N+j)$, with the empty product interpreted as one. Its total discarded weight is at most

$$(\sqrt{N}+1) \sum_{p \mid \prod_{j=1}^{w_N}(N+j)} \log p \le (\sqrt{N}+1) w_N \log(N+w_N) = o(N \log N).$$

Before these primes are discarded, Lemma 45.2 with $y = \sqrt{N+w_N}$ gives $\frac{1}{2} N \log N - O(N) - o(N \log N)$. This proves (46.1). □

Theorem 46.2 (Persistence of the leading error)**.** *In the preceding setting, suppose in addition that*

$$M_N := \sum_{j=0}^{w_N} j\, |c_{N,j}| = o(\sqrt{N}). \tag{46.2}$$

Then, with $A = e^{1/2}/\sqrt{\pi}$ and $\theta_N = 2\sqrt{N} - \pi/4$,

$$T_N - \gamma = -A N^{-3/4} \sin \theta_N + o(N^{-3/4}). \tag{46.3}$$

On the set $\mathcal{S} = \{N : |\sin \theta_N| \ge 1/2\}$, which has natural density $2/3$,

$$q_N |T_N - \gamma| \longrightarrow \infty \qquad (N \to \infty,\ N \in \mathcal{S}). \tag{46.4}$$

Proof. For $0 \le j \le w_N$, the bound $L_n(1) = O(n^{-1/4})$ implies

$$|R_{N+j} - R_N| \le \sum_{k=N+1}^{N+j} \frac{|L_k(1)|}{k} = O(j N^{-5/4}).$$

Because the weights sum to one, $T_N - R_N = O(M_N N^{-5/4}) = o(N^{-3/4})$. Apply Proposition 44.3.

For a fixed interval $I \subset [0, 2\pi)$, the elementary count

$$\#\{n \le X : (2\sqrt{n} - \pi/4) \bmod 2\pi \in I\} = \frac{|I|}{2\pi} X + O_I(\sqrt{X})$$

follows by squaring the endpoints of the phase windows and summing their lengths; there are $O(\sqrt{X})$ rounding errors. Finite unions give the same assertion. The subset on which $|\sin\theta| \ge 1/2$ has length $4\pi/3$ per period. On this subset the error in (46.3) is at least $A N^{-3/4}/4$ eventually. Combine this with (46.1). □

Corollary 46.3 (Fixed local extrapolation)**.** *Every fixed finite rational stencil $T_N = \sum_{j=0}^{w} c_j R_{N+j}$, $\sum_j c_j = 1$, satisfies*

$$T_N - \gamma = -A N^{-3/4} \sin \theta_N + O(N^{-5/4}), \qquad \log \operatorname{den}(T_N) \ge \tfrac{1}{2} N \log N - O(N).$$

Consequently its primitive linear forms do not tend to zero along the full sequence.

Proof. The analytic estimate is the preceding telescoping argument. Clear the fixed coefficient denominator D. Away from its finitely many prime divisors, the residue argument uses $\sum c_j = 1$ unchanged. The fixed window loses only $O_w(\sqrt{N} \log N) = O_w(N)$ prime weight. □

The same scale has two separate explanations. The derivative of the analytic phase is $\theta_N' = 1/\sqrt{N}$, so changing that phase by order one requires about $\sqrt{N}$ index steps. Arithmetically, the residues at primes above $\sqrt{N}$ remain identical across most shorter windows. Theorems 46.1 and 46.2 make both statements precise. They do not rule out exceptional subsequences, long windows, or filters with large weighted moment.

47 Fixed rational scales and logarithm cancellation

Using several scales can cancel logarithms exactly. The issue is whether it also cancels enough of the reduced denominator.

Theorem 47.1 (Fixed-scale denominator bound)**.** *Let $x_1, \dots, x_r$ be fixed distinct positive rational numbers, and let $c_1, \dots, c_r$ be fixed nonzero integers. Define*

$$S_N = \sum_{j=1}^{r} c_j R_N(x_j).$$

Then

$$\frac{1}{3} N \log N - O(N) \le \log \operatorname{den}(S_N) \le N \log N + O(N). \tag{47.1}$$

In particular, no fixed nonzero scale combination has subfactorial reduced denominator in the sense $\log \operatorname{den}(S_N) = o(N \log N)$.

Proof. Choose a positive integer D such that all Dx_j are integers, and put

$$P_m = \sum_{j=1}^{r} c_j x_j^m, \qquad A_m = D^m P_m = \sum_{j=1}^{r} c_j (Dx_j)^m \in \mathbb{Z}.$$

Because the largest positive x_j dominates, $A_m \neq 0$ for every sufficiently large m. Moreover $\log \max(1, |A_m|) = O(m+1)$.

For $\sqrt{N} < p \le N$, $p \nmid D$, and $m = \lfloor N/p \rfloor$, Theorem 45.1 gives

$$p^{m+1} S_N \equiv -\frac{P_m}{m\, m!} \pmod{p}. \tag{47.2}$$

Thus the full exponent $m+1$ remains whenever $p \nmid A_m$.

Set $M = \lfloor N^{1/3} \rfloor$ and use primes in $(N/(M+1), N]$. For all sufficiently large N these primes exceed $\sqrt{N}$, and $1 \le m \le M$. Their total potential contribution is, by Lemma 45.2,

$$N \log(M+1) + O(N) = \tfrac{1}{3} N \log N + O(N).$$

For a nonzero A_m, the entire contribution lost to exceptional primes in its block is at most

$$(m+1) \sum_{p \mid A_m} \log p \le (m+1) \log |A_m| = O((m+1)^2).$$

Summing to M costs $O(M^3) = O(N)$. There are only finitely many zero A_m; discarding their complete blocks also costs $O(N)$. Primes dividing the fixed integer D are eventually below the cutoff. This proves the lower bound. Every term has denominator dividing $D^N N! d_N$, which proves the upper bound. □

Corollary 47.2 (Log-balanced combinations)**.** *Under the hypotheses of Theorem 47.1, assume*

$$B := \sum_{j=1}^{r} c_j \neq 0, \qquad \prod_{j=1}^{r} x_j^{c_j} = 1. \tag{47.3}$$

Then $U_N = S_N / B \to \gamma$. There exist a constant $c > 0$ and a set $\mathcal{T} \subset \mathbb{N}$ of positive lower density such that

$$|U_N - \gamma| \ge c N^{-3/4} \quad (N \in \mathcal{T} \text{ sufficiently large}),$$

and

$$\operatorname{den}(U_N) |U_N - \gamma| \longrightarrow \infty \quad (N \to \infty,\ N \in \mathcal{T}).$$

Proof. The logarithms cancel by positivity of the scales. The tail formula gives

$$U_N - \gamma = N^{-3/4} F(\sqrt{N}) + O(N^{-5/4}),$$

where

$$F(t) = -\frac{1}{B} \sum_{j=1}^{r} c_j A(x_j) \sin(2\sqrt{x_j} t - \pi/4).$$

The frequencies are distinct and positive, so F is nonzero. For any fixed nonzero real a, comparison with the integral, using the integrable-on-finite-interval derivative bound, yields $\sum_{n \le X} e^{ia\sqrt{n}} = O_a(\sqrt{X})$. Expanding the square therefore gives

$$\frac{1}{X} \sum_{n \le X} F(\sqrt{n})^2 \longrightarrow \frac{1}{2B^2} \sum_{j=1}^{r} c_j^2 A(x_j)^2 > 0.$$

Since F is bounded, a positive lower density of these values has absolute value bounded below by a fixed positive constant. On a slightly smaller threshold set the error remainder is harmless. Finally, division of a rational number by a nonzero integer cannot decrease its reduced denominator, so $\operatorname{den}(U_N) \ge \operatorname{den}(S_N)$. Apply (47.1). □

Example 47.3 (A real cancellation that does not solve the problem)**.** The combination

$$U_N = 2R_N(2) - R_N(4)$$

converges to γ, since $2\log 2 - \log 4 = 0$. Its first moment $P_1 = 2 \cdot 2 - 4 = 0$ kills the leading residue for primes in $(N/2, N]$. It need not remove every power of those primes. For $m \ge 2$, $P_m = 2^{m+1} - 4^m \ne 0$. Theorem 47.1 shows that this genuine cancellation still leaves factorial-scale reduced denominators.

48 Integer strides and a summability transition

One might attempt to improve reconstruction by retaining only every s-th factorial value. For fixed integer $s \ge 1$ and $x > 0$, define

$$b_n^{(s)}(x) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{x^j}{(sj)!}. \tag{48.1}$$

These are the signed forward differences of the samples of $g_{s,x}(z) = x^z/\Gamma(1+sz)$ at $z = 0, 1, 2, \ldots$. The formally predicted derivative is $s\gamma + \log x$. For $s = 1$, $b_n^{(1)}(x) = L_n(x)$.

Lemma 48.1 (Exact stride generating function)**.** *For $|t| < 1$,*

$$\mathcal{B}_{s,x}(t) := \sum_{n \ge 0} b_n^{(s)}(x) t^n = \frac{1}{1-t} \mathcal{A}_s\left(-\frac{xt}{1-t}\right), \qquad \mathcal{A}_s(w) = \sum_{j \ge 0} \frac{w^j}{(sj)!}. \tag{48.2}$$

If $y^s = w$ and $\omega = e^{2\pi i/s}$, then

$$\mathcal{A}_s(w) = \frac{1}{s} \sum_{k=0}^{s-1} e^{\omega^k y}. \tag{48.3}$$

Proof. The binomial generating identity $\sum_{n \ge j} \binom{n}{j} t^n = t^j/(1-t)^{j+1}$ gives (48.2); the absolute double sum converges on smaller disks. Averaging the exponential series over the s roots of unity retains exactly the powers divisible by s, proving (48.3). □

Theorem 48.2 (Failure of ordinary Newton reconstruction)**.** *For $s \geq 2$ and $x > 0$, the sequence $b_n^{(s)}(x)$ is not $O(n^K)$ for any fixed $K \geq 0$. Consequently*

$$\sum_{n\geq 0}(-1)^n b_n^{(s)}(x)\binom{z}{n}$$

fails the term test at every $z \notin \mathbb{Z}_{\geq 0}$, and $-\sum_{n\geq 1} b_n^{(s)}(x)/n$ also fails the term test. At $z \in \mathbb{Z}_{\geq 0}$ the Newton series terminates and recovers the sample, as it must.

Proof. Choose $0 < \phi < \pi/2$ and approach $t = 1$ along $t = 1 - re^{i\phi}$, $r \downarrow 0$. These points lie inside the unit disk, and $1 - |t| \asymp r$. The number $w = -xt/(1-t)$ has modulus $x/r + O(1)$ and argument tending to $\pi - \phi$. Among its s-th roots there is a unique root of largest real part. That real part is

$$x^{1/s} r^{-1/s} \cos((\pi - \phi)/s) + o(r^{-1/s}) > 0.$$

The other exponentials in (48.3) are exponentially smaller in modulus, so they cannot cancel this dominant term. Thus $|\mathcal{B}_{s,x}(t)|$ grows faster than every power of $1/r$. If $b_n^{(s)}(x) = O(n^K)$, its power series would instead satisfy $|\mathcal{B}_{s,x}(t)| = O((1-|t|)^{-K-1})$, a contradiction.

For $z \notin \mathbb{Z}_{\geq 0}$, $|\binom{z}{n}| \asymp_z n^{-\Re z - 1}$. The term test would therefore force polynomial growth of $b_n^{(s)}(x)$, already excluded. The derivative series is the case of the factor $1/n$. □

Ordinary divergence does not determine Abel summability. The next identity gives the latter exactly, without asserting an unjustified interchange at the boundary.

Theorem 48.3 (Three Abel regimes)**.** *For $0 < t < 1$, define*

$$\mathcal{D}_{s,x}(t) = -\sum_{n\geq 1} \frac{b_n^{(s)}(x)}{n} t^n, \qquad u = \left(\frac{xt}{1-t}\right)^{1/s}, \qquad \rho_k = e^{(2k+1)\pi i/s}.$$

With the entire convention $\mathrm{Ein}(z) = \int_0^z (1 - e^{-v})\, dv/v$, one has

$$\mathcal{D}_{s,x}(t) = \sum_{k=0}^{s-1} \mathrm{Ein}(-\rho_k u) - \log(1 + u^s/x). \tag{48.4}$$

Consequently:

1. *$s = 1$: the ordinary derivative series converges absolutely and $\lim_{t\uparrow 1} \mathcal{D}_{1,x}(t) = \gamma + \log x$.*
2. *$s = 2$: the ordinary derivative series diverges, but $\lim_{t\uparrow 1} \mathcal{D}_{2,x}(t) = 2\gamma + \log x$.*
3. *$s \geq 3$: the radial Abel limit does not exist. More precisely, with $\theta = \pi/s$, $c = \cos\theta$, and $a = \sin\theta$,*

$$\mathcal{D}_{s,x}(t) = -\frac{2e^{cu}}{u}\cos(au - \theta) + O_{s,x}(e^{cu}/u^2), \qquad u \to \infty. \tag{48.5}$$

Proof. Inside the disk, differentiation gives $\mathcal{D}'_{s,x}(t) = -(\mathcal{B}_{s,x}(t) - 1)/t$. Substitute $t = u^s/(x + u^s)$ and use (48.3). The result is

$$\frac{d\mathcal{D}}{du} = \frac{s - \sum_k e^{\rho_k u}}{u} - \frac{s u^{s-1}}{x + u^s},$$

which is the derivative of (48.4). Both sides vanish at $u = 0$.

When $s = 1$, the formula is $\mathrm{Ein}(u) - \log(1 + u/x)$, whose limit is $\gamma + \log x$. When $s = 2$, the roots are $i, -i$, and

$$\mathrm{Ein}(iu) + \mathrm{Ein}(-iu) = 2\int_0^u \frac{1 - \cos v}{v}\, dv = 2(\gamma + \log u - \mathrm{Ci}(u)).$$

Here $\mathrm{Ci}(u) = -\int_u^\infty \cos v\, dv/v = O(1/u)$, the latter bound following by integration by parts. Subtracting $\log(1 + u^2/x)$ yields $2\gamma + \log x$.

For $s \geq 3$, the two roots with largest real part are $e^{i\theta}$ and $e^{-i\theta}$, with $c > 0$. For either root ρ, integration by parts on $[1, u]$ gives

$$\mathrm{Ein}(-\rho u) = -\frac{e^{\rho u}}{\rho u} + O(e^{cu}/u^2) + O(\log u).$$

The remaining roots have strictly smaller real parts; their terms and the real logarithm are absorbed in the displayed exponential remainder. The dominant pair sums to (48.5). Taking sequences with $au - \theta$ equal to even or odd multiples of π gives unbounded values of both signs. Hence a finite radial limit is impossible. □

Stride	Ordinary derivative series	Radial Abel limit	Meaning
1	Absolutely convergent	$\gamma + \log x$	Rational approximants with a proved denominator cost.
2	Terms fail to vanish	$2\gamma + \log x$	Summation recovers the derivative but supplies no small integer form.
≥ 3	Terms fail to vanish	Does not exist	Even this boundary regularization fails.

The stride-two identity is an analytic recovery theorem. It is not a transcendence theorem for an Abel sum of rational coefficients.

49 What a transcendence proof must still supply

The preceding results genuinely constrain possible descendants of this sampling construction. They do not select between γ algebraic and γ transcendental. We now state a standard elementary arithmetic target so that the outstanding implication is visible.

For an integer polynomial P, write $H(P) = \max|\text{coefficients of } P|$.

Lemma 49.1 (Algebraic value floor)**.** *Let α be algebraic of degree r, with primitive irreducible polynomial $f(X) = a\prod_{i=1}^{r}(X - \alpha_i)$, $\alpha_1 = \alpha$. Put $M(f) = |a|\prod_i \max(1, |\alpha_i|)$. For $0 \neq P \in \mathbb{Z}[X]$ of degree d with $P(\alpha) \neq 0$,*

$$|P(\alpha)| \geq M(f)^{-d}\big((d+1)H(P)\big)^{-(r-1)}. \tag{49.1}$$

Proof. The nonzero resultant is an integer, so $1 \leq |a|^d \prod_i |P(\alpha_i)|$. For $i \neq 1$, $|P(\alpha_i)| \leq (d+1)H(P)\max(1, |\alpha_i|)^d$. Rearrange and use $\max(1, |\alpha|) \geq 1$. □

Proposition 49.2 (Sufficient polynomial-value target)**.** *Suppose that for each integer $d \geq 1$ there are $\varepsilon_d > 0$ and integer polynomials $P_{d,k}$ of degree at most d with $H(P_{d,k}) \to \infty$ such that*

$$0 < |P_{d,k}(\gamma)| \leq H(P_{d,k})^{-(d-1+\varepsilon_d)} \tag{49.2}$$

for infinitely many k. Then γ is transcendental.

Proof. If γ were algebraic of degree r, take $d = r$. The fixed-degree lower bound (49.1) is a positive constant times $H(P_{r,k})^{-(r-1)}$, contradicting (49.2) as the height tends to infinity along the indicated subsequence. □

Status of the target. No family satisfying (49.2) is constructed here. The strict lower inequality is an essential arithmetic requirement, not something implied by a nonzero coefficient vector or by a numerical residual. The proposition is a standard resultant criterion, not counted as a new breakthrough or as a solution in a different vocabulary.

Within the sampling framework, the rigorous next questions are more specific than "find another representation of gamma":

Problem 49.3 (Growing support and simultaneous cancellation)**.** Construct explicitly specified growing sampling/filter families that control both their reduced arithmetic height and their true analytic error, while retaining a nonzero target coefficient. Determine whether windows of order $\sqrt{N}$ or larger can cancel the prime residues in (45.1) at sufficiently many blocks without making the coefficient cost larger than the error gain.

Problem 49.4 (Higher-degree nonvanishing)**.** If sampling-based integer polynomials in γ are constructed, prove nonvanishing and a height estimate of the form (49.2) degree by degree. Taking powers of the same linear form alone merely multiplies both its error and its height costs.

The short-window theorem does not say that every successful filter must use a long window. Very large coefficients can in principle create a much smaller analytic error while the denominator remains large; this possibility is outside Theorem 46.2. Similarly, the fixed-scale theorem leaves N-dependent scales and supports open. The restrictions in the proved results must not be silently removed.

50 Exact computations and theorem status

The following values are computed from exact fractions. The reduced denominator digit count is exact; the displayed decimal approximants are rounded and serve only to illustrate the slow oscillatory convergence.

N	$R_N(1)$	Digits of q_N	$\log_{10} q_N$
10	0.674852187570	10	9.660134
20	0.488066607111	27	26.151034
50	0.542071017225	84	83.940879
100	0.566497249360	197	196.353971

For example,

$$R_{10}(1) = \frac{3085618559}{4572288000}, \qquad R_{20}(1) = \frac{691055323045378490384005 63}{14159037167814523944960000 0}.$$

The executable appendix verifies the prime residues and fixed-window congruences over an explicit finite range using integer and fraction arithmetic. Finite checks supplement, and do not replace, the proofs.

Assertion	Logical status
Newton–Laguerre reconstruction	Known identity; independently justified here, with attribution.
Exact large-prime residue and $\frac{1}{2}$ denominator bound	Proved, including reduction to lowest terms.
Short growing integer windows	Denominator bound proved without a coefficient-size assumption.
Persistence of the tail	Proved under the stated weighted-moment condition.
Fixed rational-scale $\frac{1}{3}$ bound	Proved; exponent not claimed optimal.
Stride summability transition	Proved for fixed positive real scale and positive integer stride.
Polynomial-value implication	Proved conditional implication; its hypotheses for γ are unproved.
Irrationality or transcendence of γ	Not proved.

The change of viewpoint succeeds in turning some vague denominator concerns into exact congruences and a growing-family theorem. It does not yield the desired contradiction under algebraicity. The remaining work is an arithmetic construction with simultaneous height control and nonvanishing, not an additional analytic identity.

51 A guide to the remaining latitude

Theorems in this sampling part apply to particular sampling operations. The following distinctions prevent overreading them.

1. *A large denominator is not a proof of failure by itself.* An even smaller nonzero error could still suffice. The combined obstruction uses both the denominator theorem and the tail estimate.
2. *A positive-density obstruction is not an all-subsequence obstruction.* An exceptionally selected subsequence is not ruled out by the phase argument.
3. *Unbounded coefficients are covered only where stated.* The growing-window denominator theorem allows them. The persistence of the tail additionally needs (46.2).
4. *The arithmetic scale parameters are fixed in Theorem 47.1.* Its constants are not asserted uniform for an increasing list of rational scales or weights.
5. *Abel recovery is an analytic statement.* No theorem here transfers transcendence from a generating function to its boundary sum.
6. *An open criterion is not a proved arithmetic estimate.* Proposition 49.2 becomes a transcendence proof only after the polynomial family and both inequalities in (49.2) have been established.

Part VI
Inverse gamma interpolation and factorial reduction

52 Newton interpolation of the principal inverse gamma function

Ulgenes [25, Conjecture 3.1] proposed an explicit inverse-gamma expansion with logarithmic factorial nodes and left its convergence on the entire real principal-branch interval conjectural. We prove that precise statement below. The proof uses the established conformal geometry of $\log\Gamma$ [28]. Newton expansion of Stieltjes-class functions under reciprocal-node divergence is a classical method, appearing in van Herk [29, Theorem 48]; we provide the needed kernel argument in full rather than asserting a new general interpolation principle.

52.1 The conjecture and the result

Let α be the unique positive zero of the digamma function, and put

$$a = \log\Gamma(\alpha), \qquad c = -a.$$

Then $a < 0$ and $c > 0$: $\log\Gamma$ has its strict minimum at $\alpha \in (1,2)$, whereas $\Gamma(1) = \Gamma(2) = 1$. For real $y > a$, let $F(y) > \alpha$ be the unique solution of

$$\log\Gamma(F(y)) = y.$$

Consequently $F(\log x) = \operatorname{inv}\Gamma_0(x)$ for $x > \Gamma(\alpha)$, with the principal real branch convention of Ulgenes. Define the interpolation nodes

$$y_j = \log((j+1)!), \qquad j = 0, 1, 2, \dots.$$

In particular $y_0 = 0$ and $F(y_j) = j + 2$.

Theorem 52.1 (Ulgenes's Conjecture 3.1)**.** *For every real $x > \Gamma(\alpha)$,*

$$\operatorname{inv}\Gamma_0(x) = 2 + \sum_{n=1}^{\infty} C_n \prod_{i=1}^{n} \log\frac{x}{i!},$$

$$C_n = \sum_{k=0}^{n} k \left[\left(\prod_{i=1}^{k} \log\frac{(k+1)!}{i!} \right) \left(\prod_{i=1}^{n-k} \log\frac{(k+1)!}{(k+1+i)!} \right) \right]^{-1}.$$

Empty products equal one. The series is absolutely convergent at every point in this real interval. The conjecture's additional $n = 0$ summand is zero, so this is exactly its displayed formula.

We give all the interpolation and convergence arguments. The only special-function input is Pedersen's conformal-mapping theorem for $\log\Gamma$, together with the standard Herglotz representation for Pick functions.

Lemma 52.2 (The correct Pick function)**.** *F extends holomorphically to the upper half-plane, maps it into itself, and extends holomorphically through (a, ∞) with real values there. There are $b \geq 0$ and a positive Borel measure μ supported on $[c, \infty)$ such that*

$$F(y) = 2 + by + \int_c^{\infty} \frac{y}{s(s+y)}\, d\mu(s), \qquad y > a, \tag{52.1}$$

where

$$\int_c^{\infty} \frac{d\mu(s)}{1+s^2} < \infty.$$

Proof. Pedersen [28, Section 2, equation (2)] states that the branch of $\log\Gamma$ holomorphic in the upper half-plane maps that half-plane conformally onto

$$V = \mathbb{C} \setminus \bigcup_{k=0}^{\infty} \left([\log|\Gamma(x_k)|, \infty) - ik\pi\right),$$

where $x_0 = \alpha$ and the remaining x_k are the negative real critical points of Γ. Since V contains the entire upper half-plane, its conformal inverse restricts to a holomorphic map $F : \mathbb{H} \to \mathbb{H}$.

For any $u > \alpha$, $\psi(u) > 0$. The real inverse of $\log\Gamma$ therefore has a holomorphic local extension at $y = \log\Gamma(u)$, with derivative $1/\psi(u) > 0$. It maps nearby points immediately above the real axis into the upper half-plane. By the injectivity of $\log\Gamma$ on that half-plane, this local inverse is the same as the inverse just constructed. These local inverses give the claimed holomorphic continuation through (a, ∞).

The Herglotz representation now has a measure ν supported on $(-\infty, a]$, since F extends holomorphically with real values through (a, ∞):

$$F(z) = A + bz + \int_{-\infty}^{a} \left(\frac{1}{t - z} - \frac{t}{1 + t^2}\right) d\nu(t), \qquad b \geq 0.$$

Subtract its value at $z = 0$, which is $F(0) = 2$, and use $s = -t$. The resulting positive pushforward measure μ gives (52.1). The integral after subtraction is absolutely convergent, because $s \geq c > 0$ and its integrand is $O_y((1+s^2)^{-1})$. No assertion about the linear coefficient b is needed below. □

Lemma 52.3 (Exact Newton remainder)**.** *Let P_N be the polynomial of degree at most N interpolating $F(y_j) = j + 2$ at $y_0, \ldots, y_N$. For $N \geq 1$ and real $y > a$,*

$$F(y) - P_N(y) = (-1)^N \prod_{j=0}^{N} (y - y_j) \int_c^{\infty} \frac{d\mu(s)}{(s + y) \prod_{j=0}^{N} (s + y_j)}. \tag{52.2}$$

Proof. Write L_N for interpolation at $y_0, \ldots, y_N$. For the Cauchy kernel $h_s(y) = 1/(s + y)$, elementary divided differences give

$$h_s(y) - L_N h_s(y) = \frac{(-1)^{N+1} \prod_{j=0}^{N} (y - y_j)}{(s + y) \prod_{j=0}^{N} (s + y_j)}.$$

For example, this identity follows immediately by subtracting the two rational functions, observing the $N + 1$ interpolation zeros, and comparing residues at $y = -s$. Since

$$\frac{y}{s(s + y)} = \frac{1}{s} - h_s(y),$$

its remainder has the opposite sign. Interpolation reproduces the affine term $2 + by$ when $N \geq 1$. Applying the finite linear operator L_N under the convergent integral in (52.1) proves the formula. This interchange involves only a finite linear combination of absolutely convergent integrals, so it requires no infinite-series interchange. □

Convergence. The assertion is immediate when y is one of the interpolation nodes. Fix henceforth real $y > a$ different from all nodes. First note that

$$\sum_{j=1}^{\infty} \frac{1}{y_j} = \infty. \tag{52.3}$$

Indeed $y_j = \log((j + 1)!) \leq (j + 1)\log(j + 1)$, and the sum of the reciprocals of the right-hand side diverges.

Choose $J \geq 0$ so that $y_j > y$ for all $j > J$. For every $s \geq c$ and $j > J$,

$$\frac{|y - y_j|}{s + y_j} = 1 - \frac{s + y}{s + y_j} \in (0, 1),$$

because $s + y \geq c + y > 0$. For each fixed s, the sum $\sum_{j>J}(s + y)/(s + y_j)$ diverges by (52.3). Consequently

$$\prod_{j=J+1}^{N} \frac{|y - y_j|}{s + y_j} \longrightarrow 0.$$

The absolute value of the remainder integrand, including its exterior polynomial factor, is bounded for $N \geq \max(J, 1)$ by

$$\frac{1}{s + y} \prod_{j=0}^{J} \frac{|y - y_j|}{s + y_j}.$$

Here every factor has a denominator bounded away from zero on $[c, \infty)$. Moreover, the $j = 0$ factor is $|y|/s$, so this bound is $O_y(s^{-2})$ at infinity. It is therefore at most $K_y/(1 + s^2)$ for a finite constant K_y. Dominated convergence in (52.2) proves $P_N(y) \to F(y)$. □

Identification of the coefficients. The Newton form of P_N is

$$P_N(y) = 2 + \sum_{n=1}^{N} F[y_0, \ldots, y_n] \prod_{j=0}^{n-1} (y - y_j),$$

where the divided difference is

$$F[y_0, \ldots, y_n] = \sum_{k=0}^{n} \frac{k + 2}{\prod_{\substack{0 \leq j \leq n \\ j \neq k}} (y_k - y_j)} = \sum_{k=0}^{n} \frac{k}{\prod_{\substack{0 \leq j \leq n \\ j \neq k}} (y_k - y_j)}.$$

The last equality holds because an order-n divided difference of the constant function 2 vanishes for $n \geq 1$. For $j < k$, putting $i = j + 1$ gives

$$y_k - y_j = \log \frac{(k + 1)!}{i!}, \qquad i = 1, \ldots, k,$$

and for $j > k$, putting $i = j - k$ gives

$$y_k - y_j = \log \frac{(k + 1)!}{(k + 1 + i)!}, \qquad i = 1, \ldots, n - k.$$

All these factors are nonzero. Thus the divided difference is exactly C_n in the theorem. Finally, setting $y = \log x$ gives

$$\prod_{j=0}^{n-1} (y - y_j) = \prod_{i=1}^{n} \log \frac{x}{i!},$$

which identifies the conjectured series term for term. □

Absolute convergence. For $n \geq 2$, applying a divided difference directly to (52.1) yields

$$C_n = (-1)^{n-1} \int_c^{\infty} \frac{d\mu(s)}{\prod_{j=0}^{n} (s + y_j)}.$$

For $n = 1$ the same formula holds with the additional term b. Therefore $(-1)^{n-1} C_n \geq 0$ for every $n \geq 1$. At a fixed real nonnode y, all sufficiently late factors $y - y_j$ are negative. The signs of $C_n \prod_{j=0}^{n-1} (y - y_j)$ are consequently constant for all sufficiently large n. The already proved convergence of the Newton partial sums then implies absolute convergence. At a node the series terminates and the assertion is automatic. □

Proposition 52.4 (Normal convergence and an explicit error bound)**.** *The Newton series converges absolutely and locally uniformly on the complex half-plane $\Re y > a$. For any compact subset K of this half-plane, put*

$$\delta_K = \min_{y\in K}(\Re y + c) > 0.$$

There is a finite constant A_K such that, for all $N \geq 2$,

$$\sup_{y\in K} |F(y) - P_N(y)| \leq A_K(\log(N+2))^{-\delta_K}.$$

In particular, at a real point $x > \Gamma(\alpha)$ the error of the conjectured series through order N is

$$O_x\big((\log(N+2))^{-\log(x/\Gamma(\alpha))}\big).$$

Here absolute convergence always means absolute convergence of the outer Newton series, whose coefficients C_n are the indicated finite sums. No absolute convergence of a flattened double series is asserted.

Proof. Reflection and the local inverses used above give a holomorphic extension of F to $\mathbb{C} \setminus (-\infty, a]$. The Herglotz representation and the exact remainder continue to hold there, in particular on the half-plane under consideration.

Fix a compact K in this half-plane. For j sufficiently large, uniformly in $y \in K$,

$$\begin{aligned}\log\frac{|y_j - y|}{y_j + c} &= \Re\log\left(1 - \frac{y+c}{y_j+c}\right)\\ &\leq -\frac{\delta_K}{y_j+c} + \frac{B_K}{(y_j+c)^2}\end{aligned}$$

for a finite constant B_K. Since $s \geq c$, the same upper bound holds with $y_j + s$ in the denominator on the left. By Stirling's formula, or the elementary integral bounds for $\log(n!)$,

$$y_j = (j+1)\log(j+1) - (j+1) + O(\log(j+1)),$$

and hence

$$\sum_{j=1}^{N}\frac{1}{y_j+c} = \log\log(N+2) + O(1), \qquad \sum_{j=1}^{\infty}\frac{1}{(y_j+c)^2} < \infty.$$

For completeness, the first assertion follows by subtracting $1/(j\log j)$ for $j \geq 2$: the resulting difference is $O(1/(j\log^2 j))$, an absolutely summable sequence. It follows that the tail product in the exact remainder is bounded by a constant times $(\log(N+2))^{-\delta_K}$, uniformly in $y \in K$ and $s \geq c$.

The finite prefix, together with $1/(s+y)$, is bounded uniformly on K by a constant times $(1+s^2)^{-1}$. Indeed $|s+y| \geq s - c + \delta_K$, and the $j = 0$ factor is $|y|/s$. Integration gives the asserted error bound, and therefore local uniform convergence.

To verify local uniform *absolute* convergence as well, choose a real nonnode r with $a < r < \min_{y\in K}\Re y$. For all sufficiently large j,

$$|y_j - y| \leq y_j - r \qquad (y \in K),$$

because the difference of the squared right and left sides is $2y_j(\Re y - r) + r^2 - |y|^2$, which is eventually positive uniformly on K. The finitely many earlier factors can be absorbed into a constant. Thus the absolute values of the Newton summands on K are bounded by a fixed constant times the absolute summands at r. The latter sum converges by the preceding proof, and the Weierstrass test finishes the argument. □

52.2 Scope and attribution

This establishes Ulgenes's displayed formula on its full conjectured interval, with normal convergence and an explicit error bound. The proof applies classical Newton–Stieltjes interpolation to Pedersen's inverse-function geometry. It makes no claim about the arithmetic nature of Euler's constant.

53 A counterexample to the rational-root criterion for factorial reduction

For $a, b \in \mathbb{Z}[n]$, use the convention

$$\mathrm{PCF}[a,b] = a(0) + \cfrac{b(1)}{a(1) + \cfrac{b(2)}{a(2) + \ddots}}.$$

The canonical numerator and denominator sequences satisfy

$$u_n = a(n)u_{n-1} + b(n)u_{n-2} \quad (n \geq 1), \qquad \begin{aligned} p_{-1} &= 1, \quad p_0 = a(0), \\ q_{-1} &= 0, \quad q_0 = 1. \end{aligned} \tag{53.1}$$

Write $G_n = \gcd(p_n, q_n) > 0$. In the balanced-degree case $\deg b = 2 \deg a$, Ben David et al. [30] define the factorial-reduction parameter by

$$\log \lambda = \limsup_{n\to\infty} \frac{1}{n} \log \frac{G_n}{(n!)^{\deg a}}. \tag{53.2}$$

Factorial reduction means $\lambda > 0$. Their Conjecture 1.1 says that a polynomial b admits some a with this property exactly when b factors completely into linear polynomials over $\mathbb{Q}$. We use that conjecture as stated, together with its canonical initialization and its definition (53.2).

Theorem 53.1. *Set*

$$a(n) = n^2 + 7n + 7, \tag{53.3}$$

$$\begin{aligned} b(n) &= ((n+4)^2 - 2)((n+2)^2 - 2) \\ &= n^4 + 12n^3 + 48n^2 + 72n + 28. \end{aligned} \tag{53.4}$$

Then the following assertions hold.

1. *All coefficients are positive integers,* $\deg b = 2 \deg a$, *and both leading coefficients are* 1.
2. *Neither quadratic factor in* (53.4) *splits over* $\mathbb{Q}$, *and* $\gcd_{\mathbb{Q}[n]}(a, b) = 1$.
3. *The canonical convergents have factorial reduction, with the actual limit in* (53.2) *equal to* 0*; in particular* $\lambda = 1$.
4. *The continued fraction converges to the positive irrational number*

$$\mathrm{PCF}[a,b] = \frac{35 + 49\sqrt{5}}{11}.$$

Consequently Conjecture 1.1 of [30], as stated, is false.

54 Exact canonical convergents

Let $F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$ be the Fibonacci numbers. Define

$$C_0 = 1, \qquad C_n = \prod_{j=1}^{n} ((j+3)^2 - 2) \quad (n \geq 1), \tag{54.1}$$

$$P_n = (n+7)F_{n+1} + (2n+9)F_n, \tag{54.2}$$

$$Q_n = (3n+14)F_{n+1} - (n+1)F_n. \tag{54.3}$$

Proposition 54.1. *For every $n \geq 0$, the sequences initialized in (53.1) are*

$$p_n = C_n P_n, \qquad q_n = \frac{C_n}{14} Q_n. \tag{54.4}$$

All denominators q_n are positive.

Proof. Put $H_n = (n+3)^2 - 2$ and $z_n = u_n / C_n$. For $n \geq 2$, division of (53.1) by C_n reduces the required recurrence to

$$H_n z_n = a(n) z_{n-1} + ((n+4)^2 - 2) z_{n-2}. \tag{54.5}$$

Let r be either root of $r^2 - r - 1 = 0$. Then

$$z_n = r^n (n + 4 + r) \tag{54.6}$$

satisfies (54.5). For completeness, with $k = n + 4$, the identity to check is

$$((k-1)^2 - 2) r^2 (k + r) = (k^2 - k - 5) r (k - 1 + r) + (k^2 - 2)(k - 2 + r).$$

Substituting $r^2 = r + 1$ makes both sides

$$(k^2 - 2k - 1)\big((k+2)r + k + 1\big).$$

Write $\varphi = (1 + \sqrt{5})/2$ and $\psi = (1 - \sqrt{5})/2$. Let $L_n = \varphi^n + \psi^n$ be the Lucas numbers and set

$$X_n = (n+4)L_n + L_{n+1}, \qquad Y_n = (n+4)F_n + F_{n+1}.$$

Binet's identities show that X_n and Y_n are linear combinations of the two solutions (54.6). The elementary identities $L_n = 2F_{n+1} - F_n$ and $L_{n+1} = F_{n+1} + 2F_n$ give

$$P_n = \frac{X_n + 5Y_n}{2}, \qquad Q_n = \frac{3X_n + Y_n}{2}.$$

Thus P_n, Q_n satisfy (54.5), and the proposed p_n, q_n satisfy the original recurrence for $n \geq 2$. Their first two values are

$$(p_0, q_0) = (7, 1), \qquad (p_1, q_1) = (266, 15).$$

These agree with the canonical initial conditions because $a(0) = 7$, $a(1) = 15$, and $b(1) = 161$. Uniqueness of solutions of the second-order recurrence proves (54.4). Finally, $C_n > 0$, and for $n \geq 1$ the inequality $F_{n+1} \geq F_n > 0$ gives $Q_n \geq (2n+13)F_n > 0$. Also $q_0 = 1$. □

55 Upper and lower bounds for the greatest common divisor

Proposition 55.1. *For every $n \geq 1$, put $D_n = n^2 + 9n + 19$. Then*

$$G_n = \frac{C_n}{14}\, d_n, \qquad d_n = \gcd(14P_n, Q_n), \tag{55.1}$$

$$1 \leq d_n \leq 98 D_n. \tag{55.2}$$

In particular,

$$\frac{C_n}{14} \leq G_n \leq 7C_n(n^2 + 9n + 19). \tag{55.3}$$

Proof. The first factor of C_n is 14, so $C_n/14$ is a positive integer for $n \geq 1$. Formula (54.4) gives (55.1). Moreover,

$$\begin{pmatrix} 14P_n \\ Q_n \end{pmatrix} = \begin{pmatrix} 14(n+7) & 14(2n+9) \\ 3n+14 & -(n+1) \end{pmatrix} \begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix}.$$

The determinant of this integer matrix is $-98D_n$. Multiplication by the adjugate shows that d_n divides both $98D_nF_n$ and $98D_nF_{n+1}$. Since consecutive Fibonacci numbers are relatively prime, $d_n \mid 98D_n$. This proves (55.2) and (55.3). □

The product in (54.1) has the useful form

$$\frac{C_n}{(n!)^2} = \left(\frac{(n+1)(n+2)(n+3)}{6}\right)^2 \prod_{m=4}^{n+3} \left(1 - \frac{2}{m^2}\right). \tag{55.4}$$

The infinite product on the right converges to a positive number: every factor is positive, and the series of absolute values of their logarithms converges. Consequently

$$\log C_n - 2\log(n!) = 6\log n + O(1).$$

Combining this with (55.3) yields the two-sided estimate

$$6\log n + O(1) \;\leq\; \log \frac{G_n}{(n!)^2} \;\leq\; 8\log n + O(1),$$

where the bounded terms on the two sides may differ. Division by n proves

$$\lim_{n\to\infty} \frac{1}{n} \log \frac{G_n}{(n!)^2} = 0, \qquad \lambda = 1. \tag{55.5}$$

This also explains the arithmetic mechanism: the large common factor C_n leaves reduced denominators of exponential size, despite the irrational roots of the polynomial b.

56 Irrational limit and coprime coefficient polynomials

By (54.4), the convergent is $14P_n/Q_n$. Since $F_{n+1}/F_n \to \varphi$, formulas (54.2)–(54.3) give

$$\lim_{n\to\infty} \frac{p_n}{q_n} = 14\,\frac{\varphi + 2}{3\varphi - 1} = \frac{35 + 49\sqrt{5}}{11}.$$

Thus the limit exists and is irrational. The convergence condition $A^2 + 4B > 0$ used in [30] also holds, since the leading coefficients here are $A = B = 1$.

Each quadratic factor of b has discriminant 8 and is irreducible over $\mathbb{Q}$. To check coprimality of the coefficient polynomials, write $b = b_1 b_2$, where

$$b_1 = n^2 + 8n + 14, \qquad b_2 = n^2 + 4n + 2.$$

A common root of a and b_1 would satisfy $b_1 - a = n + 7 = 0$, whereas $a(-7) = 7 \neq 0$. A common root of a and b_2 would satisfy $b_2 - a = -3n - 5 = 0$, whereas $a(-5/3) = -17/9 \neq 0$. Therefore $\gcd_{\mathbb{Q}[n]}(a, b) = 1$. This completes the proof of Theorem 53.1.

57 Scope, versions, and equivalence transformations

The target is the rational-root condition in the *published 2024* Conjecture 1.1. The 2021 preprint [31] instead required a factorization into two rational polynomials of equal degree. Our b satisfies that weaker condition and does not refute that version.

Scalar polynomial inflation replaces a coefficient pair by

$$a(n) \longmapsto c(n)a(n), \qquad b(n) \longmapsto c(n)c(n-1)b(n).$$

This transformation is discussed in [30, Appendix C]. If our pair arose this way from a polynomial pair with nonconstant $c \in \mathbb{Q}[n]$, then c would divide both a and b, a contradiction. Here coprimality is in $\mathbb{Q}[n]$; the integer values $a(n)$ and $b(n)$ need not be coprime at every integer n.

We claim no inequivalence under all rational recurrence transformations: the construction explicitly uses Fibonacci solutions. The degree pattern is $(2, 4)$, so the separate $(1, 2)$ classification conjecture remains untouched. No arithmetic conclusion about Euler's constant follows.

A Self-contained exact verification code

The following Python 3 code uses only the standard library. It reproduces the rational table and checks all eligible prime residues for $1 \le N \le 120$ at four rational scales. It also checks the congruence for the fixed integer stencil $2R_N - R_{N+1}$ whenever no prime block is crossed. All assertions concern exact rational numbers. The executed checks returned 6,548 successful prime-residue comparisons and 1,582 successful stencil comparisons.

```
from fractions import Fraction as F
from math import comb, factorial, isqrt, log10

def primes_to(n):
    out = []
    for p in range(2, n + 1):
        if all(p % q for q in out if q * q <= p):
            out.append(p)
    return out

def R(n, x=F(1)):
    h = sum((F(1, k) for k in range(1, n + 1)), F(0))
    return -h + sum(
        (F((-1)**(k-1) * comb(n, k), k * factorial(k))
         * x**k for k in range(1, n + 1)), F(0))

def mod_fraction(a, p):
    assert a.denominator % p
    return (a.numerator * pow(a.denominator, -1, p)) % p

def vp_int(a, p):
    assert a != 0
    a, v = abs(a), 0
    while a % p == 0:
        a //= p
        v += 1
    return v

def vp(a, p):
    return vp_int(a.numerator, p) - vp_int(a.denominator, p)

scales = [F(1, 2), F(1), F(3, 2), F(2)]
residue_checks = 0
stencil_checks = 0
for n in range(1, 121):
    for x in scales:
        rn = R(n, x)
        for p in primes_to(n):
            if p*p <= n or (x.numerator*x.denominator) % p == 0:
                continue
            m = n // p
            expected = mod_fraction(-x**m / (m * factorial(m)), p)
            assert mod_fraction(p**(m+1) * rn, p) == expected
            assert expected != 0
            assert vp(rn, p) == -(m + 1)
            residue_checks += 1
    tn = 2*R(n) - R(n+1)
    for p in primes_to(n):
        if p*p <= n+1 or n//p != (n+1)//p:
            continue
        m = n//p
        expected = mod_fraction(F(-1, m*factorial(m)), p)
        assert mod_fraction(p**(m+1)*tn, p) == expected
        assert vp(tn, p) == -(m+1)
        stencil_checks += 1

print('Exact residue checks:', residue_checks)
print('Exact stencil checks:', stencil_checks)
for n in [10, 20, 50, 100]:
    v = R(n)
    print(n, format(float(v), '.12f'),
          len(str(v.denominator)), log10(v.denominator))
```

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